

# CHAPTER 10

## ***Canonical Forms***

### **10.1 Introduction**

---

Let  $T$  be a linear operator on a vector space of finite dimension. As seen in [Chapter 6](#),  $T$  may not have a diagonal matrix representation. However, it is still possible to “simplify” the matrix representation of  $T$  in a number of ways. This is the main topic of this chapter. In particular, we obtain the primary decomposition theorem, and the triangular, Jordan, and rational canonical forms.

We comment that the triangular and Jordan canonical forms exist for  $T$  if and only if the characteristic polynomial  $\Delta(t)$  of  $T$  has all its roots in the base field  $K$ . This is always true if  $K$  is the complex field  $\mathbf{C}$  but may not be true if  $K$  is the real field  $\mathbf{R}$ .

We also introduce the idea of a *quotient space*. This is a very powerful tool, and it will be used in the proof of the existence of the triangular and rational canonical forms.

### **10.2 Triangular Form**

---

Let  $T$  be a linear operator on an  $n$ -dimensional vector space  $V$ . Suppose  $T$  can be represented by the triangular matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ & a_{22} & \cdots & a_{2n} \\ & & \cdots & \cdots \\ & & & a_{nn} \end{bmatrix}$$

Then the characteristic polynomial  $\Delta(t)$  of  $T$  is a product of linear factors; that is,

$$\Delta(t) = \det(tI - A) = (t - a_{11})(t - a_{22}) \cdots (t - a_{nn})$$

The converse is also true and is an important theorem (proved in [Problem 10.28](#)).

**THEOREM 10.1:** Let  $T: V \rightarrow V$  be a linear operator whose characteristic polynomial factors into linear polynomials. Then there exists a basis of  $V$  in which  $T$  is represented by a triangular matrix.

**THEOREM 10.1:** (Alternative Form) Let  $A$  be a square matrix whose characteristic polynomial factors into linear polynomials. Then  $A$  is similar to a triangular matrix—that is, there exists an invertible matrix  $P$  such that  $P^{-1}AP$  is triangular.

We say that an operator  $T$  can be brought into triangular form if it can be represented by a triangular matrix. Note that in this case, the eigenvalues of  $T$  are precisely those entries appearing on the main diagonal. We give an application of this remark.

**EXAMPLE 10.1** Let  $A$  be a square matrix over the complex field  $\mathbf{C}$ . Suppose  $\lambda$  is an eigenvalue of  $A^2$ . Show that  $\sqrt{\lambda}$  or  $-\sqrt{\lambda}$  is an eigenvalue of  $A$ .

By [Theorem 10.1](#),  $A$  and  $A^2$  are similar, respectively, to triangular matrices of the form

$$B = \begin{bmatrix} \mu_1 & * & \dots & * \\ & \mu_2 & \dots & * \\ & & \dots & \dots \\ & & & \mu_n \end{bmatrix} \quad \text{and} \quad B^2 = \begin{bmatrix} \mu_1^2 & * & \dots & * \\ & \mu_2^2 & \dots & * \\ & & \dots & \dots \\ & & & \mu_n^2 \end{bmatrix}$$

Because similar matrices have the same eigenvalues,  $\lambda = \mu_i^2$  for some  $i$ . Hence,  $\mu_i = \sqrt{\lambda}$  or  $\mu_i = -\sqrt{\lambda}$  is an eigenvalue of  $A$ .

## 10.3 Invariance

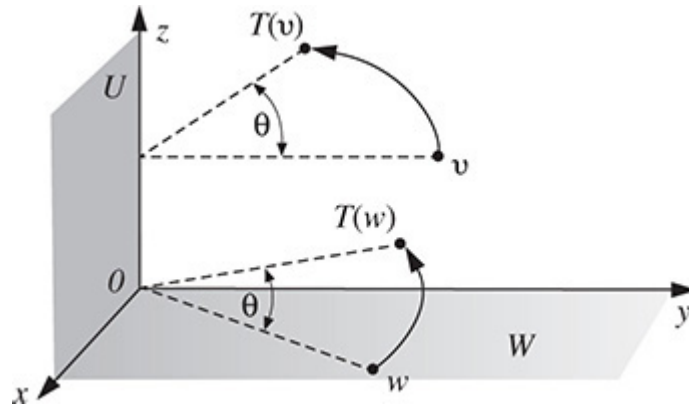
---

Let  $T: V \rightarrow V$  be linear. A subspace  $W$  of  $V$  is said to be *invariant under  $T$*  or  *$T$ -invariant* if  $T$  maps  $W$  into itself—that is, if  $v \in W$  implies  $T(v) \in W$ . In this case,  $T$  restricted to  $W$  defines a linear operator on  $W$ ; that is,  $T$  induces a linear operator  $\hat{T}: W \rightarrow W$  defined by  $\hat{T}(w) = T(w)$  for every  $w \in W$ .

### EXAMPLE 10.2

- (a) Let  $T: \mathbf{R}^3 \rightarrow \mathbf{R}^3$  be the following linear operator, which rotates each vector  $v$  about the  $z$ -axis by an angle  $\theta$  (shown in [Fig. 10-1](#)):

$$T(x, y, z) = (x \cos \theta - y \sin \theta, \ x \sin \theta + y \cos \theta, \ z)$$



**Figure 10-1**

Observe that each vector  $w = (a, b, 0)$  in the  $xy$ -plane  $W$  remains in  $W$  under the mapping  $T$ ; hence,  $W$  is  $T$ -invariant. Observe also that the  $z$ -axis  $U$  is invariant under  $T$ . Furthermore, the restriction of  $T$  to  $W$  rotates each vector about the origin  $O$ , and the restriction of  $T$  to  $U$  is the identity mapping of  $U$ .

- (b) Nonzero eigenvectors of a linear operator  $T: V \rightarrow V$  may be characterized as generators of  $T$ -invariant one-dimensional subspaces. Suppose  $T(u) = \lambda u$ ,  $u \neq 0$ . Then  $W = \{ku, k \in K\}$ , the one-dimensional subspace generated by  $u$ , is invariant under  $T$  because

$$T(ku) = kT(u) = k(\lambda u) = k\lambda u \in W$$

Conversely, suppose  $\dim U = 1$  and  $u \neq 0$  spans  $U$ , and  $U$  is invariant under  $T$ . Then  $T(u) \in U$  and so  $T(u)$  is a multiple of  $u$ —that is,  $T(u) = \mu u$ . Hence,  $u$  is an eigenvector of  $T$ .

The next theorem (proved in [Problem 10.3](#)) gives us an important class of invariant subspaces.

**THEOREM 10.2:** Let  $T: V \rightarrow V$  be any linear operator, and let  $f(t)$  be any polynomial. Then the kernel of  $f(t)$  is invariant under  $T$ .

The notion of invariance is related to matrix representations ([Problem 10.5](#)) as follows.

**THEOREM 10.3:** Suppose  $W$  is an invariant subspace of  $T: V \rightarrow V$ . Then  $T$  has a blockmatrix representation  $\begin{bmatrix} A & B \\ 0 & C \end{bmatrix}$ , where  $A$  is a matrix representation of the restriction  $\hat{T}$  of  $T$  to  $W$ .

## 10.4 Invariant Direct-Sum Decompositions

---

A vector space  $V$  is termed the *direct sum* of subspaces  $W_1, \dots, W_r$  written

$$V = W_1 \oplus W_2 \oplus \dots \oplus W_r$$

if every vector  $v \in V$  can be written uniquely in the form

$$v = w_1 + w_2 + \dots + w_r, \quad \text{with} \quad w_i \in W_i$$

The following theorem (proved in [Problem 10.7](#)) holds.

**THEOREM 10.4:** Suppose  $W_1, W_2, \dots, W_r$  are subspaces of  $V$ , and suppose

$$B_1 = \{w_{11}, w_{12}, \dots, w_{1n_1}\}, \quad \dots, \quad B_r = \{w_{r1}, w_{r2}, \dots, w_{rn_r}\}$$

are bases of  $W_1, W_2, \dots, W_r$  respectively. Then  $V$  is the direct sum of the  $W_i$  if and only if the union  $B = B_1 \cup \dots \cup B_r$  is a basis of  $V$ .

Now suppose  $T: V \rightarrow V$  is linear and  $V$  is the direct sum of (nonzero)  $T$ -invariant subspaces  $W_1, W_2, \dots, W_r$ ; that is,

$$V = W_1 \oplus \dots \oplus W_r \quad \text{and} \quad T(W_i) \subseteq W_i, \quad i = 1, \dots, r$$

Let  $T_i$  denote the restriction of  $T$  to  $W_i$ . Then  $T$  is said to be *decomposable* into the operators  $T_i$  or  $T$  is said to be the *direct sum* of the  $T_i$ , written  $T = T_1 \oplus \dots \oplus T_r$ . Also, the subspaces  $W_1, \dots, W_r$  are said to *reduce*  $T$  or to form a  *$T$ -invariant direct-sum decomposition* of  $V$ .

Consider the special case where two subspaces  $U$  and  $W$  reduce an operator  $T: V \rightarrow V$ ; say  $\dim U = 2$  and  $\dim W = 3$ , and suppose  $\{u_1, u_2\}$  and  $\{w_1, w_2, w_3\}$  are bases of  $U$  and  $W$ , respectively. If  $T_1$  and  $T_2$  denote the restrictions of  $T$  to  $U$  and  $W$ , respectively, then

$$\begin{aligned} T_1(u_1) &= a_{11}u_1 + a_{12}u_2 & T_2(w_1) &= b_{11}w_1 + b_{12}w_2 + b_{13}w_3 \\ T_1(u_2) &= a_{21}u_1 + a_{22}u_2 & T_2(w_2) &= b_{21}w_1 + b_{22}w_2 + b_{23}w_3 \\ & & T_2(w_3) &= b_{31}w_1 + b_{32}w_2 + b_{33}w_3 \end{aligned}$$

Accordingly, the following matrices  $A, B, M$  are the matrix representations of  $T_1, T_2, T$ , respectively,

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}, \quad M = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}$$

The block diagonal matrix  $M$  results from the fact that  $\{u_1, u_2, w_1, w_2, w_3\}$  is a basis of  $V$  ([Theorem 10.4](#)), and that  $T(u_i) = T_1(u_i)$  and  $T(w_j) = T_2(w_j)$ .

A generalization of the above argument gives us the following theorem.

**THEOREM 10.5:** Suppose  $T: V \rightarrow V$  is linear and suppose  $V$  is the direct sum of  $T$ -invariant subspaces, say,  $W_1, \dots, W_r$ . If  $A_i$  is a matrix representation of the restriction of  $T$  to  $W_i$ , then  $T$  can be represented by the block diagonal matrix:

$$M = \text{diag}(A_1, A_2, \dots, A_r)$$

## 10.5 Primary Decomposition

---

The following theorem shows that any operator  $T: V \rightarrow V$  is decomposable into operators whose minimum polynomials are powers of irreducible polynomials. This is the first step in obtaining a canonical form for  $T$ .

**THEOREM 10.6:** (Primary Decomposition Theorem) Let  $T: V \rightarrow V$  be a linear operator with minimal polynomial

$$m(t) = f_1(t)^{n_1} f_2(t)^{n_2} \cdots f_r(t)^{n_r}$$

where the  $f_i(t)$  are distinct monic irreducible polynomials. Then  $V$  is the direct sum of  $T$ -invariant subspaces  $W_1, \dots, W_r$  where  $W_i$  is the kernel of  $f_i(T)^{n_i}$ . Moreover,  $f_i(t)^{n_i}$  is the minimal polynomial of the restriction of  $T$  to  $W_i$ .

The above polynomials  $f_i(t)^{n_i}$  are relatively prime. Therefore, the above fundamental theorem follows ([Problem 10.11](#)) from the next two theorems (proved in Problems 10.9 and 10.10, respectively).

**THEOREM 10.7:** Suppose  $T: V \rightarrow V$  is linear, and suppose  $f(t) = g(t)h(t)$  are polynomials such that  $f(t) = \mathbf{0}$  and  $g(t)$  and  $h(t)$  are relatively prime. Then  $V$  is the direct sum of the  $T$ -invariant subspace  $U$  and  $W$ , where  $U = \text{Ker } g(T)$  and  $W = \text{Ker } h(T)$ .

**THEOREM 10.8:** In [Theorem 10.7](#), if  $f(t)$  is the minimal polynomial of  $T$  [and  $g(t)$  and  $h(t)$  are monic], then  $g(t)$  and  $h(t)$  are the minimal polynomials of the restrictions of  $T$  to  $U$  and  $W$ , respectively.

We will also use the primary decomposition theorem to prove the following useful characterization of diagonalizable operators (see [Problem 10.12](#) for the proof).

**THEOREM 10.9:** A linear operator  $T: V \rightarrow V$  is diagonalizable if and only if its minimal polynomial  $m(t)$  is a product of distinct linear polynomials.

**THEOREM 10.9:** (Alternative Form) A matrix  $A$  is similar to a diagonal matrix if and only if its minimal polynomial is a product of distinct linear polynomials.

**EXAMPLE 10.3** Suppose  $A \neq I$  is a square matrix for which  $A^3 = I$ . Determine whether or not  $A$  is similar to a diagonal matrix if  $A$  is a matrix over: (i) the real field  $\mathbf{R}$ , (ii) the complex field  $\mathbf{C}$ .

Because  $A^3 = I$ ,  $A$  is a zero of the polynomial  $f(t) = t^3 - 1 = (t - 1)(t^2 + t + 1)$ . The minimal polynomial  $m(t)$  of  $A$  cannot be  $t - 1$ , because  $A \neq I$ . Hence,

$$m(t) = t^2 + t + 1 \quad \text{or} \quad m(t) = t^3 - 1$$

Because neither polynomial is a product of linear polynomials over  $\mathbf{R}$ ,  $A$  is not diagonalizable over  $\mathbf{R}$ . On the other hand, each of the polynomials is a product of distinct linear polynomials over  $\mathbf{C}$ . Hence,  $A$  is diagonalizable over  $\mathbf{C}$ .

## 10.6 Nilpotent Operators

---

A linear operator  $T: V \rightarrow V$  is termed *nilpotent* if  $T^n = \mathbf{0}$  for some positive integer  $n$ ; we call  $k$  the *index of nilpotency* of  $T$  if  $T^k = \mathbf{0}$  but  $T^{k-1} \neq \mathbf{0}$ . Analogously, a square matrix  $A$  is termed nilpotent if  $A^n = \mathbf{0}$  for some positive integer  $n$ , and of index  $k$  if  $A^k = \mathbf{0}$  but  $A^{k-1} \neq \mathbf{0}$ . Clearly the minimum polynomial of a nilpotent operator (matrix) of index  $k$  is  $m(t) = t^k$ ; hence, 0 is its only eigenvalue.

**EXAMPLE 10.4** The following two  $r$ -square matrices will be used throughout the chapter:

$$N = N(r) = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & 0 & \dots & 0 & 0 \end{bmatrix} \quad \text{and} \quad J(\lambda) = \begin{bmatrix} \lambda & 1 & 0 & \dots & 0 & 0 \\ 0 & \lambda & 1 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & \lambda & 1 \\ 0 & 0 & 0 & \dots & 0 & \lambda \end{bmatrix}$$

The first matrix  $N$ , called a *Jordan nilpotent block*, consists of 1's above the diagonal (called the *super-diagonal*), and 0's elsewhere. It is a nilpotent matrix of index  $r$ . (The matrix  $N$  of order 1 is just the  $1 \times 1$  zero matrix  $[0]$ .)

The second matrix  $J(\lambda)$ , called a *Jordan block* belonging to the eigenvalue  $\lambda$ , consists of  $\lambda$ 's on the diagonal, 1's on the superdiagonal, and 0's elsewhere. Observe that

$$J(\lambda) = \lambda I + N$$

In fact, we will prove that any linear operator  $T$  can be decomposed into operators, each of which is the sum of a

scalar operator and a nilpotent operator.

The following (proved in [Problem 10.16](#)) is a fundamental result on nilpotent operators.

**THEOREM 10.10:** Let  $T: V \rightarrow V$  be a nilpotent operator of index  $k$ . Then  $T$  has a block diagonal matrix representation in which each diagonal entry is a Jordan nilpotent block  $N$ . There is at least one  $N$  of order  $k$ , and all other  $N$  are of orders  $\leq k$ . The number of  $N$  of each possible order is uniquely determined by  $T$ . The total number of  $N$  of all orders is equal to the nullity of  $T$ .

The proof of [Theorem 10.10](#) shows that the number of  $N$  of order  $i$  is equal to  $2m_i - m_{i+1} - m_{i-1}$ , where  $m_i$  is the nullity of  $T^i$ .

## 10.7 Jordan Canonical Form

---

An operator  $T$  can be put into Jordan canonical form if its characteristic and minimal polynomials factor into linear polynomials. This is always true if  $K$  is the complex field  $\mathbf{C}$ . In any case, we can always extend the base field  $K$  to a field in which the characteristic and minimal polynomials do factor into linear factors; thus, in a broad sense, every operator has a Jordan canonical form. Analogously, every matrix is similar to a matrix in Jordan canonical form.

The following theorem (proved in [Problem 10.18](#)) describes the *Jordan canonical form*  $J$  of a linear operator  $T$ .

**THEOREM 10.11:** Let  $T: V \rightarrow V$  be a linear operator whose characteristic and minimal polynomials are, respectively,

$$\Delta(t) = (t - \lambda_1)^{n_1} \cdots (t - \lambda_r)^{n_r} \quad \text{and} \quad m(t) = (t - \lambda_1)^{m_1} \cdots (t - \lambda_r)^{m_r}$$

where the  $\lambda_i$  are distinct scalars. Then  $T$  has a block diagonal matrix representation  $J$  in which each diagonal entry is a Jordan block  $J_{ij} = J(\lambda_i)$ . For each  $\lambda_i$ , the corresponding  $J_{ij}$  have the following properties:

- (i) There is at least one  $J_{ij}$  of order  $m_i$ ; all other  $J_{ij}$  are of order  $\leq m_i$ .
- (ii) The sum of the orders of the  $J_{ij}$  is  $n_i$ .
- (iii) The number of  $J_{ij}$  equals the geometric multiplicity of  $\lambda_i$ .
- (iv) The number of  $J_{ij}$  of each possible order is uniquely determined by  $T$ .

**EXAMPLE 10.5** Suppose the characteristic and minimal polynomials of an operator  $T$  are, respectively,

$$\Delta(t) = (t - 2)^4(t - 5)^3 \quad \text{and} \quad m(t) = (t - 2)^2(t - 5)^3$$

Then the Jordan canonical form of  $T$  is one of the following block diagonal matrices:

$$\text{diag} \left( \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix} \right) \quad \text{or} \quad \text{diag} \left( \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, [2], [2], \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix} \right)$$

The first matrix occurs if  $T$  has two independent eigenvectors belonging to the eigenvalue 2; and the second matrix occurs if  $T$  has three independent eigenvectors belonging to the eigenvalue 2.

## 10.8 Cyclic Subspaces

---

Let  $T$  be a linear operator on a vector space  $V$  of finite dimension over  $K$ . Suppose  $v \in V$  and  $v \neq 0$ . The set of all vectors of the form  $f(T)(v)$ , where  $f(t)$  ranges over all polynomials over  $K$ , is a  $T$ -invariant subspace of  $V$  called the  *$T$ -cyclic subspace of  $V$  generated by  $v$* ; we denote it by  $Z(v, T)$  and denote the restriction of  $T$  to  $Z(v, T)$  by  $T_v$ . By Problem 10.56, we could equivalently define  $Z(v, T)$  as the intersection of all  $T$ -invariant subspaces of  $V$  containing  $v$ .

Now consider the sequence

$$v, T(v), T^2(v), T^3(v), \dots$$

of powers of  $T$  acting on  $v$ . Let  $k$  be the least integer such that  $T^k(v)$  is a linear combination of those vectors that precede it in the sequence, say,

$$T^k(v) = -a_{k-1}T^{k-1}(v) - \dots - a_1T(v) - a_0v$$

Then

$$m_v(t) = t^k + a_{k-1}t^{k-1} + \dots + a_1t + a_0$$

is the unique monic polynomial of lowest degree for which  $m_v(T)(v) = 0$ . We call  $m_v(t)$  the  *$T$ -annihilator of  $v$  and  $Z(v, T)$* .

The following theorem (proved in [Problem 10.29](#)) holds.

**THEOREM 10.12:** Let  $Z(v, T)$ ,  $T_v$ ,  $m_v(t)$  be defined as above. Then

- (i) The set  $\{v, T(v), \dots, T^{k-1}(v)\}$  is a basis of  $Z(v, T)$ ; hence,  $\dim Z(v, T) = k$ .
- (ii) The minimal polynomial of  $T_v$  is  $m_v(t)$ .

(iii) The matrix representation of  $T_v$  in the above basis is just the *companion matrix*  $C(m_v)$  of  $m_v(t)$ ; that is,

$$C(m_v) = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & -a_0 \\ 1 & 0 & 0 & \dots & 0 & -a_1 \\ 0 & 1 & 0 & \dots & 0 & -a_2 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 & -a_{k-2} \\ 0 & 0 & 0 & \dots & 1 & -a_{k-1} \end{bmatrix}$$

## 10.9 Rational Canonical Form

---

In this section, we present the rational canonical form for a linear operator  $T: V \rightarrow V$ . We emphasize that this form exists even when the minimal polynomial cannot be factored into linear polynomials. (Recall that this is not the case for the Jordan canonical form.)

**LEMMA 10.13:** Let  $T: V \rightarrow V$  be a linear operator whose minimal polynomial is  $f(t)^n$ , where  $f(t)$  is a monic irreducible polynomial. Then  $V$  is the direct sum

$$V = Z(v_1, T) \oplus \dots \oplus Z(v_r, T)$$

of  $T$ -cyclic subspaces  $Z(v_j, T)$  with corresponding  $T$ -annihilators

$$m_1 = n_{11} \geq n_{12} \geq \dots \geq n_{1r_1}, \quad \dots, \quad m_s = n_{s1} \geq n_{s2} \geq \dots \geq n_{sr_s}$$

Any other decomposition of  $V$  into  $T$ -cyclic subspaces has the same number of components and the same set of  $T$ -annihilators.

We emphasize that the above lemma (proved in [Problem 10.31](#)) does not say that the vectors  $v_j$  or other  $T$ -cyclic subspaces  $Z(v_j, T)$  are uniquely determined by  $T$ , but it does

say that the set of  $T$ -annihilators is uniquely determined by  $T$ . Thus,  $T$  has a unique block diagonal matrix representation:

$$M = \text{diag}(C_1, C_2, \dots, C_r)$$

where the  $C_i$  are companion matrices. In fact, the  $C_i$  are the companion matrices of the polynomials  $f(t)^{n_i}$ .

Using the Primary Decomposition Theorem and Lemma 10.13, we obtain the following result.

**THEOREM 10.14:** Let  $T: V \rightarrow V$  be a linear operator with minimal polynomial

$$m(t) = f_1(t)^{m_1} f_2(t)^{m_2} \cdots f_s(t)^{m_s}$$

where the  $f_i(t)$  are distinct monic irreducible polynomials. Then  $T$  has a unique block diagonal matrix representation:

$$M = \text{diag}(C_{11}, C_{12}, \dots, C_{1r_1}, \dots, C_{s1}, C_{s2}, \dots, C_{sr_s})$$

where the  $C_{ij}$  are companion matrices. In particular, the  $C_{ij}$  are the companion matrices of the polynomials  $f_i(t)^{n_{ij}}$ , where

$$m_1 = n_{11} \geq n_{12} \geq \cdots \geq n_{1r_1}, \quad \dots, \quad m_s = n_{s1} \geq n_{s2} \geq \cdots \geq n_{sr_s}$$

The above matrix representation of  $T$  is called its *rational canonical form*. The polynomials  $f_i(t)^{n_{ij}}$  are called the *elementary divisors* of  $T$ .

**EXAMPLE 10.6** Let  $V$  be a vector space of dimension 8 over the rational field  $\mathbf{Q}$ , and let  $T$  be a linear operator on  $V$  whose minimal polynomial is

$$m(t) = f_1(t)f_2(t)^2 = (t^4 - 4t^3 + 6t^2 - 4t - 7)(t - 3)^2$$

Thus, because  $\dim V = 8$ , the characteristic polynomial  $\Delta(t) = f_1(t) f_2(t)^4$ . Also, the rational canonical form  $M$  of  $T$  must have one block the companion matrix of  $f_1(t)$  and one block the companion matrix of  $f_2(t)^2$ . There are two possibilities:

(a)  $\text{diag}[C(t^4 - 4t^3 + 6t^2 - 4t - 7), C((t - 3)^2), C((t - 3)^2)]$

(b)  $\text{diag}[C(t^4 - 4t^3 + 6t^2 - 4t - 7), C((t - 3)^2), C(t - 3), C(t - 3)]$

That is,

$$(a) \text{ diag} \left( \begin{bmatrix} 0 & 0 & 0 & 7 \\ 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -6 \\ 0 & 0 & 1 & 4 \end{bmatrix}, \begin{bmatrix} 0 & -9 \\ 1 & 6 \end{bmatrix}, \begin{bmatrix} 0 & -9 \\ 1 & 6 \end{bmatrix} \right), (b) \text{ diag} \left( \begin{bmatrix} 0 & 0 & 0 & 7 \\ 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -6 \\ 0 & 0 & 1 & 4 \end{bmatrix}, \begin{bmatrix} 0 & -9 \\ 1 & 6 \end{bmatrix}, [3], [3] \right)$$

## 10.10 Quotient Spaces

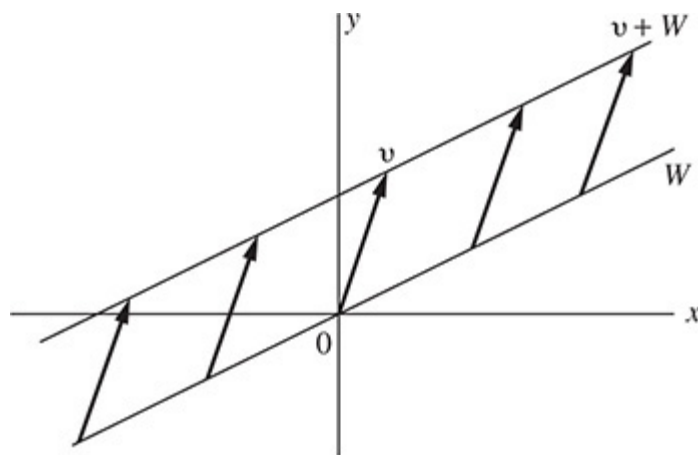
---

Let  $V$  be a vector space over a field  $K$  and let  $W$  be a subspace of  $V$ . If  $v$  is any vector in  $V$ , we write  $v + W$  for the set of sums  $v + w$  with  $w \in W$ ; that is,

$$v + W = \{v + w : w \in W\}$$

These sets are called the *cosets* of  $W$  in  $V$ . We show ([Problem 10.22](#)) that these cosets partition  $V$  into mutually disjoint subsets.

**EXAMPLE 10.7** Let  $W$  be the subspace of  $\mathbf{R}^2$  defined by



**Figure 10-2**

$$W = \{(a, b) : a = b\},$$

that is,  $W$  is the line given by the equation  $x - y = 0$ . We can view  $v + W$  as a translation of the line obtained by adding the vector  $v$  to each point in  $W$ . As shown in [Fig. 10-2](#), the coset  $v + W$  is also a line, and it is parallel to  $W$ . Thus, the cosets of  $W$  in  $\mathbf{R}^2$  are precisely all the lines parallel to  $W$ .

In the following theorem, we use the cosets of a subspace  $W$  of a vector space  $V$  to define a new vector space; it is called the *quotient space* of  $V$  by  $W$  and is denoted by  $V/W$ .

**THEOREM 10.15:** Let  $W$  be a subspace of a vector space over a field  $K$ . Then the cosets of  $W$  in  $V$  form a vector space over  $K$  with the following operations of addition and scalar multiplication:

$$(i) (u + w) + (v + W) = (u + v) + W, \quad (ii) k(u + W) = ku + W, \quad \text{where } k \in K$$

We note that, in the proof of [Theorem 10.15](#) ([Problem 10.24](#)), it is first necessary to show that the operations are

well defined; that is, whenever  $u + W = u' + W$  and  $v + W = v' + W$ , then

$$(i) (u + v) + W = (u' + v') + W \quad \text{and} \quad (ii) ku + W = ku' + W \quad \text{for any } k \in K$$

In the case of an invariant subspace, we have the following useful result (proved in [Problem 10.27](#)).

**THEOREM 10.16:** Suppose  $W$  is a subspace invariant under a linear operator  $T: V \rightarrow V$ . Then  $T$  induces a linear operator  $\bar{T}$  on  $V/W$  defined by  $T(v + W) = T(v) + W$ . Moreover, if  $T$  is a zero of any polynomial, then so is  $\bar{T}$ . Thus, the minimal polynomial of  $\bar{T}$  divides the minimal polynomial of  $T$ .

## SOLVED PROBLEMS

### Invariant Subspaces

---

- 10.1.** Suppose  $T: V \rightarrow V$  is linear. Show that each of the following is invariant under  $T$ :
- (a)  $\{0\}$ , (b)  $V$ , (c) kernel of  $T$ , (d) image of  $T$ .
  - (a) We have  $T(0) = 0 \in \{0\}$ ; hence,  $\{0\}$  is invariant under  $T$ .
  - (b) For every  $v \in V$ ,  $T(v) \in V$ ; hence,  $V$  is invariant under  $T$ .
  - (c) Let  $u \in \text{Ker } T$ . Then  $T(u) = 0 \in \text{Ker } T$  because the kernel of  $T$  is a subspace of  $V$ . Thus,  $\text{Ker } T$  is invariant under  $T$ .
  - (d) Because  $T(v) \in \text{Im } T$  for every  $v \in V$ , it is certainly true when  $v \in \text{Im } T$ . Hence, the image of  $T$  is invariant under  $T$ .

**10.2.** Suppose  $\{W_i\}$  is a collection of  $T$ -invariant subspaces of a vector space  $V$ . Show that the intersection  $W = \cap_i W_i$  is also  $T$ -invariant.

Suppose  $v \in W$ ; then  $v \in W_i$  for every  $i$ . Because  $W_i$  is  $T$ -invariant,  $T(v) \in W_i$  for every  $i$ . Thus,  $T(v) \in W$  and so  $W$  is  $T$ -invariant.

**10.3.** Prove [Theorem 10.2](#): Let  $T: V \rightarrow V$  be linear. For any polynomial  $f(t)$ , the kernel of  $f(T)$  is invariant under  $T$ .

Suppose  $v \in \text{Ker } f(T)$ —that is,  $f(T)(v) = 0$ . We need to show that  $T(v)$  also belongs to the kernel of  $f(T)$ —that is,  $f(T)(T(v)) = (f(T) \circ T)(v) = 0$ . Because  $f(t)t = tf(t)$ , we have  $f(T) \circ T = T \circ f(T)$ . Thus, as required,

$$(f(T) \circ T)(v) = (T \circ f(T))(v) = T(f(T)(v)) = T(0) = 0$$

**10.4.** Find all invariant subspaces of  $A = \begin{bmatrix} 2 & -5 \\ 1 & -2 \end{bmatrix}$  viewed as an operator on  $\mathbf{R}^2$ .

By [Problem 10.1](#),  $\mathbf{R}^2$  and  $\{0\}$  are invariant under  $A$ . Now if  $A$  has any other invariant subspace, it must be one-dimensional. However, the characteristic polynomial of  $A$  is

$$\Delta(t) = t^2 - \text{tr}(A)t + |A| = t^2 + 1$$

Hence,  $A$  has no eigenvalues (in  $\mathbf{R}$ ) and so  $A$  has no eigenvectors. But the one-dimensional invariant subspaces correspond to the eigenvectors; thus,  $\mathbf{R}^2$  and  $\{0\}$  are the only subspaces invariant under  $A$ .

**10.5.** Prove [Theorem 10.3](#): Suppose  $W$  is  $T$ -invariant. Then  $T$  has a triangular block representation  $\begin{bmatrix} A & B \\ 0 & C \end{bmatrix}$ , where

$A$  is the matrix representation of the restriction  $\hat{T}$  of  $T$  to  $W$ .

We choose a basis  $\{w_1, \dots, w_r\}$  of  $W$  and extend it to a basis  $\{w_1, \dots, w_r, v_1, \dots, v_s\}$  of  $V$ . We have

$$\begin{aligned}\hat{T}(w_1) &= T(w_1) = a_{11}w_1 + \cdots + a_{1r}w_r \\ \hat{T}(w_2) &= T(w_2) = a_{21}w_1 + \cdots + a_{2r}w_r \\ &\dots\dots\dots \\ \hat{T}(w_r) &= T(w_r) = a_{r1}w_1 + \cdots + a_{rr}w_r \\ T(v_1) &= b_{11}w_1 + \cdots + b_{1r}w_r + c_{11}v_1 + \cdots + c_{1s}v_s \\ T(v_2) &= b_{21}w_1 + \cdots + b_{2r}w_r + c_{21}v_1 + \cdots + c_{2s}v_s \\ &\dots\dots\dots \\ T(v_s) &= b_{s1}w_1 + \cdots + b_{sr}w_r + c_{s1}v_1 + \cdots + c_{ss}v_s\end{aligned}$$

But the matrix of  $T$  in this basis is the transpose of the matrix of coefficients in the above system of equations (Section 6.2). Therefore, it has the form

$\begin{bmatrix} A & B \\ 0 & C \end{bmatrix}$ , where  $A$  is the transpose of the matrix of coefficients for the obvious subsystem. By the same argument,  $A$  is the matrix of  $\hat{T}$  relative to the basis  $\{w_j\}$  of  $W$ .

- 10.6.** Let  $\hat{T}$  denote the restriction of an operator  $T$  to an invariant subspace  $W$ . Prove
- (a) For any polynomial  $f(t)$ ,  $f(\hat{T})(w) = f(T)(w)$ .
  - (b) The minimal polynomial of  $\hat{T}$  divides the minimal polynomial of  $T$ .
  - (a) If  $f(t) = 0$  or if  $f(t)$  is a constant (i.e., of degree 1), then the result clearly holds. Assume  $\deg f = n > 1$  and that the result holds for polynomials of degree less than  $n$ . Suppose that

$$f(t) = a_n t^n + a_{n-1} t^{n-1} + \cdots + a_1 t + a_0$$

Then

$$\begin{aligned} f(\hat{T})(w) &= (a_n \hat{T}^n + a_{n-1} \hat{T}^{n-1} + \cdots + a_0 I)(w) \\ &= (a_n \hat{T}^{n-1})(\hat{T}(w)) + (a_{n-1} \hat{T}^{n-1} + \cdots + a_0 I)(w) \\ &= (a_n T^{n-1})(T(w)) + (a_{n-1} T^{n-1} + \cdots + a_0 I)(w) = f(T)(w) \end{aligned}$$

(b) Let  $m(t)$  denote the minimal polynomial of  $T$ . Then by (a),  $m(\hat{T})(w) = m(T)(w) = \mathbf{0}(w) = 0$  for every  $w \in W$ ; that is,  $\hat{T}$  is a zero of the polynomial  $m(t)$ . Hence, the minimal polynomial of  $\hat{T}$  divides  $m(t)$ .

## Invariant Direct-Sum Decompositions

---

**10.7.** Prove [Theorem 10.4](#): Suppose  $W_1, W_2, \dots, W_r$  are subspaces of  $V$  with respective bases

$$B_1 = \{w_{11}, w_{12}, \dots, w_{1n_1}\}, \quad \dots, \quad B_r = \{w_{r1}, w_{r2}, \dots, w_{rn_r}\}$$

Then  $V$  is the direct sum of the  $W_i$  if and only if the union  $B = \cup_i B_i$  is a basis of  $V$ .

Suppose  $B$  is a basis of  $V$ . Then, for any  $v \in V$ ,

$$v = a_{11}w_{11} + \cdots + a_{1n_1}w_{1n_1} + \cdots + a_{r1}w_{r1} + \cdots + a_{rn_r}w_{rn_r} = w_1 + w_2 + \cdots + w_r$$

where  $w_i = a_{i1}w_{i1} + \cdots + a_{in_i}w_{in_i} \in W_i$ . We next show that such a sum is unique. Suppose

$$v = w'_1 + w'_2 + \cdots + w'_r, \quad \text{where} \quad w'_i \in W_i$$

Because  $\{w_{i1}, \dots, w_{in_i}\}$  is a basis of  $W_i$ ,  $w'_i = b_{i1}w_{i1} + \cdots + b_{in_i}w_{in_i}$ , and so

$$v = b_{11}w_{11} + \cdots + b_{1n_1}w_{1n_1} + \cdots + b_{r1}w_{r1} + \cdots + b_{m_r}w_{m_r}$$

Because  $B$  is a basis of  $V$ ,  $a_{ij} = b_{ij}$ , for each  $i$  and each  $j$ . Hence,  $w_i = w'_i$ , and so the sum for  $v$  is unique. Accordingly,  $V$  is the direct sum of the  $W_j$ .

Conversely, suppose  $V$  is the direct sum of the  $W_j$ . Then for any  $v \in V$ ,  $v = w_1 + \cdots + w_r$ , where  $w_j \in W_j$ . Because  $\{w_{ij}\}$  is a basis of  $W_j$ , each  $w_j$  is a linear combination of the  $w_{ij}$ , and so  $v$  is a linear combination of the elements of  $B$ . Thus,  $B$  spans  $V$ . We now show that  $B$  is linearly independent. Suppose

$$a_{11}w_{11} + \cdots + a_{1n_1}w_{1n_1} + \cdots + a_{r1}w_{r1} + \cdots + a_{m_r}w_{m_r} = 0$$

Note that  $a_{i1}w_{i1} + \cdots + a_{in_i}w_{in_i} \in W_i$ . We also have that  $0 = 0 + 0 + \cdots + 0 \in W_i$ . Because such a sum for 0 is unique,

$$a_{i1}w_{i1} + \cdots + a_{in_i}w_{in_i} = 0 \quad \text{for } i = 1, \dots, r$$

The independence of the bases  $\{w_{ij}\}$  implies that all the  $a$ 's are 0. Thus,  $B$  is linearly independent and is a basis of  $V$ .

**10.8.** Suppose  $T: V \rightarrow V$  is linear and suppose  $T = T_1 \oplus T_2$  with respect to a  $T$ -invariant direct-sum decomposition  $V = U \oplus W$ . Show that

- (a)  $m(t)$  is the least common multiple of  $m_1(t)$  and  $m_2(t)$ , where  $m(t)$ ,  $m_1(t)$ ,  $m_2(t)$  are the minimum polynomials of  $T$ ,  $T_1$ ,  $T_2$ , respectively.
- (b)  $\Delta(t) = \Delta_1(t)\Delta_2(t)$ , where  $\Delta(t)$ ,  $\Delta_1(t)$ ,  $\Delta_2(t)$  are the characteristic polynomials of  $T$ ,  $T_1$ ,  $T_2$ , respectively.

(a) By [Problem 10.6](#), each of  $m_1(t)$  and  $m_2(t)$  divides  $m(t)$ . Now suppose  $f(t)$  is a multiple of both  $m_1(t)$  and  $m_2(t)$ , then  $f(T_1)(U) = 0$  and  $f(T_2)(W) = 0$ . Let  $v \in V$ , then  $v = u + w$  with  $u \in U$  and  $w \in W$ . Now

$$f(T)v = f(T)u + f(T)w = f(T_1)u + f(T_2)w = 0 + 0 = 0$$

That is,  $T$  is a zero of  $f(t)$ . Hence,  $m(t)$  divides  $f(t)$ , and so  $m(t)$  is the least common multiple of  $m_1(t)$  and  $m_2(t)$ .

(b) By [Theorem 10.5](#),  $T$  has a matrix representation

$$M = \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}, \text{ where } A \text{ and } B \text{ are matrix}$$

representations of  $T_1$  and  $T_2$ , respectively. Then, as required,

$$\Delta(t) = |tI - M| = \begin{vmatrix} tI - A & 0 \\ 0 & tI - B \end{vmatrix} = |tI - A||tI - B| = \Delta_1(t)\Delta_2(t)$$

**10.9.** Prove [Theorem 10.7](#): Suppose  $T: V \rightarrow V$  is linear, and suppose  $f(t) = g(t)h(t)$  are polynomials such that  $f(t) = 0$  and  $g(t)$  and  $h(t)$  are relatively prime. Then  $V$  is the direct sum of the  $T$ -invariant subspaces  $U$  and  $W$  where  $U = \text{Ker } g(T)$  and  $W = \text{Ker } h(T)$ .

Note first that  $U$  and  $W$  are  $T$ -invariant by [Theorem 10.2](#). Now, because  $g(t)$  and  $h(t)$  are relatively prime, there exist polynomials  $r(t)$  and  $s(t)$  such that

$$r(t)g(t) + s(t)h(t) = 1$$

Hence, for the operator  $T$ ,

$$r(T)g(T) + s(T)h(T) = I \quad (*)$$

Let  $v \in V$ ; then, by (\*),

$$v = r(T)g(T)v + s(T)h(T)v$$

But the first term in this sum belongs to  $W = \text{Ker } h(T)$ , because

$$h(T)r(T)g(T)v = r(T)g(T)h(T)v = r(T)f(T)v = r(T)0v = 0$$

Similarly, the second term belongs to  $U$ . Hence,  $V$  is the sum of  $U$  and  $W$ .

To prove that  $V = U \oplus W$ , we must show that a sum  $v = u + w$  with  $u \in U$ ,  $w \in W$ , is uniquely determined by  $v$ . Applying the operator  $r(T)g(T)$  to  $v = u + w$  and using  $g(T)u = 0$ , we obtain

$$r(T)g(T)v = r(T)g(T)u + r(T)g(T)w = r(T)g(T)w$$

Also, applying (\*) to  $w$  alone and using  $h(T)w = 0$ , we obtain

$$w = r(T)g(T)w + s(T)h(T)w = r(T)g(T)w$$

Both of the above formulas give us  $w = r(T)g(T)v$ , and so  $w$  is uniquely determined by  $v$ . Similarly  $u$  is uniquely determined by  $v$ . Hence,  $V = U \oplus W$ , as required.

**10.10.** Prove [Theorem 10.8](#): In [Theorem 10.7](#) ([Problem 10.9](#)), if  $f(t)$  is the minimal polynomial of  $T$  (and  $g(t)$  and  $h(t)$  are monic), then  $g(t)$  is the minimal polynomial of the restriction  $T_1$  of  $T$  to  $U$  and  $h(t)$  is the minimal polynomial of the restriction  $T_2$  of  $T$  to  $W$ .

Let  $m_1(t)$  and  $m_2(t)$  be the minimal polynomials of  $T_1$  and  $T_2$ , respectively. Note that  $g(T_1) = 0$  and  $h(T_2) = 0$  because  $U = \text{Ker } g(T)$  and  $W = \text{Ker } h(T)$ . Thus,

$$m_1(t) \text{ divides } g(t) \quad \text{and} \quad m_2(t) \text{ divides } h(t) \quad (1)$$

By [Problem 10.9](#),  $f(t)$  is the least common multiple of  $m_1(t)$  and  $m_2(t)$ . But  $m_1(t)$  and  $m_2(t)$  are relatively prime because  $g(t)$  and  $h(t)$  are relatively prime. Accordingly,  $f(t) = m_1(t)m_2(t)$ . We also have that. These two equations together with (1) and the fact that all the polynomials are monic imply that  $g(t) = m_1(t)$  and  $h(t) = m_2(t)$ , as required.

**10.11.** Prove the Primary Decomposition [Theorem 10.6](#): Let  $T: V \rightarrow V$  be a linear operator with minimal polynomial

$$m(t) = f_1(t)^{n_1} f_2(t)^{n_2} \cdots f_r(t)^{n_r}$$

where the  $f_i(t)$  are distinct monic irreducible polynomials. Then  $V$  is the direct sum of  $T$ -invariant subspaces  $W_1, \dots, W_r$  where  $W_i$  is the kernel of  $f_i(T)^{n_i}$ . Moreover,  $f_i(t)^{n_i}$  is the minimal polynomial of the restriction of  $T$  to  $W_i$ .

The proof is by induction on  $r$ . The case  $r = 1$  is trivial. Suppose that the theorem has been proved for  $r - 1$ . By [Theorem 10.7](#), we can write  $V$  as the direct sum of  $T$ -invariant subspaces  $W_1$  and  $V_1$ , where  $W_1$  is the kernel of  $f_1(T)^{n_1}$  and where  $V_1$  is the kernel of  $f_2(T)^{n_2} \cdots f_r(T)^{n_r}$ . By [Theorem 10.8](#), the minimal polynomials of the restrictions of  $T$  to  $W_1$  and  $V_1$  are  $f_1(t)^{n_1}$  and  $f_2(t)^{n_2} \cdots f_r(t)^{n_r}$ , respectively.

Denote the restriction of  $T$  to  $V_1$  by  $\hat{T}_1$ . By the inductive hypothesis,  $V_1$  is the direct sum of subspaces  $W_2, \dots, W_r$  such that  $W_i$  is the kernel of  $f_i(\hat{T}_1)^{n_i}$  and such that  $f_i(t)^{n_i}$  is the minimal polynomial for the restriction

of  $\hat{T}_1$  to  $W_i$ . But the kernel of  $f_i(T)^{n_i}$ , for  $i = 2, \dots, r$  is necessarily contained in  $V_1$ , because  $f_i(t)^{n_i}$  divides  $f_2(t)^{n_2} \cdots f_r(t)^{n_r}$ . Thus, the kernel of  $f_i(T)^{n_i}$  is the same as the kernel of  $f_i(T_1)^{n_i}$ , which is  $W_i$ . Also, the restriction of  $T$  to  $W_i$  is the same as the restriction of  $\hat{T}_1$  to  $W_i$  (for  $i = 2, \dots, r$ ); hence,  $f_i(t)^{n_i}$  is also the minimal polynomial for the restriction of  $T$  to  $W_i$ . Thus,  $V = W_1 \oplus W_2 \oplus W_r$  is the desired decomposition of  $T$ .

**10.12.** Prove [Theorem 10.9](#): A linear operator  $T: V \rightarrow V$  has a diagonal matrix representation if and only if its minimal polynomial  $m(t)$  is a product of distinct linear polynomials.

Suppose  $m(t)$  is a product of distinct linear polynomials, say,

$$m(t) = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_r)$$

where the  $\lambda_i$  are distinct scalars. By the Primary Decomposition Theorem,  $V$  is the direct sum of subspaces  $W_1, \dots, W_r$ , where  $W_i = \text{Ker}(T - \lambda_i I)$ . Thus, if  $v \in W_i$ , then  $(T - \lambda_i I)(v) = 0$  or  $T(v) = \lambda_i v$ . In other words, every vector in  $W_i$  is an eigenvector belonging to the eigenvalue  $\lambda_i$ . By [Theorem 10.4](#), the union of bases for  $W_1, \dots, W_r$  is a basis of  $V$ . This basis consists of eigenvectors, and so  $T$  is diagonalizable.

Conversely, suppose  $T$  is diagonalizable (i.e.,  $V$  has a basis consisting of eigenvectors of  $T$ ). Let  $\lambda_1, \dots, \lambda_s$  be the distinct eigenvalues of  $T$ . Then the operator

$$f(T) = (T - \lambda_1 I)(T - \lambda_2 I) \cdots (T - \lambda_s I)$$

maps each basis vector into 0. Thus,  $f(t) = 0$ , and hence, the minimal polynomial  $m(t)$  of  $T$  divides the polynomial

$$f(t) = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_s I)$$

Accordingly,  $m(t)$  is a product of distinct linear polynomials.

## **Nilpotent Operators, Jordan Canonical Form**

**10.13.** Let  $T: V \rightarrow V$  be linear. Suppose, for  $v \in V$ ,  $T^k(v) = 0$  but  $T^{k-1}(v) \neq 0$ . Prove

- (a) The set  $S = \{v, T(v), \dots, T^{k-1}(v)\}$  is linearly independent.
- (b) The subspace  $W$  generated by  $S$  is  $T$ -invariant.
- (c) The restriction  $\hat{T}$  of  $T$  to  $W$  is nilpotent of index  $k$ .
- (d) Relative to the basis  $\{T^{k-1}(v), \dots, T(v), v\}$  of  $W$ , the matrix of  $T$  is the  $k$ -square Jordan nilpotent block  $N_k$  of index  $k$  (see [Example 10.5](#)).

(a) Suppose

$$av + a_1 T(v) + a_2 T^2(v) + \cdots + a_{k-1} T^{k-1}(v) = 0 \quad (*)$$

Applying  $T^{k-1}$  to  $(*)$  and using  $T^k(v) = 0$ , we obtain  $aT^{k-1}(v) = 0$ ; because  $T^{k-1}(v) \neq 0$ ,  $a = 0$ . Now applying  $T^{k-2}$  to  $(*)$  and using  $T^k(v) = 0$  and  $a = 0$ , we find  $a_1 T^{k-1}(v) = 0$ ; hence,  $a_1 = 0$ . Next applying  $T^{k-3}$  to  $(*)$  and using  $T^k(v) = 0$  and  $a = a_1 = 0$ , we obtain  $a_2 T^{k-1}(v) = 0$ ; hence,  $a_2 = 0$ . Continuing this process, we find that all the  $a$ 's are 0; hence,  $S$  is independent.

(b) Let  $v \in W$ . Then

$$v = bv + b_1T(v) + b_2T^2(v) + \cdots + b_{k-1}T^{k-1}(v)$$

Using  $T^k(v) = 0$ , we have

$$T(v) = bT(v) + b_1T^2(v) + \cdots + b_{k-2}T^{k-1}(v) \in W$$

Thus,  $W$  is  $T$ -invariant.

(c) By hypothesis,  $T^k(v) = 0$ . Hence, for  $i = 0, \dots, k-1$ ,

$$\hat{T}^k(T^i(v)) = T^{k+i}(v) = 0$$

That is, applying  $\hat{T}^k$  to each generator of  $W$ , we obtain 0; hence,  $\hat{T}^k = \mathbf{0}$  and so  $\hat{T}$  is nilpotent of index at most  $k$ . On the other hand,  $\hat{T}^{k-1}(v) = T^{k-1}(v) \neq 0$ ; hence,  $T$  is nilpotent of index exactly  $k$ .

(d) For the basis  $\{T^{k-1}(v), T^{k-2}(v), \dots, T(v), v\}$  of  $W$ ,

$$\begin{array}{rcl} \hat{T}(T^{k-1}(v)) & = & T^k(v) = 0 \\ \hat{T}(T^{k-2}(v)) & = & T^{k-1}(v) \\ \hat{T}(T^{k-3}(v)) & = & T^{k-2}(v) \\ & \dots & \\ \hat{T}(T(v)) & = & T^2(v) \\ \hat{T}(v) & = & T(v) \end{array}$$

Hence, as required, the matrix of  $T$  in this basis is the  $k$ -square Jordan nilpotent block  $N_k$ .

**10.14.** Let  $T: V \rightarrow V$  be linear. Let  $U = \text{Ker } T^i$  and  $W = \text{Ker } T^{i+1}$ . Show that

(a)  $U \subseteq W$ , (b)  $T(W) \subseteq U$ .

(a) Suppose  $u \in U = \text{Ker } T^i$ . Then  $T^i(u) = 0$  and so  $T^{i+1}u = T(T^i(u)) = T(0) = 0$ . Thus,  $u \in \text{Ker } T^{i+1} = W$ .

But this is true for every  $u \in U$ ; hence,  $U \subseteq W$ .

(b) Similarly, if  $w \in W = \text{Ker } T^{i+1}$ , then  $T^{i+1}(w) = 0$ .

Thus,  $T^{i+1}(w) = T^i(T(w)) = T^i(0) = 0$  and so  $T(W) \subseteq U$ .

**10.15.** Let  $T: V \rightarrow V$  be linear. Let  $X = \text{Ker } T^{i-2}$ ,  $Y = \text{Ker } T^{i-1}$ ,  $Z = \text{Ker } T^i$ . Therefore ([Problem 10.14](#)),

$$\{u_1, \dots, u_r\}, \quad \{u_1, \dots, u_r, v_1, \dots, v_s\}, \quad \{u_1, \dots, u_r, v_1, \dots, v_s, w_1, \dots, w_t\}$$

are bases of  $X$ ,  $Y$ ,  $Z$ , respectively. Show that

$$S = \{u_1, \dots, u_r, T(w_1), \dots, T(w_t)\}$$

is contained in  $Y$  and is linearly independent.

By [Problem 10.14](#),  $T(Z) \subseteq Y$ , and hence  $S \subseteq Y$ . Now suppose  $S$  is linearly dependent. Then there exists a relation

$$a_1 u_1 + \dots + a_r u_r + b_1 T(w_1) + \dots + b_t T(w_t) = 0$$

where at least one coefficient is not zero. Furthermore, because  $\{u_i\}$  is independent, at least one of the  $b_k$  must be nonzero. Transposing, we find

$$b_1 T(w_1) + \dots + b_t T(w_t) = -a_1 u_1 - \dots - a_r u_r \in X = \text{Ker } T^{i-2}$$

Hence,

$$T^{i-2}(b_1 T(w_1) + \dots + b_t T(w_t)) = 0$$

Thus,

$$T^{i-1}(b_1 w_1 + \dots + b_t w_t) = 0, \quad \text{and so} \quad b_1 w_1 + \dots + b_t w_t \in Y = \text{Ker } T^{i-1}$$

Because  $\{u_i, v_j\}$  generates  $Y$ , we obtain a relation among the  $u_i, v_j, w_k$  where one of the coefficients (i.e., one of the  $b_k$ ) is not zero. This contradicts the fact that  $\{u_i, v_j, w_k\}$  is independent. Hence,  $S$  must also be independent.

**10.16.** Prove [Theorem 10.10](#): Let  $T: V \rightarrow V$  be a nilpotent operator of index  $k$ . Then  $T$  has a unique block diagonal matrix representation consisting of Jordan nilpotent blocks  $N$ . There is at least one  $N$  of order  $k$ , and all other  $N$  are of orders  $\leq k$ . The total number of  $N$  of all orders is equal to the nullity of  $T$ .

Suppose  $\dim V = n$ . Let  $W_1 = \text{Ker } T$ ,  $W_2 = \text{Ker } T^2$ , ...,  $W_k = \text{Ker } T^k$ . Let us set  $m_i = \dim W_i$ , for  $i = 1, \dots, k$ . Because  $T$  is of index  $k$ ,  $W_k = V$  and  $W_{k-1} \neq V$  and so  $m_{k-1} < m_k = n$ . By [Problem 10.14](#),

$$W_1 \subseteq W_2 \subseteq \dots \subseteq W_k = V$$

Thus, by induction, we can choose a basis  $\{u_1, \dots, u_n\}$  of  $V$  such that  $\{u_1, \dots, u_{m_i}\}$  is a basis of  $W_i$ .

We now choose a new basis for  $V$  with respect to which  $T$  has the desired form. It will be convenient to label the members of this new basis by pairs of indices. We begin by setting

$$v(1, k) = u_{m_{k-1}+1}, \quad v(2, k) = u_{m_{k-1}+2}, \quad \dots, \quad v(m_k - m_{k-1}, k) = u_{m_k}$$

and setting

$$v(1, k-1) = Tv(1, k), \quad v(2, k-1) = Tv(2, k), \quad \dots, \quad v(m_k - m_{k-1}, k-1) = Tv(m_k - m_{k-1}, k)$$

By the preceding problem,

$$S_1 = \{u_1 \dots, u_{m_{k-2}}, v(1, k-1), \dots, v(m_k - m_{k-1}, k-1)\}$$

is a linearly independent subset of  $W_{k-1}$ . We extend  $S_1$  to a basis of  $W_{k-1}$  by adjoining new elements (if necessary), which we denote by

$$v(m_k - m_{k-1} + 1, k-1), \quad v(m_k - m_{k-1} + 2, k-1), \quad \dots, \quad v(m_{k-1} - m_{k-2}, k-1)$$

Next we set

$$\begin{aligned} v(1, k-2) &= Tv(1, k-1), & v(2, k-2) &= Tv(2, k-1), & \dots, \\ v(m_{k-1} - m_{k-2}, k-2) &= Tv(m_{k-1} - m_{k-2}, k-1) \end{aligned}$$

Again by the preceding problem,

$$S_2 = \{u_1, \dots, u_{m_{k-3}}, v(1, k-2), \dots, v(m_{k-1} - m_{k-2}, k-2)\}$$

is a linearly independent subset of  $W_{k-2}$ , which we can extend to a basis of  $W_{k-2}$  by adjoining elements

$$v(m_{k-1} - m_{k-2} + 1, k-2), \quad v(m_{k-1} - m_{k-2} + 2, k-2), \quad \dots, \quad v(m_{k-2} - m_{k-3}, k-2)$$

Continuing in this manner, we get a new basis for  $V$ , which for convenient reference we arrange as follows:

$$\begin{array}{ccccccc} v(1, k) & & \dots, & v(m_k - m_{k-1}, k) & & & \\ v(1, k-1), & \dots, & v(m_k - m_{k-1}, k-1) & \dots, & v(m_{k-1} - m_{k-2}, k-1) & & \\ & & \dots & & \dots & & \\ v(1, 2), & \dots, & v(m_k - m_{k-1}, 2), & \dots, & v(m_{k-1} - m_{k-2}, 2), & \dots, & v(m_2 - m_1, 2) \\ v(1, 1), & \dots, & v(m_k - m_{k-1}, 1), & \dots, & v(m_{k-1} - m_{k-2}, 1), & \dots, & v(m_2 - m_1, 1), \dots, v(m_1, 1) \end{array}$$

The bottom row forms a basis of  $W_1$ , the bottom two rows form a basis of  $W_2$ , and so forth. But what is important for us is that  $T$  maps each vector into the vector immediately below it in the table or into 0 if the vector is in the bottom row. That is,

$$Tv(i, j) = \begin{cases} v(i, j-1) & \text{for } j > 1 \\ 0 & \text{for } j = 1 \end{cases}$$

Now it is clear [see [Problem 10.13\(d\)](#)] that  $T$  will have the desired form if the  $v(i, j)$  are ordered lexicographically: beginning with  $v(1, 1)$  and moving up the first column to  $v(1, k)$ , then jumping to  $v(2, 1)$  and moving up the second column as far as possible.

Moreover, there will be exactly  $m_k - m_{k-1}$  diagonal entries of order  $k$ . Also, there will be

$$\begin{array}{ll} (m_{k-1} - m_{k-2}) - (m_k - m_{k-1}) = 2m_{k-1} - m_k - m_{k-2} & \text{diagonal entries of order } k-1 \\ \dots\dots\dots & \dots\dots\dots \\ 2m_2 - m_1 - m_3 & \text{diagonal entries of order 2} \\ 2m_1 - m_2 & \text{diagonal entries of order 1} \end{array}$$

as can be read off directly from the table. In particular, because the numbers  $m_1, \dots, m_k$  are uniquely determined by  $T$ , the number of diagonal entries of each order is uniquely determined by  $T$ . Finally, the identity

$$m_1 = (m_k - m_{k-1}) + (2m_{k-1} - m_k - m_{k-2}) + \dots + (2m_2 - m_1 - m_3) + (2m_1 - m_2)$$

shows that the nullity  $m_1$  of  $T$  is the total number of diagonal entries of  $T$ .

**10.17.** Let  $A = \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$  and  $B = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ .

The reader can verify that  $A$  and  $B$  are both nilpotent of index 3; that is,  $A^3 = 0$  but  $A^2 \neq 0$ , and  $B^3 = 0$  but  $B^2 \neq 0$ . Find the nilpotent matrices  $M_A$  and  $M_B$  in canonical

form that are similar to  $A$  and  $B$ , respectively.

Because  $A$  and  $B$  are nilpotent of index 3,  $M_A$  and  $M_B$  must each contain a Jordan nilpotent block of order 3, and none greater than 3. Note that  $\text{rank}(A) = 2$  and  $\text{rank}(B) = 3$ , so  $\text{nullity}(A) = 5 - 2 = 3$  and  $\text{nullity}(B) = 5 - 3 = 2$ . Thus,  $M_A$  must contain three diagonal blocks, which must be one of order 3 and two of order 1; and  $M_B$  must contain two diagonal blocks, which must be one of order 3 and one of order 2. Namely,

$$M_A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad M_B = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

**10.18.** Prove [Theorem 10.11](#) on the Jordan canonical form for an operator  $T$ .

By the primary decomposition theorem,  $T$  is decomposable into operators  $T_1, \dots, T_r$ ; that is,  $T = T_1 \oplus \dots \oplus T_r$  where  $(t - \lambda_i)^{m_i}$  is the minimal polynomial of  $T_i$ . Thus, in particular,

$$(T_1 - \lambda_1 I)^{m_1} = \mathbf{0}, \quad \dots, \quad (T_r - \lambda_r I)^{m_r} = \mathbf{0}$$

Set  $N_i = T_i - \lambda_i I$ . Then, for  $i = 1, \dots, r$ ,

$$T_i = N_i + \lambda_i I, \quad \text{where} \quad N_i^{m_i} = \mathbf{0}$$

That is,  $T_i$  is the sum of the scalar operator  $\lambda_i I$  and a nilpotent operator  $N_i$ , which is of index  $m_i$  because  $(t - \lambda_i)^{m_i}$  is the minimal polynomial of  $T_i$ .

Now, by [Theorem 10.10](#) on nilpotent operators, we can choose a basis so that  $N_i$  is in canonical form. In this basis,  $T_i = N_i + \lambda_i I$  is represented by a block diagonal matrix  $M_i$  whose diagonal entries are the matrices  $J_{ij}$ . The direct sum  $J$  of the matrices  $M_i$  is in Jordan canonical form and, by [Theorem 10.5](#), is a matrix representation of  $T$ .

Last, we must show that the blocks  $J_{ij}$  satisfy the required properties. Property (i) follows from the fact that  $N_i$  is of index  $m_i$ . Property (ii) is true because  $T$  and  $J$  have the same characteristic polynomial. Property (iii) is true because the nullity of  $N_i = T_i - \lambda_i I$  is equal to the geometric multiplicity of the eigenvalue  $\lambda_i$ . Property (iv) follows from the fact that the  $T_i$  and hence the  $N_i$  are uniquely determined by  $T$ .

**10.19.** Determine all possible Jordan canonical forms  $J$  for a linear operator  $T: V \rightarrow V$  whose characteristic polynomial  $\Delta(t) = (t - 2)^5$  and whose minimal polynomial  $m(t) = (t - 2)^2$ .

$J$  must be a  $5 \times 5$  matrix, because  $\Delta(t)$  has degree 5, and all diagonal elements must be 2, because 2 is the only eigenvalue. Moreover, because the exponent of  $t - 2$  in  $m(t)$  is 2,  $J$  must have one Jordan block of order 2, and the others must be of order 2 or 1. Thus, there are only two possibilities:

$$J = \text{diag}\left(\begin{bmatrix} 2 & 1 \\ & 2 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ & 2 \end{bmatrix}, [2]\right) \quad \text{or} \quad J = \text{diag}\left(\begin{bmatrix} 2 & 1 \\ & 2 \end{bmatrix}, [2], [2], [2]\right)$$

**10.20.** Determine all possible Jordan canonical forms for a linear operator  $T: V \rightarrow V$  whose characteristic

polynomial  $\Delta(t) = (t - 2)^3(t - 5)^2$ . In each case, find the minimal polynomial  $m(t)$ .

Because  $t - 2$  has exponent 3 in  $\Delta(t)$ , 2 must appear three times on the diagonal. Similarly, 5 must appear twice. Thus, there are six possibilities:

$$\begin{aligned} \text{(a) } & \text{diag}\left(\begin{bmatrix} 2 & 1 & \\ & 2 & 1 \\ & & 2 \end{bmatrix}, \begin{bmatrix} 5 & 1 \\ & 5 \end{bmatrix}\right), \quad \text{(b) } \text{diag}\left(\begin{bmatrix} 2 & 1 & \\ & 2 & 1 \\ & & 2 \end{bmatrix}, [5], [5]\right), \\ \text{(c) } & \text{diag}\left(\begin{bmatrix} 2 & 1 \\ & 2 \end{bmatrix}, [2], \begin{bmatrix} 5 & 1 \\ & 5 \end{bmatrix}\right), \quad \text{(d) } \text{diag}\left(\begin{bmatrix} 2 & 1 \\ & 2 \end{bmatrix}, [2], [5], [5]\right), \\ \text{(e) } & \text{diag}([2], [2], [2], \begin{bmatrix} 5 & 1 \\ & 5 \end{bmatrix}), \quad \text{(f) } \text{diag}([2], [2], [2], [5], [5]) \end{aligned}$$

The exponent in the minimal polynomial  $m(t)$  is equal to the size of the largest block. Thus,

$$\begin{aligned} \text{(a) } & m(t) = (t - 2)^3(t - 5)^2, \quad \text{(b) } m(t) = (t - 2)^3(t - 5), \\ \text{(c) } & m(t) = (t - 2)^2(t - 5)^2, \\ \text{(d) } & m(t) = (t - 2)^2(t - 5), \quad \text{(e) } m(t) = (t - 2)(t - 5)^2, \quad \text{(f) } \\ & m(t) = (t - 2)(t - 5) \end{aligned}$$

## Quotient Space and Triangular Form

**10.21.** Let  $W$  be a subspace of a vector space  $V$ . Show that the following are equivalent:

$$\text{(i) } u \in v + W, \text{ (ii) } u - v \in W, \text{ (iii) } v \in u + W.$$

Suppose  $u \in v + W$ . Then there exists  $w_0 \in W$  such that  $u = v + w_0$ . Hence,  $u - v = w_0 \in W$ . Conversely, suppose  $u - v \in W$ . Then  $u - v = w_0$  where  $w_0 \in W$ . Hence,  $u = v + w_0 \in v + W$ . Thus, (i) and (ii) are equivalent.

We also have  $u - v \in W$  iff  $-(u - v) = v - u \in W$  iff  $v \in u + W$ . Thus, (ii) and (iii) are also equivalent.

**10.22.** Prove the following: The cosets of  $W$  in  $V$  partition  $V$  into mutually disjoint sets. That is,

(a) Any two cosets  $u + W$  and  $v + W$  are either identical or disjoint.

(b) Each  $v \in V$  belongs to a coset; in fact,  $v \in v + W$ .

Furthermore,  $u + W = v + W$  if and only if  $u - v \in W$ , and so  $(v + w) + W = v + W$  for any  $w \in W$ .

Let  $v \in V$ . Because  $0 \in W$ , we have  $v = v + 0 \in v + W$ , which proves (b).

Now suppose the cosets  $u + W$  and  $v + W$  are not disjoint; say, the vector  $x$  belongs to both  $u + W$  and  $v + W$ . Then  $u - x \in W$  and  $x - v \in W$ . The proof of (a) is complete if we show that  $u + W = v + W$ . Let  $u + w_0$  be any element in the coset  $u + W$ . Because  $u - x$ ,  $x - v$ ,  $w_0$  belongs to  $W$ ,

$$(u + w_0) - v = (u - x) + (x - v) + w_0 \in W$$

Thus,  $u + w_0 \in v + W$ , and hence the coset  $u + W$  is contained in the coset  $v + W$ . Similarly,  $v + W$  is contained in  $u + W$ , and so  $u + W = v + W$ .

The last statement follows from the fact that  $u + W = v + W$  if and only if  $u \in v + W$ , and, by [Problem 10.21](#), this is equivalent to  $u - v \in W$ .

**10.23.** Let  $W$  be the solution space of the homogeneous equation  $2x + 3y + 4z = 0$ . Describe the cosets of  $W$  in  $\mathbf{R}^3$ .

$W$  is a plane through the origin  $O = (0, 0, 0)$ , and the cosets of  $W$  are the planes parallel to  $W$ .

Equivalently, the cosets of  $W$  are the solution sets of the family of equations

$$2x + 3y + 4z = k, \quad k \in \mathbf{R}$$

In fact, the coset  $v + W$ , where  $v = (a, b, c)$ , is the solution set of the linear equation

$$2x + 3y + 4z = 2a + 3b + 4c \quad \text{or} \quad 2(x - a) + 3(y - b) + 4(z - c) = 0$$

**10.24.** Suppose  $W$  is a subspace of a vector space  $V$ . Show that the operations in [Theorem 10.15](#) are well defined; namely, show that if  $u + W = u' + W$  and  $v + W = v' + W$ , then

$$(a) \quad (u + v) + W = (u' + v') + W \quad \text{and} \quad (b) \quad ku + W = ku' + W \quad \text{for any } k \in K$$

(a) Because  $u + W = u' + W$  and  $v + W = v' + W$ , both  $u - u'$  and  $v - v'$  belong to  $W$ . But then  $(u + v) - (u' + v') = (u - u') + (v - v') \in W$ . Hence,  $(u + v) + W = (u' + v') + W$ .

(b) Also, because  $u - u' \in W$  implies  $k(u - u') \in W$ , then  $ku - ku' = k(u - u') \in W$ ; accordingly,  $ku + W = ku' + W$ .

**10.25.** Let  $V$  be a vector space and  $W$  a subspace of  $V$ . Show that the natural map  $\eta: V \rightarrow V/W$ , defined by  $\eta(v) = v + W$ , is linear.

For any  $u, v \in V$  and any  $k \in K$ , we have

$$\eta(u + v) = (u + v) + W = u + W + v + W = \eta(u) + \eta(v)$$

and

$$\eta(kv) = kv + W = k(v + W) = k\eta(v)$$

Accordingly,  $\eta$  is linear.

**10.26.** Let  $W$  be a subspace of a vector space  $V$ . Suppose  $\{w_1, \dots, w_r\}$  is a basis of  $W$  and the set of cosets  $\{\bar{v}_1, \dots, \bar{v}_s\}$ , where  $\bar{v}_j = v_j + W$ , is a basis of the quotient space. Show that the set of vectors  $B = \{v_1, \dots, v_s, w_1, \dots, w_r\}$  is a basis of  $V$ . Thus,  $\dim V = \dim W + \dim(V/W)$ .

Suppose  $u \in V$ . Because  $\{\bar{v}_j\}$  is a basis of  $V/W$ ,

$$\bar{u} = u + W = a_1 \bar{v}_1 + a_2 \bar{v}_2 + \dots + a_s \bar{v}_s$$

Hence,  $u = a_1 v_1 + \dots + a_s v_s + w$ , where  $w \in W$ . Since  $\{w_i\}$  is a basis of  $W$ ,

$$u = a_1 v_1 + \dots + a_s v_s + b_1 w_1 + \dots + b_r w_r$$

Accordingly,  $B$  spans  $V$ .

We now show that  $B$  is linearly independent.

Suppose

$$c_1 v_1 + \dots + c_s v_s + d_1 w_1 + \dots + d_r w_r = 0 \tag{1}$$

Then

$$c_1 \bar{v}_1 + \dots + c_s \bar{v}_s = \bar{0} = W$$

Because  $\{\bar{v}_j\}$  is independent, the  $c$ 's are all 0.

Substituting into (1), we find  $d_1 w_1 + \dots + d_r w_r = 0$ .

Because  $\{w_i\}$  is independent, the  $d$ 's are all 0. Thus,  $B$  is linearly independent and therefore a basis of  $V$ .

**10.27.** Prove [Theorem 10.16](#): Suppose  $W$  is a subspace invariant under a linear operator  $T: V \rightarrow V$ . Then  $T$  induces a linear operator  $\bar{T}$  on  $V/W$  defined by  $\{\bar{v}_j\}$ .

Moreover, if  $T$  is a zero of any polynomial, then so is  $\bar{T}$ . Thus, the minimal polynomial of  $\bar{T}$  divides the minimal polynomial of  $T$ .

We first show that  $\bar{T}$  is well defined; that is, if  $u + W = v + W$ , then  $\bar{T}(u + W) = \bar{T}(v + W)$ . If  $u + W = v + W$ , then  $u - v \in W$ , and, as  $W$  is  $T$ -invariant,  $T(u - v) = T(u) - T(v) \in W$ . Accordingly,

$$\bar{T}(u + W) = T(u) + W = T(v) + W = \bar{T}(v + W)$$

as required.

We next show that  $\bar{T}$  is linear. We have

$$\begin{aligned}\bar{T}((u + W) + (v + W)) &= \bar{T}(u + v + W) = T(u + v) + W = T(u) + T(v) + W \\ &= T(u) + W + T(v) + W = \bar{T}(u + W) + \bar{T}(v + W)\end{aligned}$$

Furthermore,

$$\bar{T}(k(u + W)) = \bar{T}(ku + W) = T(ku) + W = kT(u) + W = k(T(u) + W) = k\bar{T}(u + W)$$

Thus,  $\bar{T}$  is linear.

Now, for any coset  $u + W$  in  $V/W$ ,

$$\bar{T}^2(u + W) = T^2(u) + W = T(T(u)) + W = \bar{T}(T(u) + W) = \bar{T}(\bar{T}(u + W)) = \bar{T}^2(u + W)$$

Hence,  $\overline{\bar{T}^2} = \bar{T}^2$ . Similarly,  $\overline{\bar{T}^n} = \bar{T}^n$  for any  $n$ . Thus, for any polynomial

$$f(t) = a_n t^n + \cdots + a_0 = \sum a_i t^i$$

$$\begin{aligned}\overline{f(T)}(u + W) &= f(T)(u) + W = \sum a_i T^i(u) + W = \sum a_i (T^i(u) + W) \\ &= \sum a_i \bar{T}^i(u + W) = \sum a_i \bar{T}^i(u + W) = (\sum a_i \bar{T}^i)(u + W) = f(\bar{T})(u + W)\end{aligned}$$

and so  $\overline{f(T)} = f(\bar{T})$ . Accordingly, if  $T$  is a root of  $f(t)$  then  $\overline{f(T)} = \mathbf{0} = W = f(\bar{T})$ ; that is,  $\bar{T}$  is also a root of  $f(t)$ . The

theorem is proved.

**10.28.** Prove [Theorem 10.1](#): Let  $T: V \rightarrow V$  be a linear operator whose characteristic polynomial factors into linear polynomials. Then  $V$  has a basis in which  $T$  is represented by a triangular matrix.

The proof is by induction on the dimension of  $V$ . If  $\dim V = 1$ , then every matrix representation of  $T$  is a  $1 \times 1$  matrix, which is triangular.

Now suppose  $\dim V = n > 1$  and that the theorem holds for spaces of dimension less than  $n$ . Because the characteristic polynomial of  $T$  factors into linear polynomials,  $T$  has at least one eigenvalue and so at least one nonzero eigenvector  $v$ , say  $T(v) = a_{11}v$ . Let  $W$  be the one-dimensional subspace spanned by  $v$ . Set  $\bar{V} = V/W$ . Then ([Problem 10.26](#))  $\dim \bar{V} = \dim V - \dim W = n - 1$ . Note also that  $W$  is invariant under  $T$ . By [Theorem 10.16](#),  $T$  induces a linear operator  $\bar{T}$  on  $\bar{V}$  whose minimal polynomial divides the minimal polynomial of  $T$ . Because the characteristic polynomial of  $T$  is a product of linear polynomials, so is its minimal polynomial, and hence, so are the minimal and characteristic polynomials of  $\bar{T}$ . Thus,  $\bar{V}$  and  $\bar{T}$  satisfy the hypothesis of the theorem. Hence, by induction, there exists a basis  $\{\bar{v}_2, \dots, \bar{v}_n\}$  of  $\bar{V}$  such that

$$\begin{aligned}\bar{T}(\bar{v}_2) &= a_{22}\bar{v}_2 \\ \bar{T}(\bar{v}_3) &= a_{32}\bar{v}_2 + a_{33}\bar{v}_3 \\ &\dots\dots\dots \\ \bar{T}(\bar{v}_n) &= a_{n2}\bar{v}_2 + a_{n3}\bar{v}_3 + \dots + a_{nn}\bar{v}_n\end{aligned}$$

Now let  $v_2, \dots, v_n$  be elements of  $V$  that belong to the cosets  $\bar{v}_2, \dots, \bar{v}_n$ , respectively. Then  $\{v, v_2, \dots, v_n\}$  is a basis of  $V$  ([Problem 10.26](#)). Because  $\bar{T}(\bar{v}_2) = a_{22}\bar{v}_2$ , we have

$$\bar{T}(\bar{v}_2) - a_{22}\bar{v}_2 = 0, \quad \text{and so} \quad T(v_2) - a_{22}v_2 \in W$$

But  $W$  is spanned by  $v$ ; hence,  $T(v_2) - a_{22}v_2$  is a multiple of  $v$ , say,

$$T(v_2) - a_{22}v_2 = a_{21}v, \quad \text{and so} \quad T(v_2) = a_{21}v + a_{22}v_2$$

Similarly, for  $i = 3, \dots, n$

$$T(v_i) - a_{i2}v_2 - a_{i3}v_3 - \cdots - a_{in}v_i \in W, \quad \text{and so} \quad T(v_i) = a_{i1}v + a_{i2}v_2 + \cdots + a_{in}v_i$$

Thus,

[illegible]

and hence the matrix of  $T$  in this basis is triangular.

## Cyclic Subspaces, Rational Canonical Form

**10.29.** Prove [Theorem 10.12](#): Let  $Z(\nu, T)$  be a  $T$ -cyclic subspace,  $T_\nu$  the restriction of  $T$  to  $Z(\nu, T)$ , and  $m_\nu(t) = t^k + a_{k-1}t^{k-1} + \cdots + a_0$  the  $T$ -annihilator of  $\nu$ . Then,

(i) The set  $\{\nu, T(\nu), \dots, T^{k-1}(\nu)\}$  is a basis of  $Z(\nu, T)$ ; hence,  $\dim Z(\nu, T) = k$ .

(ii) The minimal polynomial of  $T_\nu$  is  $m_\nu(t)$ .

(iii) The matrix of  $T_\nu$  in the above basis is the companion matrix  $C = C(m_\nu)$  of  $m_\nu(t)$  [which has 1's below the diagonal, the negative of the coefficients  $a_0, a_1, \dots, a_{k-1}$  of  $m_\nu(t)$  in the last column, and 0's elsewhere].

(i) By definition of  $m_\nu(t)$ ,  $T^k(\nu)$  is the first vector in the sequence  $\nu, T(\nu), T^2(\nu), \dots$  that is a linear combination of those vectors that precede it in the sequence; hence, the set  $B = \{\nu, T(\nu), \dots, T^{k-1}(\nu)\}$  is linearly independent. We now only have to show that  $Z(\nu, T) = L(B)$ , the linear span of  $B$ . By the above,  $T^k(\nu) \in L(B)$ . We prove by induction that  $T^n(\nu) \in L(B)$  for every  $n$ . Suppose  $n > k$  and  $T^{n-1}(\nu) \in L(B)$ —that is,  $T^{n-1}(\nu)$  is a linear combination of  $\nu, \dots, T^{k-1}(\nu)$ . Then  $T^n(\nu) = T(T^{n-1}(\nu))$  is a linear combination of  $T(\nu), \dots, T^k(\nu)$ . But  $T^k(\nu) \in L(B)$ ; hence,  $T^n(\nu) \in L(B)$  for every  $n$ . Consequently,  $f(T)(\nu) \in L(B)$  for any polynomial  $f(t)$ . Thus,  $Z(\nu, T) = L(B)$ , and so  $B$  is a basis, as claimed.

(ii) Suppose  $m(t) = t^s + b_{s-1}t^{s-1} + \dots + b_0$  is the minimal polynomial of  $T_\nu$ . Then, because  $\nu \in Z(\nu, T)$ ,

$$0 = m(T_\nu)(\nu) = m(T)(\nu) = T^s(\nu) + b_{s-1}T^{s-1}(\nu) + \dots + b_0\nu$$

Thus,  $T^s(\nu)$  is a linear combination of  $\nu, T(\nu), \dots, T^{s-1}(\nu)$ , and therefore  $k \leq s$ . However,  $m_\nu(T) = \mathbf{0}$  and so  $m_\nu(T_\nu) = \mathbf{0}$ . Then  $m(t)$  divides  $m_\nu(t)$ , and so  $s \leq k$ . Accordingly,  $k = s$  and hence  $m_\nu(t) = m(t)$ .

(iii)

$$\begin{array}{rcl} T_\nu(\nu) & = & T(\nu) \\ T_\nu(T(\nu)) & = & T^2(\nu) \\ & \dots & \\ T_\nu(T^{k-2}(\nu)) & = & T^{k-1}(\nu) \\ T_\nu(T^{k-1}(\nu)) & = & T^k(\nu) = -a_0\nu - a_1T(\nu) - a_2T^2(\nu) - \dots - a_{k-1}T^{k-1}(\nu) \end{array}$$

By definition, the matrix of  $T_\nu$  in this basis is the transpose of the matrix of coefficients of the above system of equations; hence, it is  $C$ , as required.

**10.30.** Let  $T: V \rightarrow V$  be linear. Let  $W$  be a  $T$ -invariant subspace of  $V$  and  $\bar{T}$  the induced operator on  $V/W$ . Prove

- (a) The  $T$ -annihilator of  $v \in V$  divides the minimal polynomial of  $T$ .
- (b) The  $\bar{T}$ -annihilator of  $\bar{v}$  divides the minimal polynomial of  $T$ .
- (a) The  $T$ -annihilator of  $v \in V$  is the minimal polynomial of the restriction of  $T$  to  $Z(v, T)$ ; therefore, by [Problem 10.6](#), it divides the minimal polynomial of  $T$ .
- (b) The  $\bar{T}$ -annihilator of  $\bar{v}_2$  divides the minimal polynomial of  $\bar{T}$ , which divides the minimal polynomial of  $T$  by [Theorem 10.16](#).

**Remark:** In the case where the minimum polynomial of  $T$  is  $f(t)^n$ , where  $f(t)$  is a monic irreducible polynomial, then the  $T$ -annihilator of  $v \in V$  and the  $\bar{T}$ -annihilator of  $\bar{v}$  are of the form  $f(t)^m$ , where  $m \leq n$ .

**10.31.** Prove Lemma 10.13: Let  $T: V \rightarrow V$  be a linear operator whose minimal polynomial is  $f(t)^n$ , where  $f(t)$  is a monic irreducible polynomial. Then  $V$  is the direct sum of  $T$ -cyclic subspaces  $Z_i = Z(v_i, T)$ ,  $i = 1, \dots, r$ , with corresponding  $T$ -annihilators

$$f(t)^{n_1}, f(t)^{n_2}, \dots, f(t)^{n_r}, \quad n = n_1 \geq n_2 \geq \dots \geq n_r$$

Any other decomposition of  $V$  into the direct sum of  $T$ -cyclic subspaces has the same number of components and the same set of  $T$ -annihilators.

The proof is by induction on the dimension of  $V$ . If  $\dim V = 1$ , then  $V$  is  $T$ -cyclic and the lemma holds.

Now suppose  $\dim V > 1$  and that the lemma holds for those vector spaces of dimension less than that of  $V$ .

Because the minimal polynomial of  $T$  is  $f(t)^n$ , there exists  $v_1 \in V$  such that  $f(T)^{n-1}(v_1) \neq 0$ ; hence, the  $T$ -annihilator of  $v_1$  is  $f(t)^n$ . Let  $Z_1 = Z(v_1, T)$  and recall that  $Z_1$  is  $T$ -invariant. Let  $\bar{V} = V/Z_1$  and let  $\bar{T}$  be the linear operator on  $\bar{V}$  induced by  $T$ . By [Theorem 10.16](#), the minimal polynomial of  $\bar{T}$  divides  $f(t)^n$ ; hence, the hypothesis holds for  $\bar{V}$  and  $\bar{T}$ . Consequently, by induction,  $\bar{V}$  is the direct sum of  $\bar{T}$ -cyclic subspaces; say,

$$\bar{V} = Z(\bar{v}_2, \bar{T}) \oplus \cdots \oplus Z(\bar{v}_r, \bar{T})$$

where the corresponding  $\bar{T}$ -annihilators are  $f(t)^{n_2}, \dots, f(t)^{n_r}$ ,  $n \geq n_2 \geq n_r$ .

We claim that there is a vector  $v_2$  in the coset  $\bar{v}_2$  whose  $T$ -annihilator is  $f(t)^{n_2}$ , the  $\bar{T}$ -annihilator of  $\bar{v}_2$ . Let  $w$  be any vector in  $\bar{V} = V/Z_1$ . Then  $f(t)^{n_2}(w) \in Z_1$ . Hence, there exists a polynomial  $g(t)$  for which

$$f(T)^{n_2}(w) = g(T)(v_1) \tag{1}$$

Because  $f(t)^n$  is the minimal polynomial of  $T$ , we have, by (1),

$$0 = f(T)^n(w) = f(T)^{n-n_2}g(T)(v_1)$$

But  $f(t)^n$  is the  $T$ -annihilator of  $v_1$ ; hence,  $f(t)^n$  divides  $f(t)^n - n_2 g(t)$ , and so  $g(t) = f(t)^{n_2} h(t)$  for some polynomial  $h(t)$ . We set

$$v_2 = w - h(T)(v_1)$$

Because  $w - v_2 = h(T)(v_1) \in Z_1$ ,  $v_2$  also belongs to the coset  $\bar{v}_2$ . Thus, the  $T$ -annihilator of  $v_2$  is a multiple of the  $\bar{T}$ -annihilator of  $\bar{v}_2$ . On the other hand, by (1),

$$f(T)^{n_2}(v_2) = f(T)^{n_2}(w - h(T)(v_1)) = f(T)^{n_2}(w) - g(T)(v_1) = 0$$

Consequently, the  $T$ -annihilator of  $v_2$  is  $f(t)^{n_2}$ , as claimed.

Similarly, there exist vectors  $v_3, \dots, v_r \in V$  such that  $v_i \in \bar{v}_i$  and that the  $T$ -annihilator of  $v_i$  is  $f(t)^{n_i}$ , the  $\bar{T}$ -annihilator of  $\bar{v}_i$ . We set

$$Z_2 = Z(v_2, T), \quad \dots, \quad Z_r = Z(v_r, T)$$

Let  $d$  denote the degree of  $f(t)$ , so that  $f(t)^{n_i}$  has degree  $dn_i$ . Then, because  $f(t)^{n_i}$  is both the  $T$ -annihilator of  $v_i$  and the  $\bar{T}$ -annihilator of  $\bar{v}_i$ , we know that

$$\{v_i, T(v_i), \dots, T^{dn_i-1}(v_i)\} \quad \text{and} \quad \{\bar{v}_i, \bar{T}(\bar{v}_i), \dots, \bar{T}^{dn_i-1}(\bar{v}_i)\}$$

are bases for  $Z(v_i, T)$  and  $Z(\bar{v}_i, \bar{T})$ , respectively, for  $i = 2, \dots, r$ . But  $\bar{V} = Z(\bar{v}_2, \bar{T}) \oplus \dots \oplus Z(\bar{v}_r, \bar{T})$ ; hence,

$$\{\bar{v}_2, \dots, \bar{T}^{dn_2-1}(\bar{v}_2), \dots, \bar{v}_r, \dots, \bar{T}^{dn_r-1}(\bar{v}_r)\}$$

is a basis for  $\bar{V}$ . Therefore, by [Problem 10.26](#) and the relation  $\bar{T}^i(\bar{v}) = \overline{T^i(v)}$  (see [Problem 10.27](#)),

$$\{v_1, \dots, T^{dn_1-1}(v_1), v_2, \dots, T^{dn_2-1}(v_2), \dots, v_r, \dots, T^{dn_r-1}(v_r)\}$$

is a basis for  $V$ . Thus, by [Theorem 10.4](#),  $V = Z(v_1, T) \oplus \dots \oplus Z(v_r, T)$ , as required.

It remains to show that the exponents  $n_1, \dots, n_r$  are uniquely determined by  $T$ . Because  $d = \text{degree of } f(t)$ ,

$$\dim V = d(n_1 + \dots + n_r) \quad \text{and} \quad \dim Z_i = dn_i, \quad i = 1, \dots, r$$

Also, if  $s$  is any positive integer, then (Problem 10.59)  $f(T)^s(Z_i)$  is a cyclic subspace generated by  $f(T)^s(v_i)$ , and it has dimension  $d(n_i - s)$  if  $n_i > s$  and dimension 0 if  $n_i \leq s$ .

Now any vector  $v \in V$  can be written uniquely in the form  $v = w_1 + \dots + w_r$  where  $w_i \in Z_i$ . Hence, any vector in  $f(T)^s(V)$  can be written uniquely in the form

$$f(T)^s(v) = f(T)^s(w_1) + \dots + f(T)^s(w_r)$$

where  $f(T)^s(w_i) \in f(T)^s(Z_i)$ . Let  $t$  be the integer, dependent on  $s$ , for which

$$n_1 > s, \quad \dots, \quad n_t > s, \quad n_{t+1} \geq s$$

Then

$$f(T)^s(V) = f(T)^s(Z_1) \oplus \dots \oplus f(T)^s(Z_t)$$

and so

$$\dim[f(T)^s(V)] = d[(n_1 - s) + \dots + (n_t - s)] \quad (2)$$

The numbers on the left of (2) are uniquely determined by  $T$ . Set  $s = n - 1$ , and (2) determines the number of  $n_i$  equal to  $n$ . Next set  $s = n - 2$ , and (2) determines the number of  $n_i$  (if any) equal to  $n - 1$ . We repeat the process until we set  $s = 0$  and determine the number

of  $n_i$  equal to 1. Thus, the  $n_i$  are uniquely determined by  $T$  and  $V$ , and the lemma is proved.

**10.32.** Let  $V$  be a seven-dimensional vector space over  $\mathbf{R}$ , and let  $T: V \rightarrow V$  be a linear operator with minimal polynomial  $m(t) = (t^2 - 2t + 5)(t - 3)^3$ . Find all possible rational canonical forms  $M$  of  $T$ .

Because  $\dim V = 7$ , there are only two possible characteristic polynomials,  $\Delta_1(t) = (t^2 - 2t + 5)^2(t - 3)^3$  or  $\Delta_1(t) = (t^2 - 2t + 5)(t - 3)^5$ . Moreover, the sum of the orders of the companion matrices must add up to 7. Also, one companion matrix must be  $C(t^2 - 2t + 5)$  and one must be  $C((t - 3)^3) = C(t^3 - 9t^2 + 27t - 27)$ . Thus,  $M$  must be one of the following block diagonal matrices:

$$(a) \quad \text{diag} \left( \begin{bmatrix} 0 & -5 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & -5 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 27 \\ 1 & 0 & -27 \\ 0 & 1 & 9 \end{bmatrix} \right),$$

$$(b) \quad \text{diag} \left( \begin{bmatrix} 0 & -5 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 27 \\ 1 & 0 & -27 \\ 0 & 1 & 9 \end{bmatrix}, \begin{bmatrix} 0 & -9 \\ 1 & 6 \end{bmatrix} \right),$$

$$(c) \quad \text{diag} \left( \begin{bmatrix} 0 & -5 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 27 \\ 1 & 0 & -27 \\ 0 & 1 & 9 \end{bmatrix}, [3], [3] \right)$$

## Projections

---

**10.33.** Suppose  $V = W_1 \oplus \cdots \oplus W_r$ . The *projection* of  $V$  into its subspace  $W_k$  is the mapping  $E: V \rightarrow V$  defined by  $E(v) = w_k$ , where  $v = w_1 + \cdots + w_r$ ,  $w_i \in W_i$ . Show that (a)  $E$  is linear, (b)  $E^2 = E$ .

(a) Because the sum  $v = w_1 + \cdots + w_r$ ,  $w_i \in W$  is uniquely determined by  $v$ , the mapping  $E$  is well defined. Suppose, for  $u \in V$ ,  $u = w'_1 + \cdots + w'_r$ ,  $w'_i \in W_i$ . Then

$$v + u = (w_1 + w'_1) + \cdots + (w_r + w'_r) \quad \text{and} \quad kv = kw_1 + \cdots + kw_r, \quad kw_i, w_i + w'_i \in W_i$$

are the unique sums corresponding to  $v + u$  and  $kv$ . Hence,

$$E(v + u) = w_k + w'_k = E(v) + E(u) \quad \text{and} \quad E(kv) = kw_k + kE(v)$$

and therefore  $E$  is linear.

(b) We have that

$$w_k = 0 + \cdots + 0 + w_k + 0 + \cdots + 0$$

is the unique sum corresponding to  $w_k \in W_k$ ; hence,  $E(w_k) = w_k$ . Then, for any  $v \in V$ ,

$$E^2(v) = E(E(v)) = E(w_k) = w_k = E(v)$$

Thus,  $E^2 = E$ , as required.

**10.34.** Suppose  $E: V \rightarrow V$  is linear and  $E^2 = E$ . Show that (a)  $E(u) = u$  for any  $u \in \text{Im } E$  (i.e., the restriction of  $E$  to its image is the identity mapping); (b)  $V$  is the direct sum of the image and kernel of  $E: V = \text{Im } E \oplus \text{Ker } E$ ; (c)  $E$  is the projection of  $V$  into  $\text{Im } E$ , its image. Thus, by the preceding problem, a linear mapping  $T: V \rightarrow V$  is a projection if and only if  $T^2 = T$ ; this characterization of a projection is frequently used as its definition.

(a) If  $u \in \text{Im } E$ , then there exists  $v \in V$  for which  $E(v) = u$ ; hence, as required,

$$E(u) = E(E(v)) = E^2(v) = E(v) = u$$

(b) Let  $v \in V$ . We can write  $v$  in the form  $v = E(v) + v - E(v)$ . Now  $E(v) \in \text{Im } E$  and, because

$$E(v - E(v)) = E(v) - E^2(v) = E(v) - E(v) = 0$$

$v - E(v) \in \text{Ker } E$ . Accordingly,  $V = \text{Im } E + \text{Ker } E$ .

Now suppose  $w \in \text{Im } E \cap \text{Ker } E$ . By (i),  $E(w) = w$  because  $w \in \text{Im } E$ . On the other hand,  $E(w) = 0$  because  $w \in \text{Ker } E$ . Thus,  $w = 0$ , and so  $\text{Im } E \cap \text{Ker } E = \{0\}$ . These two conditions imply that  $V$  is the direct sum of the image and kernel of  $E$ .

(c) Let  $v \in V$  and suppose  $v = u + w$ , where  $u \in \text{Im } E$  and  $w \in \text{Ker } E$ . Note that  $E(u) = u$  by (i), and  $E(w) = 0$  because  $w \in \text{Ker } E$ . Hence,

$$E(v) = E(u + w) = E(u) + E(w) = u + 0 = u$$

That is,  $E$  is the projection of  $V$  into its image.

**10.35.** Suppose  $V = U \oplus W$  and suppose  $T: V \rightarrow V$  is linear. Show that  $U$  and  $W$  are both  $T$ -invariant if and only if  $TE = ET$ , where  $E$  is the projection of  $V$  into  $U$ .

Observe that  $E(v) \in U$  for every  $v \in V$ , and that (i)  $E(v) = v$  iff  $v \in U$ , (ii)  $E(v) = 0$  iff  $v \in W$ . Suppose  $ET = TE$ . Let  $u \in U$ . Because  $E(u) = u$ ,

$$T(u) = T(E(u)) = (TE)(u) = (ET)(u) = E(T(u)) \in U$$

Hence,  $U$  is  $T$ -invariant. Now let  $w \in W$ . Because  $E(w) = 0$ ,

$$E(T(w)) = (ET)(w) = (TE)(w) = T(E(w)) = T(0) = 0, \quad \text{and so} \quad T(w) \in W$$

Hence,  $W$  is also  $T$ -invariant.

Conversely, suppose  $U$  and  $W$  are both  $T$ -invariant. Let  $v \in V$  and suppose  $v = u + w$ , where  $u \in U$  and  $w \in W$ . Then  $T(u) \in U$  and  $T(w) \in W$ ; hence,  $E(T(u)) = T(u)$  and  $E(T(w)) = 0$ . Thus,

$$\begin{aligned}(ET)(v) &= (ET)(u + w) = (ET)(u) + (ET)(w) = E(T(u)) + E(T(w)) = T(u) \\ (TE)(v) &= (TE)(u + w) = T(E(u + w)) = T(u)\end{aligned}$$

and

That is,  $(ET)(v) = (TE)(v)$  for every  $v \in V$ ; therefore,  $ET = TE$ , as required.

## SUPPLEMENTARY PROBLEMS

### Invariant Subspaces

---

- 10.36.** Suppose  $W$  is invariant under  $T: V \rightarrow V$ . Show that  $W$  is invariant under  $f(t)$  for any polynomial  $f(t)$ .
- 10.37.** Show that every subspace of  $V$  is invariant under  $I$  and  $\mathbf{0}$ , the identity and zero operators.
- 10.38.** Let  $W$  be invariant under  $T_1: V \rightarrow V$  and  $T_2: V \rightarrow V$ . Prove  $W$  is also invariant under  $T_1 + T_2$  and  $T_1 T_2$ .
- 10.39.** Let  $T: V \rightarrow V$  be linear. Prove that any eigenspace,  $E_\lambda$  is  $T$ -invariant.
- 10.40.** Let  $V$  be a vector space of odd dimension (greater than 1) over the real field  $\mathbf{R}$ . Show that any linear operator on  $V$  has an invariant subspace other than  $V$  or  $\{0\}$ .

**10.41.** Determine the invariant subspace of  $A = \begin{bmatrix} 2 & -4 \\ 5 & -2 \end{bmatrix}$  viewed as a linear operator on (a)  $\mathbf{R}^2$ , (b)  $\mathbf{C}^2$ .

**10.42.** Suppose  $\dim V = n$ . Show that  $T: V \rightarrow V$  has a triangular matrix representation if and only if there exist  $T$ -invariant subspaces  $W_1 \subset W_2 \subset \cdots \subset W_n = V$  for which  $\dim W_k = k$ ,  $k = 1, \dots, n$ .

## Invariant Direct Sums

---

**10.43.** The subspaces  $W_1, \dots, W_r$  are said to be *independent* if  $w_1 + \cdots + w_r = 0$ ,  $w_i \in W_i$ , implies that each  $w_i = 0$ . Show that  $\text{span}(W_i) = W_1 \oplus \cdots \oplus W_r$  if and only if the  $W_i$  are independent. [Here  $\text{span}(W_i)$  denotes the linear span of the  $W_i$ .]

**10.44.** Show that  $V = W_1 \oplus \cdots \oplus W_r$  if and only if (i)  $V = \text{span}(W_i)$  and (ii) for  $k = 1, 2, \dots, r$ ,  $W_k \cap \text{span}(W_1, \dots, W_{k-1}, W_{k+1}, \dots, W_r) = \{0\}$ .

**10.45.** Show that  $\text{span}(W_i) = W_1 \oplus \cdots \oplus W_r$  if and only if  $\dim [\text{span}(W_i)] = \dim W_1 + \cdots + \dim W_r$ .

**10.46.** Suppose the characteristic polynomial of  $T: V \rightarrow V$  is  $\Delta(t) = f_1(t)^{n_1} f_2(t)^{n_2} \cdots f_r(t)^{n_r}$ , where the  $f_i(t)$  are distinct monic irreducible polynomials. Let  $V = W_1 \oplus \cdots \oplus W_r$  be the primary decomposition of  $V$  into  $T$ -invariant subspaces. Show that  $f_i(t)^{n_i}$  is the characteristic polynomial of the restriction of  $T$  to  $W_i$ .

## Nilpotent Operators

---

- 10.47.** Suppose  $T_1$  and  $T_2$  are nilpotent operators that commute (i.e.,  $T_1 T_2 = T_2 T_1$ ). Show that  $T_1 + T_2$  and  $T_1 T_2$  are also nilpotent.
- 10.48.** Suppose  $A$  is a supertriangular matrix (i.e., all entries on and below the main diagonal are 0). Show that  $A$  is nilpotent.
- 10.49.** Let  $V$  be the vector space of polynomials of degree  $\leq n$ . Show that the derivative operator on  $V$  is nilpotent of index  $n + 1$ .
- 10.50.** Show that any Jordan nilpotent block matrix  $N$  is similar to its transpose  $N^T$  (the matrix with 1's below the diagonal and 0's elsewhere).
- 10.51.** Show that two nilpotent matrices of order 3 are similar if and only if they have the same index of nilpotency. Show by example that the statement is not true for nilpotent matrices of order 4.

## **Jordan Canonical Form**

---

- 10.52.** Find all possible Jordan canonical forms for those matrices whose characteristic polynomial  $\Delta(t)$  and minimal polynomial  $m(t)$  are as follows:
- (a)  $\Delta(t) = (t - 2)^4(t - 3)^2$ ,  $m(t) = (t - 2)^2(t - 3)^2$ ,  
 (b)  $\Delta(t) = (t - 7)^5$ ,  $m(t) = (t - 7)^2$ , (c)  $\Delta(t) = (t - 2)^7$ ,  
 $m(t) = (t - 2)^3$
- 10.53.** Show that every complex matrix is similar to its transpose. (*Hint:* Use its Jordan canonical form.)
- 10.54.** Show that all  $n \times n$  complex matrices  $A$  for which  $A^n = I$  but  $A_k \neq I$  for  $k < n$  are similar.

- 10.55.** Suppose  $A$  is a complex matrix with only real eigenvalues. Show that  $A$  is similar to a matrix with only real entries.

## Cyclic Subspaces

---

- 10.56.** Suppose  $T: V \rightarrow V$  is linear. Prove that  $Z(u, T)$  is the intersection of all  $T$ -invariant subspaces containing  $u$ .
- 10.57.** Let  $f(t)$  and  $g(t)$  be the  $T$ -annihilators of  $u$  and  $v$ , respectively. Show that if  $f(t)$  and  $g(t)$  are relatively prime, then  $f(t)g(t)$  is the  $T$ -annihilator of  $u + v$ .
- 10.58.** Prove that  $Z(u, T) = Z(v, T)$  if and only if  $g(T)(u) = v$  where  $g(t)$  is relatively prime to the  $T$ -annihilator of  $u$ .
- 10.59.** Let  $W = Z(u, T)$ , and suppose the  $T$ -annihilator of  $u$  is  $f(t)^n$ , where  $f(t)$  is a monic irreducible polynomial of degree  $d$ . Show that  $f(t)^s(W)$  is a cyclic subspace generated by  $f(T)^s(u)$  and that it has dimension  $d(n - s)$  if  $n > s$  and dimension 0 if  $n \leq s$ .

## Rational Canonical Form

---

- 10.60.** Find all possible rational forms for a  $6 \times 6$  matrix over  $\mathbf{R}$  with minimal polynomial:
- (a)  $m(t) = (t^2 - 2t + 3)(t + 1)^2$ , (b)  $m(t) = (t - 2)^3$ .
- 10.61.** Let  $A$  be a  $4 \times 4$  matrix with minimal polynomial  $m(t) = (t^2 + 1)(t^2 - 3)$ . Find the rational canonical form for  $A$  if  $A$  is a matrix over (a) the rational field  $\mathbf{Q}$ , (b) the real field  $\mathbf{R}$ , (c) the complex field  $\mathbf{C}$ .
- 10.62.** Find the rational canonical form for the four-square Jordan block with  $\lambda$ 's on the diagonal.

- 10.63.** Prove that the characteristic polynomial of an operator  $T: V \rightarrow V$  is a product of its elementary divisors.
- 10.64.** Prove that two  $3 \times 3$  matrices with the same minimal and characteristic polynomials are similar.
- 10.65.** Let  $C(f(t))$  denote the companion matrix to an arbitrary polynomial  $f(t)$ . Show that  $f(t)$  is the characteristic polynomial of  $C(f(t))$ .

## Projections

---

- 10.66.** Suppose  $V = W_1 \oplus \cdots \oplus W_r$ . Let  $E_i$  denote the projection of  $V$  into  $W_i$ . Prove (i)  $E_i E_j = 0$ ,  $i \neq j$ ; (ii)  $I = E_1 + \cdots + E_r$
- 10.67.** Let  $E_1, \dots, E_r$  be linear operators on  $V$  such that  
 (i)  $E_i^2 = E_i$  (i.e., the  $E_i$  are projections); (ii)  $E_i E_j = 0$ ,  $i \neq j$ ; (iii)  $I = E_1 + \cdots + E_r$   
 Prove that  $V = \text{Im } E_1 \oplus \cdots \oplus \text{Im } E_r$
- 10.68.** Suppose  $E: V \rightarrow V$  is a projection (i.e.,  $E^2 = E$ ). Prove that  $E$  has a matrix representation of the form  $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ , where  $r$  is the rank of  $E$  and  $I_r$  is the  $r$ -square identity matrix.
- 10.69.** Prove that any two projections of the same rank are similar. (*Hint:* Use the result of Problem 10.68.)
- 10.70.** Suppose  $E: V \rightarrow V$  is a projection. Prove  
 (i)  $I - E$  is a projection and  $V = \text{Im } E \oplus \text{Im } (I - E)$ , (ii)  $I + E$  is invertible (if  $1 + 1 \neq 0$ ).

## Quotient Spaces

---

**10.71.** Let  $W$  be a subspace of  $V$ . Suppose the set of cosets  $\{v_1 + W, v_2 + W, \dots, v_n + W\}$  in  $V/W$  is linearly independent. Show that the set of vectors  $\{v_1, v_2, \dots, v_n\}$  in  $V$  is also linearly independent.

**10.72.** Let  $W$  be a subspace of  $V$ . Suppose the set of vectors  $\{u_1, u_2, \dots, u_n\}$  in  $V$  is linearly independent, and that  $L(u_i) \cap W = \{0\}$ . Show that the set of cosets  $\{u_1 + W, \dots, u_n + W\}$  in  $V/W$  is also linearly independent.

**10.73.** Suppose  $V = U \oplus W$  and that  $\{u_1, \dots, u_n\}$  is a basis of  $U$ . Show that  $\{u_1 + W, \dots, u_n + W\}$  is a basis of the quotient spaces  $V/W$ . (Observe that no condition is placed on the dimensionality of  $V$  or  $W$ .)

**10.74.** Let  $W$  be the solution space of the linear equation

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = 0, \quad a_i \in K$$

and let  $v = (b_1, b_2, \dots, b_n) \in K^n$ . Prove that the coset  $v + W$  of  $W$  in  $K^n$  is the solution set of the linear equation

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b, \quad \text{where} \quad b = a_1b_1 + \dots + a_nb_n$$

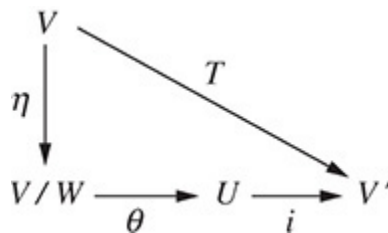
**10.75.** Let  $V$  be the vector space of polynomials over  $\mathbf{R}$  and let  $W$  be the subspace of polynomials divisible by  $t^4$  (i.e., of the form  $a_0t^4 + a_1t^5 + \dots + a_{n-4}t^n$ ). Show that the quotient space  $V/W$  has dimension 4.

**10.76.** Let  $U$  and  $W$  be subspaces of  $V$  such that  $W \subset U \subset V$ . Note that any coset  $u + W$  of  $W$  in  $U$  may also be viewed as a coset of  $W$  in  $V$ , because  $u \in U$  implies  $u \in V$ ; hence,  $U/W$  is a subset of  $V/W$ . Prove that (i)  $U/W$  is

a subspace of  $V/W$ , (ii)  $\dim(V/W) - \dim(U/W) = \dim(V/U)$ .

**10.77.** Let  $U$  and  $W$  be subspaces of  $V$ . Show that the cosets of  $U \cap W$  in  $V$  can be obtained by intersecting each of the cosets of  $U$  in  $V$  by each of the cosets of  $W$  in  $V$ :

$$V/(U \cap W) = \{(v + U) \cap (v' + W) : v, v' \in V\}$$



**10.78.** Let  $T: V \rightarrow V$  be linear with kernel  $W$  and image  $U$ . Show that the quotient space  $V/W$  is isomorphic to  $U$  under the mapping  $\theta: V/W \rightarrow U$  defined by  $\theta(v + W) = T(v)$ . Furthermore, show that  $T = i \circ \theta \circ \eta$ , where  $\eta: V \rightarrow V/W$  is the natural mapping of  $V$  into  $V/W$  (i.e.,  $\eta(v) = v + W$ ), and  $i: U \hookrightarrow V'$  is the inclusion mapping (i.e.,  $i(u) = u$ ). (See diagram.)

## ANSWERS TO SUPPLEMENTARY PROBLEMS

**10.41.** (a)  $\mathbf{R}^2$  and  $\{0\}$ , (b)  $\mathbf{C}^2$ ,  $\{0\}$ ,  $W_1 = \text{span}(2, 1 - 2i)$ ,  $W_2 = \text{span}(2, 1 + 2i)$

**10.52.**

(a)  $\text{diag}\left(\begin{bmatrix} 2 & 1 \\ & 2 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ & 2 \end{bmatrix}, \begin{bmatrix} 3 & 1 \\ & 3 \end{bmatrix}\right), \quad \text{diag}\left(\begin{bmatrix} 2 & 1 \\ & 2 \end{bmatrix}, [2], [2], \begin{bmatrix} 3 & 1 \\ & 3 \end{bmatrix}\right)$

$$(b) \operatorname{diag}\left(\begin{bmatrix} 7 & 1 \\ & 7 \end{bmatrix}, \begin{bmatrix} 7 & 1 \\ & 7 \end{bmatrix}, [7]\right), \quad \operatorname{diag}\left(\begin{bmatrix} 7 & 1 \\ & 7 \end{bmatrix}, [7], [7], [7]\right);$$

(c) Let  $M_k$  denote a Jordan block with  $\lambda = 2$  and order  $k$ . Then  $\operatorname{diag}(M_3, M_3, M_1)$ ,  $\operatorname{diag}(M_3, M_2, M_2)$ ,  $\operatorname{diag}(M_3, M_2, M_1, M_1)$ ,  $\operatorname{diag}(M_3, M_1, M_1, M_1, M_1)$

**10.60.** Let

$$A = \begin{bmatrix} 0 & -3 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & -1 \\ 1 & -2 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 0 & 8 \\ 1 & 0 & -12 \\ 0 & 1 & 6 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & -4 \\ 1 & 4 \end{bmatrix}$$

(a)  $\operatorname{diag}(A, A, B)$ ,  $\operatorname{diag}(A, B, B)$ ,  $\operatorname{diag}(A, B, -1, -1)$ ; (b)  $\operatorname{diag}(C, C)$ ,  $\operatorname{diag}(C, D, 2)$ ,  $\operatorname{diag}(C, 2, 2, 2)$

**10.61.** Let  $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ ,  $B = \begin{bmatrix} 0 & 3 \\ 1 & 0 \end{bmatrix}$ .

(a)  $\operatorname{diag}(A, B)$ , (b)  $\operatorname{diag}(A, \sqrt{3}, -\sqrt{3})$ , (c)  $\operatorname{diag}(i, -i, \sqrt{3}, -\sqrt{3})$

**10.62.** Companion matrix with the last column  $[\lambda^4, 4\lambda^3, -6\lambda^2, 4\lambda]^T$

# CHAPTER 11

## ***Linear Functionals and the Dual Space***

### **11.1 Introduction**

---

In this chapter, we study linear mappings from a vector space  $V$  into its field  $K$  of scalars. (Unless otherwise stated or implied, we view  $K$  as a vector space over itself.)

Naturally all the theorems and results for arbitrary mappings on  $V$  hold for this special case. However, we treat these mappings separately because of their fundamental importance and because the special relationship of  $V$  to  $K$  gives rise to new notions and results that do not apply in the general case.

### **11.2 Linear Functionals and the Dual Space**

---

Let  $V$  be a vector space over a field  $K$ . A mapping  $\phi: V \rightarrow K$  is termed a linear functional (or linear form) if, for every  $u, v \in V$  and every  $a, b \in K$ ,

$$\phi(au + bv) = a\phi(u) + b\phi(v)$$

In other words, a linear functional on  $V$  is a linear mapping from  $V$  into  $K$ .

### EXAMPLE 11.1

- (a) Let  $\pi_i: K^n \rightarrow K$  be the  $i$ th projection mapping; that is,  $\pi_i(a_1; a_2; \dots; a_n) = a_i$ . Then  $\pi_i$  is linear and so it is a linear functional on  $K^n$ .
- (b) Let  $V$  be the vector space of polynomials in  $t$  over  $\mathbf{R}$ . Let  $\mathbf{J}: V \rightarrow \mathbf{R}$  be the integral operator defined by  $\mathbf{J}(p(t)) = \int_0^1 p(t) dt$ . Recall that  $\mathbf{J}$  is linear; and hence, it is a linear functional on  $V$ .
- (c) Let  $V$  be the vector space of  $n$ -square matrices over  $K$ . Let  $T: V \rightarrow K$  be the trace mapping

$$T(A) = a_{11} + a_{22} + \cdots + a_{nn}, \quad \text{where} \quad A = [a_{ij}]$$

That is,  $T$  assigns to a matrix  $A$  the sum of its diagonal elements. This map is linear (Problem 11.24), and so it is a linear functional on  $V$ .

By [Theorem 5.10](#), the set of linear functionals on a vector space  $V$  over a field  $K$  is also a vector space over  $K$ , with addition and scalar multiplication defined by

$$(\phi + \sigma)(v) = \phi(v) + \sigma(v) \quad \text{and} \quad (k\phi)(v) = k\phi(v)$$

where  $\phi$  and  $\sigma$  are linear functionals on  $V$  and  $k \in K$ . This space is called the *dual space* of  $V$  and is denoted by  $V^*$ .

**EXAMPLE 11.2** Let  $V \cong K^n$ , the vector space of  $n$ -tuples, which we write as column vectors. Then the dual space  $V^*$  can be identified with the space of row vectors. In particular, any linear functional  $\phi = (a_1; \dots; a_n)$  in  $V^*$  has the representation

$$\phi(x_1, x_2, \dots, x_n) = [a_1, a_2, \dots, a_n][x_1, x_2, \dots, x_n]^T = a_1x_1 + a_2x_2 + \cdots + a_nx_n$$

Historically, the formal expression on the right was termed a *linear form*.

## 11.3 Dual Basis

---

Suppose  $V$  is a vector space of dimension  $n$  over  $K$ . By [Theorem 5.11](#), the dimension of the dual space  $V^*$  is also  $n$  (because  $K$  is of dimension 1 over

itself). In fact, each basis of  $V$  determines a basis of  $V^*$  as follows (see [Problem 11.3](#) for the proof).

**THEOREM 11.1:** Suppose  $\{v_1; \dots; v_n\}$  is a basis of  $V$  over  $K$ . Let  $\phi_1; \dots; \phi_n \in V^*$  be the linear functionals as defined by

$$\phi_i(v_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

The above basis  $\{\phi\}$  is termed the basis dual to  $\{v_i\}$  or the dual basis. The above formula, which uses the Kronecker delta  $\delta_{ij}$ , is a short way of writing

$$\begin{aligned} \phi_1(v_1) &= 1, & \phi_1(v_2) &= 0, & \phi_1(v_3) &= 0, & \dots, & \phi_1(v_n) &= 0 \\ \phi_2(v_1) &= 0, & \phi_2(v_2) &= 1, & \phi_2(v_3) &= 0, & \dots, & \phi_2(v_n) &= 0 \\ & \dots & & & & & & & \\ \phi_n(v_1) &= 0, & \phi_n(v_2) &= 0, & \dots, & \phi_n(v_{n-1}) &= 0, & \phi_n(v_n) &= 1 \end{aligned}$$

By [Theorem 5.2](#), these linear mappings  $\phi_i$  are unique and well defined

**EXAMPLE 11.3** Consider the basis  $\{v_1 = (2, 1), v_2 = (3, 1)\}$  of  $\mathbf{R}^2$ . Find the dual basis  $\{\phi_1, \phi_2\}$ . We seek linear functionals  $\phi_1(x, y) = ax + by$  and  $\phi_2(x, y) = cx + dy$  such that

$$\phi_1(v_1) = 1, \quad \phi_1(v_2) = 0, \quad \phi_2(v_1) = 0, \quad \phi_2(v_2) = 1$$

These four conditions lead to the following two systems of linear equations:

$$\left. \begin{aligned} \phi_1(v_1) &= \phi_1(2, 1) = 2a + b = 1 \\ \phi_1(v_2) &= \phi_1(3, 1) = 3a + b = 0 \end{aligned} \right\} \quad \text{and} \quad \left. \begin{aligned} \phi_2(v_1) &= \phi_2(2, 1) = 2c + d = 0 \\ \phi_2(v_2) &= \phi_2(3, 1) = 3c + d = 1 \end{aligned} \right\}$$

The solutions yield  $a = 1$ ,  $b = -3$  and  $c = 1$ ,  $d = 2$ . Hence,  $\phi_1(x, y) = x - 3y$  and  $\phi_2(x, y) = x + 2y$  form the dual basis.

The next two theorems (proved in Problems 11.4 and 11.5, respectively) give relationships between bases and their duals.

**THEOREM 11.2:** Let  $\{v_1, \dots, v_n\}$  be a basis of  $V$  and let  $\{f_1, \dots, f_n\}$  be the dual basis in  $V^*$ . Then

(i) For any vector

$$u \in V, \quad u = \phi_1(u)v_1 + \phi_2(u)v_2 + \dots + \phi_n(u)v_n.$$

(ii) For any linear functional

$$\sigma \in V^*, \quad \sigma = \sigma(v_1)\phi_1 + \sigma(v_2)\phi_2 + \dots + \sigma(v_n)\phi_n.$$

**THEOREM 11.3:** Let  $\{v_1, \dots, v_n\}$  and  $\{w_1, \dots, w_n\}$  be bases of  $V$  and let  $\{f_1, \dots, f_n\}$  and  $\{s_1, \dots, s_n\}$  be the bases of  $V^*$  dual to  $\{v_i\}$  and  $\{w_i\}$ , respectively. Suppose  $P$  is the change-of-basis matrix from  $\{v_i\}$  to  $\{w_i\}$ . Then  $(P^{-1})^T$  is the change-of-basis matrix from  $\{f_i\}$  to  $\{s_i\}$ .

## 11.4 Second Dual Space

---

We repeat: Every vector space  $V$  has a dual space  $V^*$ , which consists of all the linear functionals on  $V$ . Thus,  $V^*$  has a dual space  $V^{**}$ , called the second dual of  $V$ , which consists of all the linear functionals on  $V^*$ .

We now show that each  $v \in V$  determines a specific element  $\hat{v} \in V^{**}$ . First, for any  $\phi \in V^*$ , we define

$$\hat{v}(\phi) = \phi(v)$$

It remains to be shown that this map  $\hat{\cdot} : V^* \rightarrow K$  is linear. For any scalars  $a, b \in K$  and any linear functionals  $\phi, \rho \in V^*$ , we have

$$\hat{v}(a\phi + b\rho) = (a\phi + b\rho)(v) = a\phi(v) + b\rho(v) = a\hat{v}(\phi) + b\hat{v}(\rho)$$

That is,  $\hat{u}$  is linear and so  $\hat{u} \in V^{**}$ . The following theorem (proved in [Problem 12.7](#)) holds.

**THEOREM 11.4:** If  $V$  has finite dimensions, then the mapping  $v \mapsto \hat{v}$  is an isomorphism of  $V$  onto  $V^{**}$ .

The above mapping  $v \mapsto \hat{v}$  is called the natural mapping of  $V$  into  $V^{**}$ . We emphasize that this mapping is never onto  $V^{**}$  if  $V$  is not finite-dimensional. However, it is always linear, and moreover, it is always one-to-one.

Now suppose  $V$  does have finite dimension. By [Theorem 11.4](#), the natural mapping determines an isomorphism between  $V$  and  $V^{**}$ . Unless otherwise stated, we will identify  $V$  with  $V^{**}$  by this mapping. Accordingly, we will view  $V$  as the space of linear functionals on  $V^*$  and write  $V = V^{**}$ . We remark that if  $\{\phi_i\}$  is the basis of  $V^*$  dual to a basis  $\{v_i\}$  of  $V$ , then  $\{v_i\}$  is the basis of  $V^{**} = V$  that is dual to  $\{\phi_i\}$ .

## 11.5 Annihilators

---

Let  $W$  be a subset (not necessarily a subspace) of a vector space  $V$ . A linear functional  $\phi \in V^*$  is called an annihilator of  $W$  if  $\phi(w) = 0$  for every  $w \in W$ —that is, if  $\phi(W) = \{0\}$ . We show that the set of all such mappings, denoted by  $W^0$  and called the annihilator of  $W$ , is a subspace of  $V^*$ . Clearly,  $0 \in W^0$ : Now suppose  $\phi, \sigma \in W^0$ . Then, for any scalars  $a, b \in K$  and for any  $w \in W$ ,

$$(a\phi + b\sigma)(w) = a\phi(w) + b\sigma(w) = a0 + b0 = 0$$

Thus,  $a\phi + b\sigma \in W^0$ , and so  $W^0$  is a subspace of  $V^*$ .

In the case that  $W$  is a subspace of  $V$ , we have the following relationship between  $W$  and its annihilator  $W^0$  (see [Problem 11.11](#) for the proof).

**THEOREM 11.5:** Suppose  $V$  has finite dimension and  $W$  is a subspace of  $V$ . Then

$$(i) \dim W + \dim W^0 = \dim V \quad \text{and} \quad (ii) W^{00} = W$$

Here  $W^{00} = \{v \in V : \phi(v) = 0 \text{ for every } \phi \in W^0\}$  or, equivalently,  $W^{00} = (W^0)^0$ , where  $W^{00}$  is viewed as a subspace of  $V$  under the identification of  $V$  and  $V^{**}$ .

## 11.6 Transpose of a Linear Mapping

---

Let  $T: V \rightarrow U$  be an arbitrary linear mapping from a vector space  $V$  into a vector space  $U$ . Now for any linear functional  $\phi \in U^*$ , the composition  $\phi \circ T$  is a linear mapping from  $V$  into  $K$ :

$$\begin{array}{ccccc} V & \xrightarrow{T} & U & \xrightarrow{\phi} & K \\ & & \searrow \phi \circ T & & \nearrow \end{array}$$

That is,  $\phi \circ T \in V^*$ . Thus, the correspondence

$$\phi \mapsto \phi \circ T$$

is a mapping from  $U^*$  into  $V^*$ ; we denote it by  $T^t$  and call it the transpose of  $T$ . In other words,  $T^t: U^* \rightarrow V^*$  is defined by

$$T^t(\phi) = \phi \circ T$$

Thus,  $(T^t(\phi))(v) = \phi(T(v))$  for every  $v \in V$ .

**THEOREM 11.6:** The transpose mapping  $T^t$  defined above is linear.

Proof. For any scalars  $a, b \in K$  and any linear functionals  $\phi, \sigma \in U^*$ ,

$$T^t(a\phi + b\sigma) = (a\phi + b\sigma) \circ T = a(\phi \circ T) + b(\sigma \circ T) = aT^t(\phi) + bT^t(\sigma)$$

That is,  $T^t$  is linear, as claimed.

We emphasize that if  $T$  is a linear mapping from  $V$  into  $U$ , then  $T^t$  is a linear mapping from  $U^*$  into  $V^*$ . The same “transpose” for the mapping  $T^t$  no doubt derives from the following theorem (proved in [Problem 11.16](#)).

**THEOREM 11.7:** Let  $T: V \rightarrow U$  be linear, and let  $A$  be the matrix representation of  $T$  relative to bases  $\{v_i\}$  of  $V$  and  $\{u_i\}$  of  $U$ . Then the transpose matrix  $A^T$  is the matrix representation of  $T^t: U^* \rightarrow V^*$  relative to the bases dual to  $\{u_i\}$  and  $\{v_i\}$ .

## SOLVED PROBLEMS

### Dual Spaces and Dual Bases

**11.1.** Find the basis  $\{\phi_1, \phi_2, \phi_3\}$  that is dual to the following basis of  $\mathbf{R}^3$ :

$$\{v_1 = (1, -1, 3), \quad v_2 = (0, 1, -1), \quad v_3 = (0, 3, -2)\}$$

The linear functionals may be expressed in the form

$$\phi_1(x, y, z) = a_1x + a_2y + a_3z, \quad \phi_2(x, y, z) = b_1x + b_2y + b_3z, \quad \phi_3(x, y, z) = c_1x + c_2y + c_3z$$

By definition of the dual basis,  $f_i(v_j) = 0$  for  $i \neq j$ , but  $f_i(v_j) = 1$  for  $i = j$ .

We find  $\phi_1$  by setting  $\phi_1(v_1) = 1$ ;  $\phi_1(v_2) = 0$ ;  $\phi_1(v_3) = 0$ : This yields

$$\phi_1(1, -1, 3) = a_1 - a_2 + 3a_3 = 1, \quad \phi_1(0, 1, -1) = a_2 - a_3 = 0, \quad \phi_1(0, 3, -2) = 3a_2 - 2a_3 = 0$$

Solving the system of equations yields  $a_1 = 1$ ,  $a_2 = 0$ ,  $a_3 = 0$ . Thus,  $f_1(x; y; z) = x$ .

We find  $\phi_2$  by setting  $\phi_2(v_1) = 0$ ,  $\phi_2(v_2) = 1$ ,  $\phi_2(v_3) = 0$ . This yields

$$\phi_2(1, -1, 3) = b_1 - b_2 + 3b_3 = 0, \quad \phi_2(0, 1, -1) = b_2 - b_3 = 1, \quad \phi_2(0, 3, -2) = 3b_2 - 2b_3 = 0$$

Solving the system of equations yields  $c_1 = 7$ ,  $c_2 = 2$ ,  $c_3 = 3$ . Thus,  $\phi_2(x; y; z) = 7x - 2y - 3z$ .

We find  $\phi_3$  by setting  $\phi_3(v_1) = 0$ ,  $\phi_3(v_2) = 0$ ,  $\phi_3(v_3) = 1$ . This yields

$$\phi_3(1, -1, 3) = c_1 - c_2 + 3c_3 = 0, \quad \phi_3(0, 1, -1) = c_2 - c_3 = 0, \quad \phi_3(0, 3, -2) = 3c_2 - 2c_3 = 1$$

Solving the system of equations yields  $c_1 = 2$ ,  $c_2 = 1$ ,  $c_3 = 1$ . Thus,  $\phi_3(x; y; z) = -2x + y + z$ .

**11.2.** Let  $V = \{a + bt : a; b \in \mathbb{R}\}$ , the vector space of real polynomials of degree  $\geq 1$ . Find the basis  $\{v_1, v_2\}$  of  $V$  that is dual to the basis  $\{\phi_1, \phi_2\}$  of  $V^*$  defined by

$$\phi_1(f(t)) = \int_0^1 f(t) dt \quad \text{and} \quad \phi_2(f(t)) = \int_0^2 f(t) dt$$

Let  $v_1 = a + bt$  and  $v_2 = c + dt$ . By definition of the dual basis,

$$\phi_1(v_1) = 1, \quad \phi_1(v_2) = 0 \quad \text{and} \quad \phi_2(v_1) = 0, \quad \phi_2(v_2) = 1$$

Thus,

$$\left. \begin{aligned} \phi_1(v_1) &= \int_0^1 (a + bt) dt = a + \frac{1}{2}b = 1 \\ \phi_2(v_1) &= \int_0^2 (a + bt) dt = 2a + 2b = 0 \end{aligned} \right\} \quad \text{and} \quad \left. \begin{aligned} \phi_1(v_2) &= \int_0^1 (c + dt) dt = c + \frac{1}{2}d = 0 \\ \phi_2(v_2) &= \int_0^2 (c + dt) dt = 2c + 2d = 1 \end{aligned} \right\}$$

Solving each system yields  $a = 2$ ,  $b = -2$  and  $c = -\frac{1}{2}$ ,  $d = 1$ . Thus,  $\{v_1 = 2 - 2t, v_2 = -\frac{1}{2} + t\}$  is the basis of  $V$  that is dual to  $\{\phi_1, \phi_2\}$ .

**11.3.** Prove [Theorem 11.1](#): Suppose  $\{v_1, \dots, v_n\}$  is a basis of  $V$  over  $K$ . Let  $\phi_1, \dots, \phi_n \in V^*$  be defined by  $\phi_i(v_j) = 0$  for  $i \neq j$ , but  $\phi_i(v_i) = 1$  for  $i = j$ . Then  $\{\phi_1, \dots, \phi_n\}$  is a basis of  $V^*$ .

We first show that  $\{\phi_1, \dots, \phi_n\}$  spans  $V^*$ . Let  $\phi$  be an arbitrary element of  $V^*$ , and suppose

$$\phi(v_1) = k_1, \quad \phi(v_2) = k_2, \quad \dots, \quad \phi(v_n) = k_n$$

Set  $\sigma = k_1\phi_1 + \dots + k_n\phi_n$ . Then

$$\begin{aligned} \sigma(v_1) &= (k_1\phi_1 + \dots + k_n\phi_n)(v_1) = k_1\phi_1(v_1) + k_2\phi_2(v_1) + \dots + k_n\phi_n(v_1) \\ &= k_1 \cdot 1 + k_2 \cdot 0 + \dots + k_n \cdot 0 = k_1 \end{aligned}$$

Similarly, for  $i = 2; \dots; n$ ,

$$\sigma(v_i) = (k_1\phi_1 + \dots + k_n\phi_n)(v_i) = k_1\phi_1(v_i) + \dots + k_i\phi_i(v_i) + \dots + k_n\phi_n(v_i) = k_i$$

Thus,  $\phi(v_i) = \sigma(v_i)$  for  $i = 1; \dots; n$ . Because  $\phi$  and  $\sigma$  agree on the basis vectors,  $\phi = \sigma = k_1\phi_1 + \dots + k_n\phi_n$ . Accordingly,  $\{\phi_1, \dots, \phi_n\}$  spans  $V^*$ .

It remains to be shown that  $\{\phi_1, \dots, \phi_n\}$  is linearly independent. Suppose

$$a_1\phi_1 + a_2\phi_2 + \dots + a_n\phi_n = \mathbf{0}$$

Applying both sides to  $v_1$ , we obtain

$$\begin{aligned} 0 &= \mathbf{0}(v_1) = (a_1\phi_1 + \dots + a_n\phi_n)(v_1) = a_1\phi_1(v_1) + a_2\phi_2(v_1) + \dots + a_n\phi_n(v_1) \\ &= a_1 \cdot 1 + a_2 \cdot 0 + \dots + a_n \cdot 0 = a_1 \end{aligned}$$

Similarly, for  $i = 2; \dots ; n$ ,

$$0 = \mathbf{0}(v_i) = (a_1\phi_1 + \dots + a_n\phi_n)(v_i) = a_1\phi_1(v_i) + \dots + a_i\phi_i(v_i) + \dots + a_n\phi_n(v_i) = a_i$$

That is,  $a_1 = 0; \dots ; a_n = 0$ . Hence,  $\{\phi_1, \dots ; \phi_n\}$  is linearly independent, and so it is a basis of  $V^*$ .

**11.4.** Prove [Theorem 11.2](#): Let  $\{v_1, \dots ; v_n\}$  be a basis of  $V$  and let  $\{\phi_1, \dots ; \phi_n\}$  be the dual basis in  $V^*$ . For any  $u \in V$  and any  $s \in V^*$ , (i)  $u = \sum_i \phi_i(u)v_i$ . (ii)  $\sigma = \sum_i \phi(v_i)\phi_i$ .

Suppose

$$u = a_1v_1 + a_2v_2 + \dots + a_nv_n \tag{1}$$

Then

$$\phi_1(u) = a_1\phi_1(v_1) + a_2\phi_1(v_2) + \dots + a_n\phi_1(v_n) = a_1 \cdot 1 + a_2 \cdot 0 + \dots + a_n \cdot 0 = a_1$$

Similarly, for  $i = 2; \dots ; n$ ,

$$\phi_i(u) = a_1\phi_i(v_1) + \dots + a_i\phi_i(v_i) + \dots + a_n\phi_i(v_n) = a_i$$

That is,  $\phi_1(u) = a_1, \phi_2(u) = a_2, \dots ; \phi_n(u) = a_n$ .

Substituting these results into (1), we obtain (i).

Next we prove (ii). Applying the linear functional  $s$  to both sides of (i),

$$\begin{aligned} \sigma(u) &= \phi_1(u)\sigma(v_1) + \phi_2(u)\sigma(v_2) + \dots + \phi_n(u)\sigma(v_n) \\ &= \sigma(v_1)\phi_1(u) + \sigma(v_2)\phi_2(u) + \dots + \sigma(v_n)\phi_n(u) \\ &= (\sigma(v_1)\phi_1 + \sigma(v_2)\phi_2 + \dots + \sigma(v_n)\phi_n)(u) \end{aligned}$$

Because the above holds for every  $u \in V$ ,  $\sigma = \sigma(v_1)\phi_1 + \sigma(v_2)\phi_2 + \dots + \sigma(v_n)\phi_n$ , as claimed.

**11.5.** Prove [Theorem 11.3](#). Let  $\{v_i\}$  and  $\{w_i\}$  be bases of  $V$  and let  $\{f_i\}$  and  $\{s_i\}$  be the respective dual bases in  $V^*$ . Let  $P$  be the change-of-basis matrix from  $\{v_i\}$  to  $\{w_i\}$ : Then  $P^{-1}$  is the change-of-basis matrix from  $\{\phi_i\}$  to  $\{\sigma_i\}$ .

Suppose, for  $i = 1; \dots; n$ ,

$$w_i = a_{i1}v_1 + a_{i2}v_2 + \dots + a_{in}v_n \quad \text{and} \quad \sigma_i = b_{i1}\phi_1 + b_{i2}\phi_2 + \dots + b_{in}\phi_n$$

Then  $P = [a_{ij}]$  and  $Q = [b_{ij}]$ . We seek to prove that  $Q = (P^{-1})^T$ .

Let  $R_i$  denote the  $i$ th row of  $Q$  and let  $C_j$  denote the  $j$ th column of  $P^T$ . Then

$$R_i = (b_{i1}, b_{i2}, \dots, b_{in}) \quad \text{and} \quad C_j = (a_{j1}, a_{j2}, \dots, a_{jn})^T$$

By definition of the dual basis,

$$\begin{aligned} \sigma_i(w_j) &= (b_{i1}\phi_1 + b_{i2}\phi_2 + \dots + b_{in}\phi_n)(a_{j1}v_1 + a_{j2}v_2 + \dots + a_{jn}v_n) \\ &= b_{i1}a_{j1} + b_{i2}a_{j2} + \dots + b_{in}a_{jn} = R_i C_j = \delta_{ij} \end{aligned}$$

where  $\delta_{ij}$  is the Kronecker delta. Thus,

$$QP^T = [R_i C_j] = [\delta_{ij}] = I$$

Therefore,  $Q = (P^T)^{-1} = (P^{-1})^T$ , as claimed.

**11.6.** Suppose  $v \in V$ ,  $v \neq 0$ , and  $\dim V = n$ . Show that there exists  $\phi \in V^*$  such that  $\phi(v) \neq 0$ .

We extend  $\{v\}$  to a basis  $\{v; v_2, \dots; v_n\}$  of  $V$ . By [Theorem 5.2](#), there exists a unique linear mapping  $\phi: V \rightarrow K$  such that  $\phi(v) = 1$  and  $\phi(v_i) = 0$ ,  $i = 2; \dots; n$ . Hence,  $\phi$  has the desired property.

**11.7.** Prove [Theorem 11.4](#): Suppose  $\dim V = n$ . Then the natural mapping  $v \mapsto \hat{v}$  is an isomorphism of  $V$  onto  $V^{**}$ .

We first prove that the map  $v \mapsto \hat{v}$  is linear—that is, for any vectors  $v, w \in V$  and any scalars

$\widehat{av + bw} = a\hat{v} + b\hat{w}$ . For any linear functional  $\phi \in V^*$ ,

$$\widehat{av + bw}(\phi) = \phi(av + bw) = a\phi(v) + b\phi(w) = a\hat{v}(\phi) + b\hat{w}(\phi) = (a\hat{v} + b\hat{w})(\phi)$$

Because  $\widehat{av + bw}(\phi) = (a\hat{v} + b\hat{w})(\phi)$  for every  $\phi \in V^*$ , we have  $\widehat{av + bw} = a\hat{v} + b\hat{w}$ . Thus, the map  $v \mapsto \hat{v}$  is linear.

Now suppose  $v \in V, v \neq 0$ . Then, by [Problem 11.6](#), there exists  $\phi \in V^*$  for which  $\phi(v) \neq 0$ . Hence,  $\hat{v}(\phi) = \phi(v) \neq 0$ , and thus  $\hat{v} \neq 0$ . Because  $v \neq 0$  implies  $\hat{v} \neq 0$ , the map  $v \mapsto \hat{v}$  is nonsingular and hence an isomorphism (Theorem 5.64).

Now  $\dim V = \dim V^* = \dim V^{**}$ , because  $V$  has finite dimension. Accordingly, the mapping  $v \mapsto \hat{v}$  is an isomorphism of  $V$  onto  $V^{**}$ .

## Annihilators

---

**11.8.** Show that if  $\phi \in V^*$  annihilates a subset  $S$  of  $V$ , then  $\phi$  annihilates the linear span  $L(S)$  of  $S$ . Hence,  $S^0 = {}^{1/2}\text{span}(S)^0$ .

Suppose  $v \in \text{span}(S)$ . Then there exists  $w_1, \dots, w_r \in S$  for which  $v = a_1w_1 + a_2w_2 + \dots + a_rw_r$ .

$$\phi(v) = a_1\phi(w_1) + a_2\phi(w_2) + \dots + a_r\phi(w_r) = a_10 + a_20 + \dots + a_r0 = 0$$

Because  $v$  was an arbitrary element of  $\text{span}(S)$ ,  $\phi$  annihilates  $\text{span}(S)$ , as claimed.

**11.9.** Find a basis of the annihilator  $W^0$  of the subspace  $W$  of  $\mathbf{R}^4$  spanned by

$$v_1 = (1, 2, -3, 4) \quad \text{and} \quad v_2 = (0, 1, 4, -1)$$

By [Problem 11.8](#), it suffices to find a basis of the set of linear functionals  $\phi$  such that  $\phi(v_1) = 0$ , where  $\phi(x_1, x_2, x_3, x_4) = ax_1 + bx_2 + cx_3 + dx_4$ . Thus,

$$\phi(1, 2, -3, 4) = a + 2b - 3c + 4d = 0 \quad \text{and} \quad \phi(0, 1, 4, -1) = b + 4c - d = 0$$

The system of two equations in the unknowns  $a, b, c, d$  is in echelon form with free variables  $c$  and  $d$ .

(1) Set  $c = 1, d = 0$  to obtain the solution  $a = 11, b = 4, c = 1, d = 0$ .

(2) Set  $c = 0, d = 1$  to obtain the solution  $a = 6, b = 1, c = 0, d = 1$ .

The linear functions  $\phi_1(x_i) = 11x_1 + 4x_2 + x_3$  and  $\phi_2(x_i) = 6x_1 + x_2 + x_4$  form a basis of  $W^0$ .

**11.10.** Show that (a) For any subset  $S$  of  $V$ ;  $S \subseteq S^{00}$ . (b) If  $S_1 \subseteq S_2$ , then  $S_2^0 \subseteq S_1^0$

(a) Let  $v \in S$ . Then for every linear functional  $\phi \in S^0$ ,  $\hat{v}\phi = f(v) = 0$ . Hence,  $\hat{v}(\phi) = \phi(v) = 0$ . Therefore, under the identification of  $V$  and  $V^{**}$ ,  $v \in S^{00}$ . Accordingly,  $S \subseteq S^{00}$ .

(b) Let  $\phi \in S_2^0$ . Then  $\phi(v) = 0$  for every  $v \in S_2$ . But  $S_1 \subseteq S_2$ , hence,  $\phi$  annihilates every element of  $S_1$  (i.e.,  $\phi \in S_1^0$ ). Therefore,  $S_2^0 \subseteq S_1^0$ .

**11.11.** Prove [Theorem 11.5](#): Suppose  $V$  has finite dimension and  $W$  is a subspace of  $V$ . Then

(i)  $\dim W + \dim W^0 = \dim V$ , (ii)  $W^{00} = W$ .

(i) Suppose  $\dim V = n$  and  $\dim W = r \leq n$ . We want to show that  $\dim W^0 = n - r$ . We choose a basis  $\{w_1,$

$\dots; w_r\}$  of  $W$  and extend it to a basis of  $V$ , say  $\{w_1, \dots; w_r, v_1, \dots; v_{n-r}\}$ . Consider the dual basis

$$\{\phi_1, \dots, \phi_r, \sigma_1, \dots, \sigma_{n-r}\}$$

By definition of the dual basis, each of the above  $s$ 's annihilates each  $\sigma_j$ , hence,  $\sigma_1, \dots; \sigma_{n-r} \in W^0$ . We claim that  $\{\sigma_j\}$  is a basis of  $W^0$ . Now  $\{\sigma_j\}$  is part of a basis of  $V^*$ , and so it is linearly independent. We next show that  $\{\phi_j\}$  spans  $W^0$ . Let  $\sigma \in W^0$ . By [Theorem 11.2](#),

$$\begin{aligned} \sigma &= \sigma(w_1)\phi_1 + \dots + \sigma(w_r)\phi_r + \sigma(v_1)\sigma_1 + \dots + \sigma(v_{n-r})\sigma_{n-r} \\ &= 0\phi_1 + \dots + 0\phi_r + \sigma(v_1)\sigma_1 + \dots + \sigma(v_{n-r})\sigma_{n-r} \\ &= \sigma(v_1)\sigma_1 + \dots + \sigma(v_{n-r})\sigma_{n-r} \end{aligned}$$

Consequently,  $\{\sigma_1, \dots; \sigma_{n-r}\}$  spans  $W^0$  and so it is a basis of  $W^0$ . Accordingly, as required

$$\dim W^0 = n - r = \dim V - \dim W.$$

(ii) Suppose  $\dim V = n$  and  $\dim W = r$ . Then  $\dim V^* = n$  and, by (i),  $\dim W^0 = n - r$ . Thus, by (i),  $\dim W^{00} = n - (n - r) = r$ ; therefore,  $\dim W = \dim W^{00}$ . By [Problem 11.10](#),  $W \subseteq W^{00}$ . Accordingly,  $W = W^{00}$ .

**11.12.** Let  $U$  and  $W$  be subspaces of  $V$ . Prove that  $(U + W)^0 = U^0 \cap W^0$ .

Let  $\phi \in (U + W)^0$ . Then  $\phi$  annihilates  $U + W$ ; and so, in particular,  $\phi$  annihilates  $U$  and  $W$ . That is,  $\phi \in U^0$  and  $\phi \in W^0$ ; hence,  $\phi \in U^0 \cap W^0$ . Thus,  $(U + W)^0 \subseteq U^0 \cap W^0$ .

On the other hand, suppose  $\sigma \in U^0 \cap W^0$ . Then  $\sigma$  annihilates  $U$  and also  $W$ . If  $v \in U + W$ , then  $v = u +$

$w$ , where  $u \in U$  and  $w \in W$ . Hence,  $s(v) = s(u) + s(w) = 0 + 0 = 0$ . Thus,  $\sigma$  annihilates  $U + W$ ; that is,  $s \in (U + W)^0$ . Accordingly,  $U^0 + W^0 \subseteq (U + W)^0$ .

The two inclusion relations together give us the desired equality.

**Remark:** Observe that no dimension argument is employed in the proof; hence, the result holds for spaces of finite or infinite dimension.

## **Transpose of a Linear Mapping**

---

**11.13.** Let  $\phi$  be the linear functional on  $\mathbf{R}^2$  defined by  $\phi(x; y) = x - 2y$ . For each of the following linear operators  $T$  on  $\mathbf{R}^2$ , find  $(T^t(f))(x; y)$ :

(a)  $T(x, y) = (x, 0)$ , (b)  $T(x, y) = (y, x + y)$ , (c)  $T(x, y) = (2x - 3y, 5x + 2y)$

By definition,  $T^t(f) = f \circ T$ ; that is,  $(T^t(f))(v) = f(T(v))$  for every  $v$ . Hence,

(a)  $(T^t(\phi))(x, y) = \phi(T(x, y)) = \phi(x, 0) = x$

(b)  $(T^t(\phi))(x, y) = \phi(T(x, y)) = \phi(y, x + y) = y - 2(x + y) = -2x - y$

(c)  $(T^t(\phi))(x, y) = \phi(T(x, y)) = \phi(2x - 3y, 5x + 2y) = (2x - 3y) - 2(5x + 2y) = -8x - 7y$

**11.14.** Let  $T: V \rightarrow U$  be linear and let  $T^t: U^* \rightarrow V^*$  be its transpose. Show that the kernel of  $T^t$  is the annihilator of the image of  $T$ —that is,  $\text{Ker } T^t = (\text{Im } T)^0$ .

Suppose  $\phi \in \text{Ker } T^t$ ; that is,  $T^t(\phi) = 0$ . Then  $\phi(T(v)) = 0$  for every  $v \in V$ ; hence,

$$\phi(u) = \phi(T(v)) = (T^t(\phi))(v) = 0(v) = 0$$

We have that  $\phi(u) = 0$  for every  $u \in \text{Im } T$ ; hence,  $f \in (\text{Im } T)^0$ . Thus,  $\text{Ker } T^t \subseteq (\text{Im } T)^0$ .

On the other hand, suppose  $\sigma \in (\text{Im } T)^0$ ; that is,  $\sigma(\text{Im } T) = \{0\}$ . Then, for every  $v \in V$ ,

$$(T^t(\sigma))(v) = (\sigma \circ T)(v) = \sigma(T(v)) = 0 = \mathbf{0}(v)$$

We have  $(T^t(\sigma))(v) = \mathbf{0}(v)$  for every  $v \in V$ ; hence,  $T^t(\sigma) = 0$ . Thus,  $\sigma \in \text{Ker } T^t$ , and so  $(\text{Im } T)^0 \subseteq \text{Ker } T^t$ .

The two inclusion relations together give us the required equality.

**11.15.** Suppose  $V$  and  $U$  have finite dimension and  $T: V \rightarrow U$  is linear. Prove  $\text{rank}(T) = \text{rank}(T^t)$ .

Suppose  $\dim V = n$  and  $\dim U = m$ , and suppose  $\text{rank}(T) = r$ . By [Theorem 11.5](#),

$$\dim(\text{Im } T)^0 = \dim U - \dim(\text{Im } T) = m - \text{rank}(T) = m - r$$

By [Problem 11.14](#),  $\text{Ker } T^t = (\text{Im } T)^0$ . Hence, nullity  $(T^t) = m - r$ . It then follows that, as claimed,

$$\text{rank}(T^t) = \dim U^* - \text{nullity}(T^t) = m - (m - r) = r = \text{rank}(T)$$

**11.16.** Prove [Theorem 11.7](#): Let  $T: V \rightarrow U$  be linear and let  $A$  be the matrix representation of  $T$  in the bases  $\{v_j\}$  of  $V$  and  $\{u_i\}$  of  $U$ . Then the transpose matrix  $A^T$  is the matrix representation of  $T^t: U^* \rightarrow V^*$  in the bases dual to  $\{u_i\}$  and  $\{v_j\}$ .

Suppose, for  $j = 1; \dots; n$ ,

$$T(v_j) = a_{j1}u_1 + a_{j2}u_2 + \dots + a_{jm}u_m \tag{1}$$

We want to prove that, for  $i = 1; \dots; m$ ,

$$T'(\sigma_i) = a_{1i}\phi_1 + a_{2i}\phi_2 + \cdots + a_{mi}\phi_m \quad (2)$$

where  $\{\sigma_i\}$  and  $\{\phi_j\}$  are the bases dual to  $\{u_i\}$  and  $\{v_j\}$ , respectively.

Let  $v \in V$  and suppose  $v = k_1v_1 + k_2v_2 + \cdots + k_mv_m$ . Then, by (1),

$$\begin{aligned} T(v) &= k_1T(v_1) + k_2T(v_2) + \cdots + k_mT(v_m) \\ &= k_1(a_{11}u_1 + \cdots + a_{1n}u_n) + k_2(a_{21}u_1 + \cdots + a_{2n}u_n) + \cdots + k_m(a_{m1}u_1 + \cdots + a_{mn}u_n) \\ &= (k_1a_{11} + k_2a_{21} + \cdots + k_ma_{m1})u_1 + \cdots + (k_1a_{1n} + k_2a_{2n} + \cdots + k_ma_{mn})u_n \\ &= \sum_{i=1}^n (k_1a_{1i} + k_2a_{2i} + \cdots + k_ma_{mi})u_i \end{aligned}$$

Hence, for  $j = 1; \dots; n$ .

$$\begin{aligned} (T'(\sigma_j)(v)) &= \sigma_j(T(v)) = \sigma_j\left(\sum_{i=1}^n (k_1a_{1i} + k_2a_{2i} + \cdots + k_ma_{mi})u_i\right) \\ &= k_1a_{1j} + k_2a_{2j} + \cdots + k_ma_{mj} \end{aligned} \quad (3)$$

On the other hand, for  $j = 1; \dots; n$ ,

$$\begin{aligned} (a_{1j}\phi_1 + a_{2j}\phi_2 + \cdots + a_{mj}\phi_m)(v) &= (a_{1j}\phi_1 + a_{2j}\phi_2 + \cdots + a_{mj}\phi_m)(k_1v_1 + k_2v_2 + \cdots + k_mv_m) \\ &= k_1a_{1j} + k_2a_{2j} + \cdots + k_ma_{mj} \end{aligned} \quad (4)$$

Because  $v \in V$  was arbitrary, (3) and (4) imply that

$$T'(\sigma_j) = a_{1j}\phi_1 + a_{2j}\phi_2 + \cdots + a_{mj}\phi_m, \quad j = 1, \dots, n$$

which is (2). Thus, the theorem is proved.

## SUPPLEMENTARY PROBLEMS

### Dual Spaces and Dual Bases

---

**11.17.** Find (a)  $\phi + \sigma$ , (b)  $3\phi$ , (c)  $2\phi + 5\sigma$ , where  $\phi: \mathbf{R}^3 \rightarrow \mathbf{R}$  and  $\sigma: \mathbf{R}^3 \rightarrow \mathbf{R}$  are defined by

$$\phi(x, y, z) = 2x - 3y + z \quad \text{and} \quad \sigma(x, y, z) = 4x - 2y + 3z$$

**11.18.** Find the dual basis of each of the following bases of  $\mathbf{R}^3$ : (a)  $\{(1; 0; 0); (0; 1; 0); (0; 0; 1)\}$ , (b)  $\{(1; 2; 3); (1; 1; 1); (2; 4; 7)\}$ .

**11.19.** Let  $V$  be the vector space of polynomials over  $\mathbf{R}$  of degree  $\geq 2$ . Let  $\phi_1, \phi_2, \phi_3$  be the linear functionals on  $V$  defined by

$$\phi_1(f(t)) = \int_0^1 f(t) dt, \quad \phi_2(f(t)) = f'(1), \quad \phi_3(f(t)) = f(0)$$

Here  $f(t) = a + bt + ct^2 \in V$  and  $f'(t)$  denotes the derivative of  $f(t)$ . Find the basis  $\{f_1(t); f_2(t); f_3(t)\}$  of  $V$  that is dual to  $\{\phi_1, \phi_2, \phi_3\}$ .

**11.20.** Suppose  $u, v \in V$  and that  $\phi(u) = 0$  implies  $\phi(v) = 0$  for all  $\phi \in V^*$ . Show that  $v = ku$  for some scalar  $k$ .

**11.21.** Suppose  $\phi, \sigma \in V^*$  and that  $\phi(v) = 0$  implies  $\sigma(v) = 0$  for all  $v \in V$ . Show that  $\sigma = k\phi$  for some scalar  $k$ .

**11.22.** Let  $V$  be the vector space of polynomials over  $K$ . For  $a \in K$ , define  $f_a: V \rightarrow K$  by  $\phi_a(f(t)) = f(a)$ . Show that (a)  $f_a$  is linear; (b) if  $a \neq b$ , then  $\phi_a \neq \phi_b$ .

**11.23.** Let  $V$  be the vector space of polynomials of degree  $\geq 2$ . Let  $a, b, c \in K$  be distinct scalars. Let  $\phi_a, \phi_b, \phi_c$  be the linear functionals defined by  $\phi_a(f(t)) = f(a)$ ,  $\phi_b(f(t)) = f(b)$ ,  $\phi_c(f(t)) = f(c)$ . Show that  $\{\phi_a, \phi_b, \phi_c\}$  is linearly independent, and find the basis  $\{\phi_1(t); \phi_2(t); \phi_3(t)\}$  of  $V$  that is its dual.

- 11.24.** Let  $V$  be the vector space of square matrices of order  $n$ . Let  $T: V \rightarrow K$  be the trace mapping; that is,  $T(A) = a_{11} + a_{22} + \dots + a_{nn}$ , where  $A = (a_{ij})$ . Show that  $T$  is linear.
- 11.25.** Let  $W$  be a subspace of  $V$ . For any linear functional  $\phi$  on  $W$ , show that there is a linear functional  $\sigma$  on  $V$  such that  $\sigma(w) = \phi(w)$  for any  $w \in W$ ; that is,  $\phi$  is the restriction of  $\sigma$  to  $W$ .
- 11.26.** Let  $\{e_1, \dots, e_n\}$  be the usual basis of  $K^n$ . Show that the dual basis is  $\{\pi_1, \dots, \pi_n\}$  where  $\pi_i$  is the  $i$ th projection mapping; that is,  $\pi_i(a_1, \dots, a_n) = a_i$ .
- 11.27.** Let  $V$  be a vector space over  $\mathbf{R}$ . Let  $\phi_1, \phi_2 \in V^*$  and suppose  $\sigma: V \rightarrow \mathbf{R}$  defined by  $\sigma(v) = \phi_1(v)\phi_1(v)\phi_2(v)$  also belongs to  $V^*$ . Show that either  $\phi_1 = \mathbf{0}$  or  $\phi_2 = \mathbf{0}$ .

## **Annihilators**

---

- 11.28.** Let  $W$  be the subspace of  $\mathbf{R}^4$  spanned by  $(1; 2; 3; 4)$ ,  $(1; 3; 2; 6)$ ,  $(1; 4; 1; 8)$ . Find a basis of the annihilator of  $W$ .
- 11.29.** Let  $W$  be the subspace of  $\mathbf{R}^3$  spanned by  $(1; 1; 0)$  and  $(0; 1; 1)$ . Find a basis of the annihilator of  $W$ .
- 11.30.** Show that, for any subset  $S$  of  $V$ ;  $\text{span}(S)^\circ = S^{\circ\circ}$ , where  $\text{span}(S)$  is the linear span of  $S$ .
- 11.31.** Let  $U$  and  $W$  be subspaces of a vector space  $V$  of finite dimension. Prove that  $(U \cup W)^\circ = U^\circ + W^\circ$ .
- 11.32.** Suppose  $V = U \oplus W$ . Prove that  $V^\circ = U^\circ \oplus W^\circ$ .

## **Transpose of a Linear Mapping**

---

**11.33.** Let  $\phi$  be the linear functional on  $\mathbf{R}^2$  defined by  $\phi(x; y) = 3x - 2y$ . For each of the following linear mappings  $T: \mathbf{R}^3 \rightarrow \mathbf{R}^2$ , find  $(T^t(\phi))(x; y; z)$ :

(a)  $T(x, y, z) = (x + y, y + z)$ , (b)  $T(x, y, z) = (x + y + z, 2x - y)$

**11.34.** Suppose  $T_1: U \rightarrow V$  and  $T_2: V \rightarrow W$  are linear. Prove that  $(T_2 \circ T_1)^t = T_1^t \circ T_2^t$ .

**11.35.** Suppose  $T: V \rightarrow U$  is linear and  $V$  has finite dimension. Prove that  $\text{Im } T^t = (\text{Ker } T)^0$ .

**11.36.** Suppose  $T: V \rightarrow U$  is linear and  $u \in U$ . Prove that  $u \in \text{Im } T$  or there exists  $\phi \in V^*$  such that  $T^t(\phi) = 0$  and  $\phi(u) = 1$ .

**11.37.** Let  $V$  be of finite dimension. Show that the mapping  $T \mapsto T^t$  is an isomorphism from  $\text{Hom}(V; V)$  onto  $\text{Hom}(V^*; V^*)$ . (Here  $T$  is any linear operator on  $V$ .)

## Miscellaneous Problems

---

**11.38.** Let  $V$  be a vector space over  $\mathbf{R}$ . The line segment  $\overline{uv}$  joining points  $u; v \in V$  is defined by  $\overline{uv} = \{tu + (1 - t)v : 0 \leq t \leq 1\}$ . subset  $S$  of  $V$  is convex if  $u; v \in S$  implies  $\overline{uv} \subseteq S$ . Let  $\phi \in V^*$ . Let  $\phi \in V^*$ . Define

$$W^+ = \{v \in V : \phi(v) > 0\}, \quad W = \{v \in V : \phi(v) = 0\}, \quad W^- = \{v \in V : \phi(v) < 0\}$$

Prove that  $W^+$ ;  $W$ , and  $W^-$  are convex.

**11.39.** Let  $V$  be a vector space of finite dimension. A hyperplane  $H$  of  $V$  may be defined as the kernel of a nonzero linear functional  $\phi$  on  $V$ . Show that every subspace of  $V$  is the intersection of a finite number of hyperplanes.

## ANSWERS TO SUPPLEMENTARY PROBLEMS

**11.17.** (a)  $6x + 5y + 4z$ , (b)  $6x - 9y + 3z$ , (c)  $-16x + 4y - 13z$

**11.18.** (a)  $\phi_1 = x$ ;  $\phi_2 = y$ ;  $\phi_3 = z$ ; (b)  $\phi_1 = -3x - 5y - 2z$ ;  $\phi_2 = 2x + y$ ;  $\phi_3 = x + 2y + z$

**11.19.**  $f_1(t) = 3t - \frac{3}{2}t^2$ ,  $f_2(t) = -\frac{1}{2}t + \frac{3}{4}t^2$ ,  $f_3(t) = 1 - 3t + \frac{3}{2}t^2$

**11.22.** (b) Let  $f(t) = t$ . Then  $\phi_a(f(t)) = a \neq b = \phi_b(f(t))$ ; and therefore,  $\phi_a \neq \phi_b$

**11.23.**

$$\left\{ f_1(t) = \frac{t^2 - (b+c)t + bc}{(a-b)(a-c)}, f_2(t) = \frac{t^2 - (a+c)t + ac}{(b-a)(b-c)}, f_3(t) = \frac{t^2 - (a+b)t + ab}{(c-a)(c-b)} \right\}$$

**11.28.**  $\{\phi_1(x; y; z; t) = 5x - y + z; \phi_2(x, y, z, t) = 2y - t\}$

**11.29.**  $\{\phi(x; y; z) = x - y + z\}$

**11.33.** (a)  $(T^t(\phi))(x; y; z) = 3x + y - 2z$ , (b)  $(T^t(\phi))(x; y; z) = -x + 5y + 3z$

## CHAPTER 12

# *Bilinear, Quadratic, and Hermitian Forms*

## 12.1 Introduction

---

This chapter generalizes the notions of linear mappings and linear functionals. Specifically, we introduce the notion of a bilinear form. These bilinear maps also give rise to quadratic and Hermitian forms. Although quadratic forms were discussed previously, this chapter is treated independently of the previous results.

Although the field  $K$  is arbitrary, we will later specialize to the cases  $K = \mathbf{R}$  and  $K = \mathbf{C}$ . Furthermore, we may sometimes need to divide by 2. In such cases, we must assume that  $1 + 1 \neq 0$ , which is true when  $K = \mathbf{R}$  or  $K = \mathbf{C}$ .

## 12.2 Bilinear Forms

---

Let  $V$  be a vector space of finite dimension over a field  $K$ . A bilinear form on  $V$  is a mapping  $f: V \times V \rightarrow K$  such that, for all  $a, b \in K$  and all  $u_i, v_i \in V$ :

- (i)  $f(au_1 + bu_2, v) = af(u_1, v) + bf(u_2, v),$   
(ii)  $f(u; av_1 + bv_2) = af(u; v_1) + bf(u; v_2)$

We express condition (i) by saying  $f$  is linear in the first variable, and condition (ii) by saying  $f$  is *linear in the second variable*.

### EXAMPLE 12.1

(a) Let  $f$  be the dot product on  $\mathbf{R}^n$ ; that is, for  $u = (a_i)$  and  $v = (b_i)$ ,

$$f(u, v) = u \cdot v = a_1b_1 + a_2b_2 + \cdots + a_nb_n$$

Then  $f$  is a bilinear form on  $\mathbf{R}^n$ . (In fact, any inner product on a real vector space  $V$  is a bilinear form on  $V$ .)

(b) Let  $\phi$  and  $\sigma$  be arbitrarily linear functionals on  $V$ . Let  $f: V \times V \rightarrow K$  be defined by  $f(u; v) = \phi(u)\sigma(v)$ . Then  $f$  is a bilinear form, because  $\phi$  and  $\sigma$  are each linear.

(c) Let  $A = [a_{ij}]$  be any  $n \times n$  matrix over a field  $K$ . Then  $A$  may be identified with the following bilinear form  $F$  on  $K^n$ , where  $X = [x]_i$  and  $Y = [y]_j$  are column vectors of variables:

$$f(X, Y) = X^T AY = \sum_{i,j} a_{ij}x_iy_j = a_{11}x_1y_1 + a_{12}x_1y_2 + \cdots + a_{nn}x_ny_n$$

The above formal expression in the variables  $x_i, y_j$  is termed the bilinear polynomial corresponding to the matrix  $A$ . Equation (12.1) shows that, in a certain sense, every bilinear form is of this type.

## Space of Bilinear Forms

---

Let  $B(V)$  denote the set of all bilinear forms on  $V$ . A vector space structure is placed on  $B(V)$ , where for any  $f, g \in B(V)$  and any  $k \in K$ , we define  $f + g$  and  $kf$  as follows:

$$(f + g)(u, v) = f(u, v) + g(u, v) \quad \text{and} \quad (kf)(u, v) = kf(u, v)$$

The following theorem (proved in [Problem 12.4](#)) applies.

**THEOREM 12.1:** Let  $V$  be a vector space of dimension  $n$  over  $K$ . Let  $\{\phi_1, \dots, \phi_n\}$  be any basis of the dual space  $V^*$ . Then  $\{f_{ij} : i, j = 1; \dots; n\}$  is a

basis of  $B(V)$ , where  $f_{ij}$  is defined by  $f_{ij}(u; v) = \phi_i(u)\phi_j(v)$ . Thus, in particular,  $\dim B(V) = n^2$ .

## 12.3 Bilinear Forms and Matrices

---

Let  $f$  be a bilinear form on  $V$  and let  $S = \{u_1, \dots, u_n\}$  be a basis of  $V$ . Suppose  $u, v \in V$  and

$$u = a_1u_1 + \cdots + a_nu_n \quad \text{and} \quad v = b_1u_1 + \cdots + b_nu_n$$

Then

$$f(u, v) = f(a_1u_1 + \cdots + a_nu_n, b_1u_1 + \cdots + b_nu_n) = \sum_{i,j} a_ib_jf(u_i, u_j)$$

Thus,  $f$  is completely determined by the  $n^2$  values  $f(u_i, u_j)$ .

The matrix  $A = [a_{ij}]$  where  $a_{ij} = f(u_i, u_j)$  is called the matrix representation of  $f$  relative to the basis  $S$  or, simply, the “matrix of  $f$  in  $S$ .” It “represents”  $f$  in the sense that, for all  $u, v \in V$ ,

$$f(u, v) = \sum_{i,j} a_ib_jf(u_i, u_j) = [u]_S^T A [v]_S$$

[As usual,  $[u]_S$  denotes the coordinate (column) vector of  $u$  in the basis  $S$ .]

### Change of Basis, Congruent Matrices

---

We now ask, how does a matrix representing a bilinear form transform when a new basis is selected? The answer is given in the following theorem (proved in [Problem 12.5](#)).

**THEOREM 12.2:** Let  $P$  be a change-of-basis matrix from one basis  $S$  to another basis  $S^0$ . If  $A$  is the matrix representing a bilinear form  $f$  in the original basis  $S$ , then  $B = P^TAP$  is the matrix representing  $f$  in the new basis  $S^0$ .

The above theorem motivates the following definition.

**DEFINITION:** A matrix  $B$  is congruent to a matrix  $A$ , written  $B \simeq A$ , if there exists a nonsingular matrix  $P$  such that  $B = P^TAP$ .

Thus, by [Theorem 12.2](#), matrices representing the same bilinear form are congruent. We remark that congruent matrices have the same rank, because  $P$  and  $P^T$  are nonsingular; hence, the following definition is well defined.

**DEFINITION:** The rank of a bilinear form  $f$  on  $V$ , written  $\text{rank}(f)$ , is the rank of any matrix representation of  $f$ . We say  $f$  is degenerate or nondegenerate according to whether  $\text{rank}(f) < \dim V$  or  $\text{rank}(f) = \dim V$ .

## 12.4 Alternating Bilinear Forms

---

Let  $f$  be a bilinear form on  $V$ . Then  $f$  is called

- (i) alternating if  $f(v, v) = 0$  for every  $v \in V$ ;
- (ii) skew-symmetric if  $f(u, v) = -f(v, u)$  for every  $u, v \in V$ .

Now suppose (i) is true. Then (ii) is true, because, for any  $u, v \in V$ ,

$$0 = f(u + v, u + v) = f(u, u) + f(u, v) + f(v, u) + f(v, v) = f(u, v) + f(v, u)$$

On the other hand, suppose (ii) is true and also  $1 + 1 \neq 0$ . Then (i) is true, because, for every  $v \in V$ , we have  $f(v, v) = f-(v, v)$ . In other words, alternating and skew-symmetric are equivalent when  $1 + 1 \neq 0$ .

The main structure theorem of alternating bilinear forms (proved in [Problem 12.23](#)) is as follows.

**THEOREM 12.3:** Let  $f$  be an alternating bilinear form on  $V$ . Then there exists a basis of  $V$  in which  $f$  is represented by a block diagonal matrix  $M$  of the form

$$M = \text{diag} \left( \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, [0], [0], \dots [0] \right)$$

Moreover, the number of nonzero blocks is uniquely determined by  $f$  [because it is equal to  $\frac{1}{2} \text{rank}(f)$ ].

In particular, the above theorem shows that any alternating bilinear form must have even rank.

## 12.5 Symmetric Bilinear Forms, Quadratic Forms

---

This section investigates the important notions of symmetric bilinear forms and quadratic forms and their representation by means of symmetric matrices. The only restriction on the field  $K$  is that  $1 + 1 \neq 0$ . In Section 12.6, we will restrict  $K$  to be the real field  $\mathbb{R}$ , which yields important special results.

### Symmetric Bilinear Forms

---

Let  $f$  be a bilinear form on  $V$ . Then  $f$  is said to be symmetric if, for every  $u, v \in V$ ,

$$f(u, v) = f(v, u)$$

One can easily show that  $f$  is symmetric if and only if any matrix representation  $A$  of  $f$  is a symmetric matrix.

The main result for symmetric bilinear forms (proved in [Problem 12.10](#)) is as follows. (We emphasize that we are assuming that  $1 + 1 \neq 0$ .)

**THEOREM 12.4:** Let  $f$  be a symmetric bilinear form on  $V$ . Then  $V$  has a basis  $\{v_1, \dots, v_n\}$  in which  $f$  is represented by a diagonal matrix—that is, where  $f(v_i, v_j) = 0$  for  $i \neq j$ .

**THEOREM 12.4:** (Alternative Form) Let  $A$  be a symmetric matrix over  $K$ . Then  $A$  is congruent to a diagonal matrix; that is, there exists a nonsingular matrix  $P$  such that  $P^TAP$  is diagonal.

## **Diagonalization Algorithm**

---

Recall that a nonsingular matrix  $P$  is a product of elementary matrices. Accordingly, one way of obtaining the diagonal form  $D = P^TAP$  is by a sequence of elementary row operations and the same sequence of elementary column operations. This same sequence of elementary row operations on the identity matrix  $I$  will yield  $P^T$ . This algorithm is formalized below.

**ALGORITHM 12.1:** (Congruence Diagonalization of a Symmetric Matrix) The input is a symmetric matrix  $A = [a_{ij}]$  of order  $n$ .

**Step 1.** Form the  $n \times 2n$  (block) matrix  $M = [A_1, I]$ , where  $A_1 = A$  is the left half of  $\mathbf{M}$  and the identity matrix  $I$  is the right half of  $M$ .

**Step 2.** Examine the entry  $a_{11}$ . There are three cases.

*Case I:*  $a_{11} \neq 0$ . (Use  $a_{11}$  as a pivot to put 0's below  $a_{11}$  in  $M$  and to the right of  $a_{11}$  in  $A_1$ .) For  $i = 2; \dots; n$ :

(a) Apply the row operation "Replace  $R_i$  by  $-a_{i1}R_1 + a_{11}R_i$ ."

(b) Apply the corresponding column operation "Replace  $C_i$  by  $a_{i1}C_1 + a_{11}C_i$ ."

These operations reduce the matrix  $M$  to the form

$$M \sim \begin{bmatrix} a_{11} & 0 & * & * \\ 0 & A_1 & * & * \end{bmatrix} \quad (*)$$

*Case II:*  $a_{11} = 0$  but  $a_{kk} \neq 0$ , for some  $k > 1$ .

(a) Apply the row operation "Interchange  $R_1$  and  $R_k$ ."

(b) Apply the corresponding column operation "Interchange  $C_1$  and  $C_k$ ."

(These operations bring  $a_{kk}$  into the first diagonal position, which reduces the matrix to Case I.)

*Case III:* All diagonal entries  $a_{ii} = 0$  but some  $a_{ij} \neq 0$ .

(a) Apply the row operation “Replace  $R_i$  by  $R_j + R_i$ .”

(b) Apply the corresponding column operation “Replace  $C_i$  by  $C_j + C_i$ .”

(These operations bring  $2a_{ij}$  into the  $i$ th diagonal position, which reduces the matrix to **Case II.**)

Thus,  $M$  is finally reduced to the form (\*), where  $A_2$  is a symmetric matrix of order less than  $A$ .

**Step 3.** Repeat Step  $\epsilon$  with each new matrix  $A_k$  (by neglecting the first row and column of the preceding matrix) until  $A$  is diagonalized. Then  $M$  is transformed into the form  $M^0 = [D, Q]$ , where  $D$  is diagonal.

**Step 4.** Set  $P = Q^T$ . Then  $D = P^TAP$ .

**Remark 1:** We emphasize that in Step 2, the row operations will change both sides of  $M$ , but the column operations will only change the left half of  $M$ .

**Remark 2:** The condition  $1 + 1 \neq 0$  is used in Case III, where we assume that  $2a_{ij} \neq 0$  when  $a_{ij} \neq 0$ .

The justification for the above algorithm appears in [Problem 12.9](#).

**EXAMPLE 12.2** Let  $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 5 & -4 \\ -3 & -4 & 8 \end{bmatrix}$ . Apply Algorithm 9.1 to find a nonsingular matrix  $P$  such that  $D = P^TAP$  is diagonal.

First form the block matrix  $M = [A, I]$  that is, let

$$M = [A, I] = \left[ \begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 2 & 5 & -4 & 0 & 1 & 0 \\ -3 & -4 & 8 & 0 & 0 & 1 \end{array} \right]$$

Apply the row operations “Replace  $R_2$  by  $-2R_1 + R_2$ ” and “Replace  $R_3$  by  $3R_1 + R_3$ ” to  $M$ , and then apply the corresponding column operations “Replace  $C_2$  by  $-2C_1 + C_2$ ” and “Replace  $C_3$  by  $3C_1 + C_3$ ” to obtain

$$\left[ \begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & 1 & 2 & -2 & 1 & 0 \\ 0 & 2 & -1 & 3 & 0 & 1 \end{array} \right] \quad \text{and then} \quad \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & -2 & 1 & 0 \\ 0 & 2 & -1 & 3 & 0 & 1 \end{array} \right]$$

Next apply the row operation “Replace  $R_3$  by  $2R_2 + R_3$ ” and then the corresponding column operation “Replace  $C_3$  by  $2C_2 + C_3$ ” to obtain

$$\left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & -2 & 1 & 0 \\ 0 & 0 & -5 & 7 & -2 & 1 \end{array} \right] \quad \text{and then} \quad \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & -5 & 7 & -2 & 1 \end{array} \right]$$

Now  $A$  has been diagonalized. Set

$$P = \begin{bmatrix} 1 & -2 & 7 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and then} \quad D = P^{-1}AP = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -5 \end{bmatrix}$$

We emphasize that  $P$  is the transpose of the right half of the final matrix.

## Quadratic Forms

---

We begin with a definition.

**DEFINITION A:** A mapping  $q: V \rightarrow K$  is a quadratic form if  $q(v) = f(v, v)$  for some symmetric bilinear form  $f$  on  $V$ .

If  $1 + 1 \neq 0$  in  $K$ , then the bilinear form  $f$  can be obtained from the quadratic form  $q$  by the following polar form of  $f$ :

$$f(u, v) = \frac{1}{2}[q(u + v) - q(u) - q(v)]$$

Now suppose  $f$  is represented by a symmetric matrix  $A = [a_{ij}]$ , and  $1 + 1 \neq 0$ . Letting  $X = [x]_i$  denote a column vector of variables,  $q$  can be represented in the form

$$q(X) = f(X, X) = X^T A X = \sum_{i,j} a_{ij} x_i x_j = \sum_i a_{ii} x_i^2 + 2 \sum_{i < j} a_{ij} x_i x_j$$

The above formal expression in the variables  $x_i$  is also called a quadratic form. Namely, we have the following second definition.

**DEFINITION B:** A *quadratic form*  $q$  in variables  $x_1, x_2, \dots, x_n$  is a polynomial such that every term has degree two; that is,

$$q(x_1, x_2, \dots, x_n) = \sum_i c_i x_i^2 + \sum_{i < j} d_{ij} x_i x_j$$

Using  $1 + 1 \neq 0$ , the quadratic form  $q$  in Definition B determines a symmetric matrix  $A = [a_{ij}]$  where  $a_{ij} = a_{ji} = \frac{1}{2} d_{ij}$ . Thus, Definitions A and B are essentially the same. If the matrix representation  $A$  of  $q$  is diagonal, then  $q$  has the diagonal representation

$$q(X) = X^T A X = a_{11} x_1^2 + a_{22} x_2^2 + \dots + a_{nn} x_n^2$$

That is, the quadratic polynomial representing  $q$  will contain no “cross product” terms. Moreover, by [Theorem 12.4](#), every quadratic form has such a representation (when  $1 + 1 \neq 0$ ).

## 12.6 Real Symmetric Bilinear Forms, Law of Inertia

---

This section treats symmetric bilinear forms and quadratic forms on vector spaces  $V$  over the real field  $\mathbf{R}$ . The special nature of  $\mathbf{R}$  permits an independent theory. The main result (proved in [Problem 12.14](#)) is as follows.

**THEOREM 12.5:** Let  $f$  be a symmetric form on  $V$  over  $\mathbf{R}$ . Then there exists a basis of  $V$  in which  $f$  is represented by a diagonal matrix. Every other diagonal matrix representation of  $f$  has the same number  $\mathbf{p}$  of positive entries and the same number  $\mathbf{n}$  of negative entries.

The above result is sometimes called the Law of Inertia or Sylvester's Theorem. The rank and signature of the symmetric bilinear form  $f$  are denoted and defined by

$$\text{rank}(f) = \mathbf{p} + \mathbf{n} \quad \text{and} \quad \text{sig}(f) = \mathbf{p} - \mathbf{n}$$

These are uniquely defined by [Theorem 12.5](#).

A real symmetric bilinear form  $f$  is said to be

- (i) *positive definite* if  $q(v) = f(v, v) > 0$  for every  $v \neq 0$ ,
- (ii) *nonnegative semidefinite* if  $q(v) = f(v, v) \geq 0$  for every  $v$ .

**EXAMPLE 12.3** Let  $f$  be the dot product on  $\mathbf{R}^n$ . Recall that  $f$  is a symmetric bilinear form on  $\mathbf{R}^n$ . We note that  $f$  is also positive definite. That is, for any  $u = (a_j) \neq 0 \in \mathbf{R}^n$

$$f(u, u) = a_1^2 + a_2^2 + \cdots + a_n^2 > 0$$

Section 12.5 and [Chapter 13](#) tell us how to diagonalize a real quadratic form  $q$  or, equivalently, a real symmetric matrix  $A$  by means of an orthogonal transition matrix  $P$ . If  $P$  is merely nonsingular, then  $q$  can be represented in diagonal form with only 1's and -1's as nonzero coefficients. Namely, we have the following corollary.

**COROLLARY 12.6:** Any real quadratic form  $q$  has a unique representation in the form

$$q(x_1, x_2, \dots, x_n) = x_1^2 + \dots + x_p^2 - x_{p+1}^2 - \dots - x_r^2$$

where  $r = p + n$  is the rank of the form.

**COROLLARY 12.6:** (Alternative Form) Any real symmetric matrix  $A$  is congruent to the unique diagonal matrix

$$D = \text{diag}(I_p, -I_n, 0)$$

where  $r = p + n$  is the rank of  $A$ .

## 12.7 Hermitian Forms

---

Let  $V$  be a vector space of finite dimension over the complex field  $\mathbf{C}$ . A *Hermitian form on  $V$*  is a mapping  $f: V \times V \rightarrow \mathbf{C}$  such that, for all  $a, b \in \mathbf{C}$  and all  $u, v \in V$ ,

$$(i) f(au_1 + bu_2, v) = af(u_1, v) + bf(u_2, v),$$

$$(ii) f(u, v) = \overline{f(v, u)}.$$

(As usual,  $\bar{k}$  denotes the complex conjugate of  $k \in \mathbf{C}$ .)

Using (i) and (ii), we get

$$\begin{aligned} f(u, av_1 + bv_2) &= \overline{f(av_1 + bv_2, u)} = \overline{af(v_1, u) + bf(v_2, u)} \\ &= \overline{af(v_1, u)} + \overline{bf(v_2, u)} = \bar{a}\overline{f(v_1, u)} + \bar{b}\overline{f(v_2, u)} = \bar{a}f(u, v_1) + \bar{b}f(u, v_2) \end{aligned}$$

That is,

$$(iii) f(u, av_1 + bv_2) = \bar{a}f(u, v_1) + \bar{b}f(u, v_2).$$

As before, we express condition (i) by saying  $f$  is linear in the first variable. On the other hand, we express condition (iii) by saying  $f$  is “conjugate linear” in the second variable. Moreover, condition (ii) tells us that  $f(v, v) = \overline{f(v, v)}$ , and hence,  $f(v, v)$  is real for every  $v \in V$ .

The results of Sections 12.5 and 12.6 for symmetric forms have their analogues for Hermitian forms. Thus, the mapping  $q : V \rightarrow \mathbb{C}$ , defined by  $q(v) = f(v, v)$ , is called the *Hermitian quadratic form* or complex quadratic form associated with the Hermitian form  $f$ . We can obtain  $f$  from  $q$  by the polar form

$$f(u, v) = \frac{1}{4} [q(u + v) - q(u - v)] + \frac{1}{4} [q(u + iv) - q(u - iv)]$$

Now suppose  $S = \{u_1, \dots, u_n\}$  is a basis of  $V$ . The matrix  $H = [h_{ij}]$  where  $h_{ij} = f(u_i, u_j)$  is called the matrix representation of  $f$  in the basis  $S$ . By (ii),  $f(u_i, u_j) = \overline{f(u_j, u_i)}$ ; hence,  $H$  is Hermitian and, in particular, the diagonal entries of  $H$  are real. Thus, any diagonal representation of  $f$  contains only real entries.

The next theorem (to be proved in [Problem 12.47](#)) is the complex analog of [Theorem 12.5](#) on real symmetric bilinear forms.

**THEOREM 12.7:** Let  $f$  be a Hermitian form on  $V$  over  $\mathbb{C}$ . Then there exists a basis of  $V$  in which  $f$  is represented by a diagonal matrix. Every other diagonal matrix representation of  $f$  has the same number  $p$  of positive entries and the same number  $n$  of negative entries.

Again the rank and signature of the Hermitian form  $f$  are denoted and defined by

$$\text{rank}(f) = \mathbf{p} + \mathbf{n} \quad \text{and} \quad \text{sig}(f) = \mathbf{p} - \mathbf{n}$$

These are uniquely defined by [Theorem 12.7](#).

Analogously, a Hermitian form  $f$  is said to be

(i) *positive definite* if  $q(v) = f(v, v) > 0$  for every  $v \neq 0$ ,

(ii) *nonnegative semidefinite* if  $q(v) = f(v, v) \geq 0$  for every  $v$ .

**EXAMPLE 12.4** Let  $f$  be the dot product on  $\mathbb{C}^n$ ; that is, for any  $u = (z_j)$  and  $v = (w_j)$  in  $\mathbb{C}^n$ ,

$$f(u, v) = u \cdot v = z_1 \bar{w}_1 + z_2 \bar{w}_2 + \cdots + z_n \bar{w}_n$$

Then  $f$  is a Hermitian form on  $\mathbb{C}^n$ . Moreover,  $f$  is also positive definite, because, for any  $u = (z_j) \neq 0$  in  $\mathbb{C}^n$ ,

$$f(u, u) = z_1 \bar{z}_1 + z_2 \bar{z}_2 + \cdots + z_n \bar{z}_n = |z_1|^2 + |z_2|^2 + \cdots + |z_n|^2 > 0$$

## SOLVED PROBLEMS

### Bilinear Forms

**12.1.** Let  $u = (x_1, x_2, x_3)$  and  $v = (y_1, y_2, y_3)$ . Express  $f$  in matrix notation, where

$$f(u, v) = 3x_1y_1 - 2x_1y_3 + 5x_2y_1 + 7x_2y_2 - 8x_2y_3 + 4x_3y_2 - 6x_3y_3$$

Let  $A = [a_{ij}]$ , where  $a_{ij}$  is the coefficient of  $x_iy_j$ . Then

$$f(u, v) = X^T A Y = [x_1, x_2, x_3] \begin{bmatrix} 3 & 0 & -2 \\ 5 & 7 & -8 \\ 0 & 4 & -6 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

**12.2.** Let  $A$  be an  $n \times n$  matrix over  $K$ . Show that the mapping  $f$  defined by  $f(X; Y) = X^T A Y$  is a bilinear form on  $K^n$ .

For any  $a, b \in K$  and any  $X_i, Y_i \in K^n$ ,

$$f + g = (\dots, 0, a_n, \dots, a_m + b_m, \dots, a_1 + b_1, a_0 + b_0)$$

Hence,  $f$  is linear in the first variable. Also,

$$f(X, aY_1 + bY_2) = X^T A(aY_1 + bY_2) = aX^T AY_1 + bX^T AY_2 = af(X, Y_1) + bf(X, Y_2)$$

Hence,  $f$  is linear in the second variable, and so  $f$  is a bilinear form on  $K^n$ .

**12.3.** Let  $f$  be the bilinear form on  $\mathbf{R}^2$  defined by

$$f[(x_1, x_2), (y_1, y_2)] = 2x_1y_1 - 3x_1y_2 + 4x_2y_2$$

- (a) Find the matrix  $A$  of  $f$  in the basis  $\{u_1 = (1, 0); u_2 = (1, 1)\}$ .
- (b) Find the matrix  $B$  of  $f$  in the basis  $\{v_1 = (2, 1); v_2 = (1, 1)\}$ .
- (c) Find the change-of-basis matrix  $P$  from the basis  $\{u_i\}$  to the basis  $\{v_i\}$ , and verify that  $B = P^TAP$ .

(a) Set  $A = [a_{ij}]$ , where  $a_{ij} = f(u_i, u_j)$ . This yields

$$\begin{aligned} a_{11} &= f[(1, 0), (1, 0)] = 2 - 0 - 0 = 2, & a_{21} &= f[(1, 1), (1, 0)] = 2 - 0 + 0 = 2 \\ a_{12} &= f[(1, 0), (1, 1)] = 2 - 3 - 0 = -1, & a_{22} &= f[(1, 1), (1, 1)] = 2 - 3 + 4 = 3 \end{aligned}$$

Thus,  $A = \begin{bmatrix} 2 & -1 \\ 2 & 3 \end{bmatrix}$  is the matrix of  $f$  in the basis  $\{u_1, u_2\}$ .

(b) Set  $B = [b_{ij}]$ , where  $b_{ij} = f(v_i, v_j)$ . This yields

$$\begin{aligned} b_{11} &= f[(2, 1), (2, 1)] = 8 - 6 + 4 = 6, & b_{21} &= f[(1, -1), (2, 1)] = 4 - 3 - 4 = -3 \\ b_{12} &= f[(2, 1), (1, -1)] = 4 + 6 - 4 = 6, & b_{22} &= f[(1, -1), (1, -1)] = 2 + 3 + 4 = 9 \end{aligned}$$

Thus,  $B = \begin{bmatrix} 6 & 6 \\ -3 & 9 \end{bmatrix}$  is the matrix of  $f$  in the basis  $\{v_1, v_2\}$ .

- (c) Writing  $v_1$  and  $v_2$  in terms of the  $u_i$  yields  $v_1 = u_1 + u_2$  and  $v_2 = 2u_1 - u_2$ . Then

$$P = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}, \quad P^T = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix}$$

$$\text{and } P^T A P = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 6 & 6 \\ -3 & 9 \end{bmatrix} = B$$

**12.4.** Prove [Theorem 12.1](#): Let  $V$  be an  $n$ -dimensional vector space over  $K$ . Let  $\{\phi_1, \dots, \phi_n\}$  be any basis of the dual space  $V^*$ . Then  $\{f_{ij} : i, j = 1; \dots; n\}$  is a basis of  $B(V)$ , where  $f_{ij}$  is defined by  $f_{ij}(u, v) = f_i(u)f_j(v)$ . Thus,  $\dim B(V) = n^2$ .

Let  $\{u_1, \dots, u_n\}$  be the basis of  $V$  dual to  $\{\phi_i\}$ . We first show that  $\{f_{ij}\}$  spans  $B(V)$ . Let  $f \in B(V)$  and suppose  $f(u_i, u_j) = a_{ij}$ . We claim that  $f = \sum_{i,j} a_{ij} f_{ij}$ . It suffices to show that

$$f(u_s, u_t) = \left( \sum a_{ij} f_{ij} \right)(u_s, u_t) \quad \text{for } s, t = 1, \dots, n$$

We have

$$\left( \sum a_{ij} f_{ij} \right)(u_s, u_t) = \sum a_{ij} f_{ij}(u_s, u_t) = \sum a_{ij} \phi_i(u_s) \phi_j(u_t) = \sum a_{ij} \delta_{is} \delta_{jt} = a_{st} = f(u_s, u_t)$$

as required. Hence,  $\{f_{ij}\}$  spans  $B(V)$ . Next, suppose  $\sum a_{ij} f_{ij} = 0$ . Then for  $s, t = 1; \dots; n$ ,  $0 =$

$$0 = \mathbf{0}(u_s, u_t) = \left( \sum a_{ij} f_{ij} \right)(u_s, u_t) = a_{rs}$$

The last step follows as above. Thus,  $\{f_{ij}\}$  is independent, and hence is a basis of  $B(V)$ .

**12.5.** Prove [Theorem 12.2](#). Let  $P$  be the change-of-basis matrix from a basis  $S$  to a basis  $S'$ . Let  $A$  be the matrix representing a bilinear form in the basis  $S$ . Then  $B = P^T A P$  is the matrix representing  $f$  in the basis  $S'$ .

Let  $u, v \in V$ . Because  $P$  is the change-of-basis matrix from  $S$  to  $S^0$ , we have  $P[u]_{S^0} = [u]_S$  and also  $P[v]_{S^0} = [v]_S$ , hence,  $[u]_S^T = [u]_{S^0}^T P^T$ . Thus,

$$f(u, v) = [u]_S^T A [v]_S = [u]_{S^0}^T P^T A P [v]_{S^0}$$

Because  $u$  and  $v$  are arbitrary elements of  $V$ ,  $P^T A P$  is the matrix of  $f$  in the basis  $S^0$ .

## **Symmetric Bilinear Forms, Quadratic Forms**

**12.6.** Find the symmetric matrix that corresponds to each of the following quadratic forms:

(a)  $q(x; y; z) = 3x^2 + 4xy - y^2 + 8xz - 6yz + z^2$ ,

(b)  $q^0(x; y; z) = 3x^2 + xz + 2yz$ , (c)  $q^{00}(x; y; z) = 2x^2 - 5y^2 - 7z^2$

The symmetric matrix  $A = [a_{ij}]$  that represents  $q(x_1, \dots, x_n)$  has the diagonal entry  $a_{ii}$  equal to the coefficient of the square term  $x_i^2$  and the nondiagonal entries  $a_{ij}$  and  $a_{ji}$  each equal to half of the coefficient of the cross-product term  $x_i x_j$ . Thus,

$$(a) \quad A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & -1 & -3 \\ 4 & -3 & 1 \end{bmatrix}, (b) \quad A' = \begin{bmatrix} 3 & 0 & \frac{1}{2} \\ 0 & 0 & -1 \\ \frac{1}{2} & -1 & 0 \end{bmatrix}, (c) \quad A'' = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & -7 \end{bmatrix}$$

The third matrix  $A''$  is diagonal, because the quadratic form  $q''$  is diagonal; that is,  $q''$  has no cross-product terms.

**12.7.** Find the quadratic form  $q(X)$  that corresponds to each of the following symmetric matrices:

$$(a) \quad A = \begin{bmatrix} 5 & -3 \\ -3 & 8 \end{bmatrix}, (b) \quad B = \begin{bmatrix} 4 & -5 & 7 \\ -5 & -6 & 8 \\ 7 & 8 & -9 \end{bmatrix}, (c) \quad C = \begin{bmatrix} 2 & 4 & -1 & 5 \\ 4 & -7 & -6 & 8 \\ -1 & -6 & 3 & 9 \\ 5 & 8 & 9 & 1 \end{bmatrix}$$

The quadratic form  $q(X)$  that corresponds to a symmetric matrix  $M$  is defined by  $q(X) = X^T M X$ , where  $X = [x]_i$  is the column vector of unknowns.

(a) Compute as follows:

$$\begin{aligned} q(x, y) &= X^T A X = [x, y] \begin{bmatrix} 5 & -3 \\ -3 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = [5x - 3y, -3x + 8y] \begin{bmatrix} x \\ y \end{bmatrix} \\ &= 5x^2 - 3xy - 3xy + 8y^2 = 5x^2 - 6xy + 8y^2 \end{aligned}$$

As expected, the coefficient 5 of the square term  $x^2$  and the coefficient 8 of the square term  $y^2$  are the diagonal elements of  $A$ , and the coefficient 6 of the cross-product term  $xy$  is the sum of the nondiagonal elements 3 and 3 of  $A$  (or twice the nondiagonal element 3, because  $A$  is symmetric).

(b) Because  $B$  is a three-square matrix, there are three unknowns, say  $x$ ;  $y$ ;  $z$  or  $x_1, x_2, x_3$ . Then

$$q(x, y, z) = 4x^2 - 10xy - 6y^2 + 14xz + 16yz - 9z^2$$

or

$$q(x_1, x_2, x_3) = 4x_1^2 - 10x_1x_2 - 6x_2^2 + 14x_1x_3 + 16x_2x_3 - 9x_3^2$$

Here we use the fact that the coefficients of the square terms  $x_1^2, x_2^2, x_3^2$  (or  $x^2, y^2, z^2$ ) are the respective diagonal elements 4; -6; -9 of  $B$ , and the coefficient of the cross-product term  $x_i x_j$  is the sum

of the nondiagonal elements  $b_{ij}$  and  $b_{ji}$  (or twice  $b_{ij}$ , because  $b_{ij} = b_{ji}$ ).

(c) Because  $C$  is a four-square matrix, there are four unknowns. Hence,

$$q(x_1, x_2, x_3, x_4) = 2x_1^2 - 7x_2^2 + 3x_3^2 + x_4^2 + 8x_1x_2 - 2x_1x_3 \\ + 10x_1x_4 - 12x_2x_3 + 16x_2x_4 + 18x_3x_4$$

**12.8.** Let  $A = \begin{bmatrix} 1 & -3 & 2 \\ -3 & 7 & -5 \\ 2 & -5 & 8 \end{bmatrix}$  Apply Algorithm 12.1 to find a nonsingular matrix  $P$  such that  $D = P^TAP$  is diagonal, and find  $\text{sig}(A)$ , the signature of  $A$ .

First form the block matrix  $M = [A, I]$  :

$$M = [A, I] = \left[ \begin{array}{ccc|ccc} 1 & -3 & 2 & 1 & 0 & 0 \\ -3 & 7 & -5 & 0 & 1 & 0 \\ 2 & -5 & 8 & 0 & 0 & 1 \end{array} \right]$$

Using  $a_{11} = 1$  as a pivot, apply the row operations “Replace  $R_2$  by  $3R_1 + R_2$ ” and “Replace  $R_3$  by  $2R_1 + R_3$ ” to  $M$  and then apply the corresponding column operations “Replace  $C_2$  by  $3C_1 + C_2$ ” and “Replace  $C_3$  by  $2C_1 + C_3$ ” to  $A$  to obtain

$$\left[ \begin{array}{ccc|ccc} 1 & -3 & 2 & 1 & 0 & 0 \\ 0 & -2 & 1 & 3 & 1 & 0 \\ 0 & 1 & 4 & -2 & 0 & 1 \end{array} \right] \quad \text{and then} \quad \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -2 & 1 & 3 & 1 & 0 \\ 0 & 1 & 4 & -2 & 0 & 1 \end{array} \right].$$

Next apply the row operation “Replace  $R_3$  by  $R_2 + 2R_3$ ” and then the corresponding column operation “Replace  $C_3$  by  $C_2 + 2C_3$ ” to obtain

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -2 & 1 & 3 & 1 & 0 \\ 0 & 0 & 9 & -1 & 1 & 2 \end{bmatrix} \quad \text{and then} \quad \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -2 & 0 & 3 & 1 & 0 \\ 0 & 0 & 18 & -1 & 1 & 2 \end{bmatrix}$$

Now  $A$  has been diagonalized and the transpose of  $P$  is in the right half of  $M$ . Thus, set

$$P = \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} \quad \text{and then} \quad D = P^T A P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 18 \end{bmatrix}$$

Note  $D$  has  $\mathbf{p} = 2$  positive and  $\mathbf{n} = 1$  negative diagonal elements. Thus, the signature of  $A$  is  $\text{sig}(A) = \mathbf{p} - \mathbf{n} = 2 - 1 = 1$ .

**12.9.** Justify Algorithm 12.1, which diagonalizes (under congruence) a symmetric matrix  $A$ .

Consider the block matrix  $M = [A, I]$ . The algorithm applies a sequence of elementary row operations and the corresponding column operations to the left side of  $M$ , which is the matrix  $A$ . This is equivalent to premultiplying  $A$  by a sequence of elementary matrices, say,  $E_1, E_2, \dots, E_r$ , and postmultiplying  $A$  by the transposes of the  $E_i$ . Thus, when the algorithm ends, the diagonal matrix  $D$  on the left side of  $M$  is equal to

$$D = E_r \cdots E_2 E_1 A E_1^T E_2^T \cdots E_r^T = Q A Q^T, \quad \text{where} \quad Q = E_r \cdots E_2 E_1$$

On the other hand, the algorithm only applies the elementary row operations to the identity matrix  $I$  on the right side of  $M$ . Thus, when the algorithm ends, the matrix on the right side of  $M$  is equal to

$$E_r \cdots E_2 E_1 I = E_r \cdots E_2 E_1 = Q$$

Setting  $P = Q^T$ , we get  $D = P^T A P$ , which is a diagonalization of  $A$  under congruence.

**12.10.** Prove [Theorem 12.4](#): Let  $f$  be a symmetric bilinear form on  $V$  over  $K$  (where  $1 + 1 \neq 0$ ). Then  $V$  has a basis in which  $f$  is represented by a diagonal matrix.

Algorithm 12.1 shows that every symmetric matrix over  $K$  is congruent to a diagonal matrix. This is equivalent to the statement that  $f$  has a diagonal representation.

**12.11.** Let  $q$  be the quadratic form associated with the symmetric bilinear form  $f$ . Verify the polar identity  $f(u, v) = \frac{1}{2} [q(u + v) - q(u) - q(v)]$ . (Assume that  $1 + 1 \neq 0$ .)

We have

$$\begin{aligned} q(u + v) - q(u) - q(v) &= f(u + v, u + v) - f(u, u) - f(v, v) \\ &= f(u, u) + f(u, v) + f(v, u) + f(v, v) - f(u, u) - f(v, v) = 2f(u, v) \end{aligned}$$

If  $1 + 1 \neq 0$ , we can divide by 2 to obtain the required identity.

**12.12.** Consider the quadratic form  $q(x, y) = 3x^2 + 2xy + y^2$  and the linear substitution

$$x = s - 3t, \quad y = 2s + t$$

- Rewrite  $q(x, y)$  in matrix notation, and find the matrix  $A$  representing  $q(x, y)$ .
- Rewrite the linear substitution using matrix notation, and find the matrix  $P$  corresponding to the substitution.
- Find  $q(s, t)$  using direct substitution.
- Find  $q(s, t)$  using matrix notation.

(a) Here  $q(x, y) = [x, y] \begin{bmatrix} 3 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$ . Thus,  $A = \begin{bmatrix} 3 & 1 \\ 1 & -1 \end{bmatrix}$

and  $q(x) = X^T A X$ , where  $X = [x, y]^T$ .

(b) Here  $xy \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & -3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} s \\ t \end{bmatrix}$ . Thus,  $P = \begin{bmatrix} 1 & -3 \\ 2 & 1 \end{bmatrix}$  and  
 $X = \begin{bmatrix} x \\ y \end{bmatrix}, Y = \begin{bmatrix} s \\ t \end{bmatrix}$

(c) Substitute for  $x$  and  $y$  in  $q$  to obtain

$$\begin{aligned} q(s, t) &= 3(s - 3t)^2 + 2(s - 3t)(2s + t) - (2s + t)^2 \\ &= 3(s^2 - 6st + 9t^2) + 2(2s^2 - 5st - 3t^2) - (4s^2 + 4st + t^2) = 3s^2 - 32st + 20t^2 \end{aligned}$$

(d) Here  $q(X) = X^T A X$  and  $X = PY$ . Thus,  $X^T = Y^T P^T$ .  
 Therefore,

$$\begin{aligned} q(s, t) &= q(Y) = Y^T P^T A P Y = [s, t] \begin{bmatrix} 1 & 2 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} s \\ t \end{bmatrix} \\ &= [s, t] \begin{bmatrix} 3 & -16 \\ -16 & 20 \end{bmatrix} \begin{bmatrix} s \\ t \end{bmatrix} = 3s^2 - 32st + 20t^2 \end{aligned}$$

[As expected, the results in parts (c) and (d) are equal.]

**12.13.** Consider any diagonal matrix  $A = \text{diag}(a_1, \dots; a_n)$  over  $K$ . Show that for any nonzero scalars  $k_1, \dots; k_n \in K$ ,  $A$  is congruent to a diagonal matrix  $D$  with diagonal entries  $a_1 k_1^2; \dots; a_n k_n^2$ . Furthermore, show that

(a) If  $K = \mathbf{C}$ , then we can choose  $D$  so that its diagonal entries are only 1's and 0's.

(b) If  $K = \mathbf{R}$ , then we can choose  $D$  so that its diagonal entries are only 1's, -1's, and 0's.

Let  $P = \text{diag}(k_1, \dots; k_n)$ . Then, as required,

$$D = P^T A P = \text{diag}(k_i) \text{diag}(a_i) \text{diag}(k_i) = \text{diag}(a_1 k_1^2, \dots, a_n k_n^2)$$

$$(a) \text{ Let } P = \text{diag}(b_i), \text{ where } b_i = \begin{cases} 1/\sqrt{a_i} & \text{if } a_i \neq 0 \\ 1 & \text{if } a_i = 0 \end{cases}$$

Then  $P^T A P$  has the required form.

$$(b) \text{ Let } P = \text{diag}(b_i), \text{ where } b_i = \begin{cases} 1/\sqrt{|a_i|} & \text{if } a_i \neq 0 \\ 1 & \text{if } a_i = 0 \end{cases}$$

Then  $P^T A P$  has the required form.

**Remark:** We emphasize that (b) is no longer true if “congruence” is replaced by “Hermitian congruence.”

**12.14.** Prove [Theorem 12.5](#): Let  $f$  be a symmetric bilinear form on  $V$  over  $R$ . Then there exists a basis of  $V$  in which  $f$  is represented by a diagonal matrix. Every other diagonal matrix representation of  $f$  has the same number  $p$  of positive entries and the same number  $n$  of negative entries.

By [Theorem 12.4](#), there is a basis  $\{u_1, \dots, u_n\}$  of  $V$  in which  $f$  is represented by a diagonal matrix with, say,  $p$  positive and  $n$  negative entries. Now suppose  $\{w_1, \dots, w_n\}$  is another basis of  $V$ , in which  $f$  is represented by a diagonal matrix with  $p'$  positive and  $n'$  negative entries. We can assume without loss of generality that the positive entries in each matrix appear first. Because  $\text{rank}(f) = \mathbf{p} + \mathbf{n} = \mathbf{p}' + \mathbf{n}'$ , it suffices to prove that  $\mathbf{p} = \mathbf{p}'$ .

Let  $U$  be the linear span of  $u_1, \dots, u_p$  and let  $W$  be the linear span of  $w_{p'+1}, \dots, w_n$ . Then  $f(v, v) > 0$  for every nonzero  $v \in U$ , and  $f(v, v) < 0$  for every nonzero  $v \in W$ . Hence,  $U \cap W = \{0\}$ . Note that  $\dim U = p$  and  $\dim W = n - p'$ . Thus,

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W) = \mathbf{p} + (n - \mathbf{p}') - 0 = \mathbf{p} - \mathbf{p}' + n$$

But  $\dim(U + W) \geq \dim V = n$ ; hence,  $p - p^0 + n - n$  or  $p \geq p^0$ . Similarly,  $p^0 \leq p$  and therefore  $\mathbf{p} = \mathbf{p}^0$ , as required.

**Remark:** The above theorem and proof depend only on the concept of positivity. Thus, the theorem is true for any subfield  $K$  of the real field  $\mathbf{R}$  such as the rational field  $\mathbf{Q}$ .

## **Positive Definite Real Quadratic Forms**

---

**12.15.** Prove that the following definitions of a positive definite quadratic form  $q$  are equivalent:

- (a) The diagonal entries are all positive in any diagonal representation of  $q$ .
- (b)  $q(Y) > 0$ , for any nonzero vector  $Y$  in  $R^n$ .

Suppose  $q(Y) = a_1y_1^2 + a_2y_2^2 + \cdots + a_ny_n^2$ . If all the coefficients are positive, then clearly  $q(Y) > 0$  whenever  $Y \neq 0$ . Thus, (a) implies (b). Conversely, suppose (a) is not true; that is, suppose some diagonal entry  $a_k \leq 0$ . Let  $e_k = (0; \dots; 1; \dots; 0)$  be the vector whose entries are all 0 except 1 in the  $k$ th position. Then  $q(e_k) = a_k$  is not positive, and so (b) is not true. That is, (b) implies (a). Accordingly, (a) and (b) are equivalent.

**12.16.** Determine whether each of the following quadratic forms  $q$  is positive definite:

- (a)  $q(x; y; z) = x^2 + 2y^2 + 4xz + 4yz + 7z^2$
- (b)  $q(x; y; z) = x^2 + y^2 + 2xz + 4yz + 3z^2$

Diagonalize (under congruence) the symmetric matrix  $A$  corresponding to  $q$ .

- (a) Apply the operations "Replace  $R_3$  by  $2R_1 + R_3$ " and "Replace  $C_3$  by  $2C_1 + C_3$ ," and then "Replace

$R_3$  by  $R_2 + R_3$ ” and “Replace  $C_3$  by  $C_2 + C_3$ .” These yield

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 2 & -2 \\ -2 & -2 & 7 \end{bmatrix} \simeq \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & -2 \\ 0 & -2 & 3 \end{bmatrix} \simeq \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The diagonal representation of  $q$  only contains positive entries, 1, 2, 1, on the diagonal. Thus,  $q$  is positive definite.

(b) We have

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix} \simeq \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 2 \end{bmatrix} \simeq \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

There is a negative entry -2 on the diagonal representation of  $q$ . Thus,  $q$  is not positive definite.

**12.17.** Show that  $q(x, y) = ax^2 + bxy + cy^2$  is positive definite if and only if  $a > 0$  and the discriminant  $D = b^2 - 4ac < 0$ .

Suppose  $v = (x, y) \neq 0$ . Then either  $x \neq 0$  or  $y \neq 0$ ; say,  $y \neq 0$ . Let  $t = x/y$ . Then

$$q(v) = y^2[a(x/y)^2 + b(x/y) + c] = y^2(at^2 + bt + c)$$

However, the following are equivalent:

- (i)  $s = at^2 + bt + c$  is positive for every value of  $t$ .
- (ii)  $s = at^2 + bt + c$  lies above the  $t$ -axis.
- (iii)  $a > 0$  and  $D = b^2 - 4ac < 0$ .

Thus,  $q$  is positive definite if and only if  $a > 0$  and  $D < 0$ . [Remark:  $D < 0$  is the same as  $\det(A) > 0$ , where  $A$  is the symmetric matrix corresponding to  $q$ .]

**12.18.** Determine whether or not each of the following quadratic forms  $q$  is positive definite:

(a)  $q(x, y) = x^2 - 4xy + 7y^2$ , (b)  $q(x, y) = x^2 + 8xy + 5y^2$ ,

(c)  $q(x, y) = 3x^2 + 2xy + y^2$  Compute the discriminant  $D = b^2 - 4ac$ , and then use [Problem 12.17](#).

(a)  $D = 16 - 28 = -12$ . Because  $a = 1 > 0$  and  $D > 0$ ;  $q$  is positive definite.

(b)  $D = 64 - 20 = 44$ . Because  $D > 0$ ;  $q$  is not positive definite.

(c)  $D = 4 - 12 = -8$ . Because  $a = 3 > 0$  and  $D > 0$ ;  $q$  is positive definite.

## Hermitian Forms

---

**12.19.** Determine whether the following matrices are Hermitian:

(a)  $\begin{bmatrix} 2 & 2+3i & 4-5i \\ 2-3i & 5 & 6+2i \\ 4+5i & 6-2i & -7 \end{bmatrix}$ , (b)  $\begin{bmatrix} 3 & 2-i & 4+i \\ 2-i & 6 & i \\ 4+i & i & 7 \end{bmatrix}$ , (c)  $\begin{bmatrix} 4 & -3 & 5 \\ -3 & 2 & 1 \\ 5 & 1 & -6 \end{bmatrix}$

A complex matrix  $A = [a_{ij}]$  is Hermitian if  $A^* = A$ —that is, if  $a_{ij} = \bar{a}_{ji}$

(a) Yes, because it is equal to its conjugate transpose.

(b) No, even though it is symmetric.

(c) Yes. In fact, a real matrix is Hermitian if and only if it is symmetric.

**12.20.** Let  $A$  be a Hermitian matrix. Show that  $f$  is a Hermitian form on  $C^n$  where  $f$  is defined by  $f(X, Y) = X^T A \bar{Y}$ .

For all  $a, b \in C$  and all  $X_1, X_2, Y \in C^n$ ,

$$\begin{aligned} f(aX_1 + bX_2, Y) &= (aX_1 + bX_2)^T A \bar{Y} = (aX_1^T + bX_2^T) A \bar{Y} \\ &= aX_1^T A \bar{Y} + bX_2^T A \bar{Y} = af(X_1, Y) + bf(X_2, Y) \end{aligned}$$

Hence,  $f$  is linear in the first variable. Also,

$$\overline{f(X, Y)} = \overline{X^T A \bar{Y}} = \overline{(X^T A \bar{Y})^T} = \overline{\bar{Y}^T A^T X} = Y^T A^* \bar{X} = Y^T A \bar{X} = f(Y, X)$$

Hence,  $f$  is a Hermitian form on  $C^n$ .

**Remark:** We use the fact that  $X^T A \bar{Y}$  is a scalar and so it is equal to its transpose.

**12.21.** Let  $f$  be a Hermitian form on  $V$ . Let  $H$  be the matrix of  $f$  in a basis  $S = \{u_i\}$  of  $V$ . Prove the following:

- (a) (a)  $f(u, v) = [u]_S^T H [\bar{v}]_S$  for all  $u, v \in V$ .
- (b) If  $P$  is the change-of-basis matrix from  $S$  to a new basis  $S^0$  of  $V$ , then  $a_{ij} = \bar{a}_{ji}$  (or  $B = Q^* H Q$ , where  $Q = P$ ) is the matrix of  $f$  in the new basis  $S^0$ .

Note that (b) is the complex analog of [Theorem 12.2](#).

- (a) Let  $u, v \in V$  and suppose  $u = a_1 u_1 + \dots + a_n u_n$  and  $v = b_1 u_1 + \dots + b_n u_n$ . Then, as required,

$$\begin{aligned} f(u, v) &= f(a_1 u_1 + \dots + a_n u_n, b_1 u_1 + \dots + b_n u_n) \\ &= \sum_{i,j} a_i \bar{b}_j f(u_i, v_j) = [a_1, \dots, a_n] H [\bar{b}_1, \dots, \bar{b}_n]^T = [u]_S^T H [\bar{v}]_S \end{aligned}$$

- (b) Because  $P$  is the change-of-basis matrix from  $S$  to  $S^0$ , we have  $P[u]_{S^0} = [u]_S$  and  $P[v]_{S^0} = [v]_S$ , hence,  $[u]_S^T = [u]_{S^0}^T P^T$  and  $[\bar{v}]_S = \bar{P} [\bar{v}]_{S^0}$ : Thus, by (a),

$$f(u, v) = [u]_S^T H [\bar{v}]_S = [u]_{S^0}^T P^T H \bar{P} [\bar{v}]_{S^0}$$

But  $u$  and  $v$  are arbitrary elements of  $V$ ; hence,  $P^T H \bar{P}$  is the matrix of  $f$  in the basis  $S^0$ :

**12.22.** Let  $H = \begin{bmatrix} 1 & 1+i & 2i \\ 1-i & 4 & 2-3i \\ -2i & 2+3i & 7 \end{bmatrix}$ , a Hermitian matrix.

Find a nonsingular matrix  $P$  such that  $D = P^T H \bar{P}$  is diagonal. Also, find the signature of  $H$ .

Use the modified Algorithm 12.1 that applies the same row operations but the corresponding conjugate column operations. Thus, first form the block matrix  $M = [H, I]$ :

$$M = \begin{bmatrix} 1 & 1+i & 2i & 1 & 0 & 0 \\ 1-i & 4 & 2-3i & 0 & 1 & 0 \\ -2i & 2+3i & 7 & 0 & 0 & 1 \end{bmatrix}$$

Apply the row operations “Replace  $R_2$  by  $(-1 + i)R_1 + R_2$ ” and “Replace  $R_3$  by  $2iR_1 + R_3$ ” and then the corresponding conjugate column operations “Replace  $C_2$  by  $(-1 - i)C_1 + C_2$ ” and “Replace  $C_3$  by  $2iC_1 + C_3$ ” to obtain

$$\begin{bmatrix} 1 & 1+i & 2i & 1 & 0 & 0 \\ 0 & 2 & -5i & -1+i & 1 & 0 \\ 0 & 5i & 3 & 2i & 0 & 1 \end{bmatrix} \quad \text{and then} \quad \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & -5i & -1+i & 1 & 0 \\ 0 & 5i & 3 & 2i & 0 & 1 \end{bmatrix}$$

Next apply the row operation “Replace  $R_3$  by  $5iR_2 + 2R_3$ ” and the corresponding conjugate column operation “Replace  $C_3$  by  $5iC_2 + 2C_3$ ” to obtain

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & -5i & -1+i & 1 & 0 \\ 0 & 0 & -19 & 5+9i & -5i & 2 \end{bmatrix} \quad \text{and then} \quad \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & -1+i & 1 & 0 \\ 0 & 0 & -38 & 5+9i & -5i & 2 \end{bmatrix}$$

Now  $H$  has been diagonalized, and the transpose of the right half of  $M$  is  $P$ . Thus, set

$$P = \begin{bmatrix} 1 & -1+i & 5+9i \\ 0 & 1 & -5i \\ 0 & 0 & 2 \end{bmatrix}, \quad \text{and then} \quad D = P^T H \bar{P} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -38 \end{bmatrix}.$$

Note  $D$  has  $\mathbf{p} = 2$  positive elements and  $\mathbf{n} = 1$  negative elements. Thus, the signature of  $H$  is  $\text{sig}(H) = 2 - 1 = 1$ .

## Miscellaneous Problems

---

**12.23.** Prove [Theorem 12.3](#): Let  $f$  be an alternating form on  $V$ . Then there exists a basis of  $V$  in which  $f$  is represented by a block diagonal matrix  $M$  with blocks of the form  $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$  or  $0$ . The number of nonzero blocks is uniquely determined by  $f$  [because it is equal to  $\frac{1}{2} \text{rank}(f)$ ].

If  $f = 0$ , then the theorem is obviously true. Also, if  $\dim V = 1$ , then  $f(k_1 u; k_2 u) = k_1 k_2 f(u, u) = 0$  and so  $f = 0$ . Accordingly, we can assume that  $\dim V > 1$  and  $f \neq 0$ .

Because  $f \neq 0$ , there exist (nonzero)  $u_1, u_2 \in V$  such that  $f(u_1, u_2) \neq 0$ . In fact, multiplying  $u_1$  by an appropriate factor, we can assume that  $f(u_1; u_2) = 1$  and so  $f(u_2, u_1) = 1$ . Now  $u_1$  and  $u_2$  are linearly independent; because if, say,  $u_2 = k u_1$ , then  $f(u_1, u_2) = f(u_1, k u_1) = k f(u_1, u_1) = 0$ . Let  $U = \text{span}(u_1, u_2)$ ; then,

(i) The matrix representation of the restriction of  $f$  to  $U$  in the basis  $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ ,

(ii) If  $u \in U$ , say  $u = au_1 + bu_2$ , then

$$f(u, u_1) = f(au_1 + bu_2, u_1) = -b \quad \text{and} \quad f(u, u_2) = f(au_1 + bu_2, u_2) = a$$

Let  $W$  consists of those vectors  $w \in V$  such that  $f(w, u_1) = 0$  and  $f(w, u_2) = 0$ : Equivalently,

$$W = \{w \in V : f(w, u) = 0 \text{ for every } u \in U\}$$

We claim that  $V = U \oplus W$ . It is clear that  $U \cap W = \{0\}$ , and so it remains to show that  $V = U + W$ . Let  $v \in V$ . Set

$$u = f(v, u_2)u_1 - f(v, u_1)u_2 \quad \text{and} \quad w = v - u \tag{1}$$

Because  $u$  is a linear combination of  $u_1$  and  $u_2$ ,  $u \in U$ .

We show next that  $w \in W$ . By (1) and (ii),  $f(u, u_1) = f(v, u_1)$ ; hence,

$$f(w, u_1) = f(v - u, u_1) = f(v, u_1) - f(u, u_1) = 0$$

Similarly,  $f(u, u_2) = f(v, u_2)$

$$f(w, u_2) = f(v - u, u_2) = f(v, u_2) - f(u, u_2) = 0$$

Then  $w \in W$  and so, by (1),  $v = u + w$ , where  $u \in U$ . This shows that  $V = U + W$ ; therefore,  $V = U \oplus W$ .

Now the restriction of  $f$  to  $W$  is an alternating bilinear form on  $W$ . By induction, there exists a basis  $u_3, \dots, u_n$  of  $W$  in which the matrix representing  $f$

restricted to  $W$  has the desired form. Accordingly,  $u_1, u_2, u_3, \dots; u_n$  is a basis of  $V$  in which the matrix representing  $f$  has the desired form.

## SUPPLEMENTARY PROBLEMS

### Bilinear Forms

**12.24.** Let  $u = (x_1, x_2)$  and  $v = (y_1, y_2)$ . Determine which of the following are bilinear forms on  $\mathbb{R}^2$ :

- (a)  $f(u, v) = 2x_1y_2 - 3x_2y_1$ ,      (c)  $f(u, v) = 3x_2y_2$ ,      (e)  $f(u, v) = 1$ ,  
 (b)  $f(u, v) = x_1 + y_2$ ,      (d)  $f(u, v) = x_1x_2 + y_1y_2$ ,      (f)  $f(u, v) = 0$

**12.25.** Let  $f$  be the bilinear form on  $\mathbb{R}^2$  defined by

$$f[(x_1, x_2), (y_1, y_2)] = 3x_1y_1 - 2x_1y_2 + 4x_2y_1 - x_2y_2$$

- (a) Find the matrix  $A$  of  $f$  in the basis  $\{u_1 = (1; 1); u_2 = (1; 2)\}$ .  
 (b) Find the matrix  $B$  of  $f$  in the basis  $\{v_1 = (1; 1); v_2 = (3; 1)\}$ .  
 (c) Find the change-of-basis matrix  $P$  from  $\{u_i\}$  to  $\{v_i\}$ , and verify that  $B = P^TAP$ .

**12.26.** Let  $V$  be the vector space of two-square matrices over  $\mathbb{R}$ . Let  $M = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ , and let  $f(A, B) = \text{tr}(A^TMB)$ , where  $A, B \in V$  and “tr” denotes trace. (a) Show that  $f$  is a bilinear form on  $V$ . (b) Find the matrix of  $f$  in the basis

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

**12.27.** Let  $B(V)$  be the set of bilinear forms on  $V$  over  $K$ . Prove the following:

- (a) If  $f, g \in B(V)$ , then  $f + g, kf \in B(V)$  for any  $k \in K$ .
- (b) If  $\phi$  and  $\sigma$  are linear functions on  $V$ , then  $f(u, v) = \phi(u)\sigma(v)$  belongs to  $B(V)$ .

**12.28.** Let  $[f]$  denote the matrix representation of a bilinear form  $f$  on  $V$  relative to a basis  $\{u_i\}$ . Show that the mapping  $f \mapsto [f]$  is an isomorphism of  $B(V)$  onto the vector space  $V$  of  $n$ -square matrices.

**12.29.** Let  $f$  be a bilinear form on  $V$ . For any subset  $S$  of  $V$ , let

$$S^\perp = \{v \in V : f(u, v) = 0 \text{ for every } u \in S\} \text{ and } S^T = \{v \in V : f(v, u) = 0 \text{ for every } u \in S\}$$

Show that: (a)  $S^\perp$  and  $S^T$  are subspaces of  $V$ ; (b)  $S_1 \subseteq S_2$  implies  $S_2^\perp \subseteq S_1^\perp$  (c)  $S_2^T \subseteq S_1^T$ ;

**12.30.** Suppose  $f$  is a bilinear form on  $V$ . Prove that:  $\text{rank}(f) = \dim V - \dim V^\perp = \dim V - \dim V^T$ , and hence,  $\dim V^\perp = \dim V^T$ .

**12.31.** Let  $f$  be a bilinear form on  $V$ . For each  $u \in V$ , let  $U: V \rightarrow K$  and  $w: V \rightarrow K$  be defined by  $U(x) = f(x, u)$  and  $w(x) = f(u, x)$ . Prove the following:

- (a)  $U$  and  $w$  are each linear; i.e.,  $U, w \in V^*$ ,
- (b)  $u \mapsto U$  and  $u \mapsto w$  are each linear mappings from  $V$  into  $V^*$ ,
- (c)  $\text{rank}(f) = \text{rank}(u \mapsto U) = \text{rank}(u \mapsto w)$ .

**12.32.** Show that congruence of matrices (denoted by  $\sim$ ) is an equivalence relation; that is,

- (i)  $A \sim A$ ; (ii) If  $A \sim B$ , then  $B \sim A$ ; (iii) If  $A \sim B$  and  $B \sim C$ , then  $A \sim C$ .

## Symmetric Bilinear Forms, Quadratic Forms

**12.33.** Find the symmetric matrix  $A$  belonging to each of the following quadratic forms:

- (a)  $q(x, y, z) = 2x^2 - 8xy + y^2 - 16xz + 14yz + 5z^2$ , (c)  $q(x, y, z) = xy + y^2 + 4xz + z^2$   
 (b)  $q(x, y, z) = x^2 - xz + y^2$ , (d)  $q(x, y, z) = xy + yz$

**12.34.** For each of the following symmetric matrices  $A$ , find a nonsingular matrix  $P$  such that  $D = P^T A P$  is diagonal:

(a)  $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 6 \\ 2 & 6 & 7 \end{bmatrix}$ , (b)  $A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 5 & 3 \\ 1 & 3 & -2 \end{bmatrix}$ , (c)  $A = \begin{bmatrix} 1 & -1 & 0 & 2 \\ -1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 2 & -1 \end{bmatrix}$

**12.35.** Let  $q(x, y) = 2x^2 - 6xy - 3y^2$  and  $x = s + 2t$ ,  $y = 3s - t$ .

- (a) Rewrite  $q(x, y)$  in matrix notation, and find the matrix  $A$  representing the quadratic form.  
 (b) Rewrite the linear substitution using matrix notation, and find the matrix  $P$  corresponding to the substitution.  
 (c) Find  $q(s, t)$  using (i) direct substitution, (ii) matrix notation.

**12.36.** For each of the following quadratic forms  $q(x, y, z)$ , find a nonsingular linear substitution expressing the variables  $x, y, z$  in terms of variables  $r, s, t$  such that  $q(r, s, t)$  is diagonal:

- (a)  $q(x, y, z) = x^2 + 6xy + 8y^2 - 4xz + 2yz - 9z^2$ ,  
 (b)  $q(x, y, z) = 2x^2 - 3y^2 + 8xz + 12yz + 25z^2$ ,  
 (c)  $q(x, y, z) = x^2 + 2xy + 3y^2 + 4xz + 8yz + 6z^2$ .

In each case, find the rank and signature.

**12.37.** Give an example of a quadratic form  $q(x, y)$  such that  $q(u) = 0$  and  $q(v) = 0$  but  $q(u + v) \neq 0$ .

**12.38.** Let  $S(V)$  denote all symmetric bilinear forms on  $V$ . Show that

(a)  $S(V)$  is a subspace of  $B(V)$ ; (b) If  $\dim V = n$ , then  $\dim S(V) = \frac{1}{2}n(n+1)$ .

**12.39.** Consider a real quadratic polynomial

$$q(x_1, \dots, x_n) = \sum_{i,j=1}^n a_{ij}x_i x_j \text{ where } a_{ij} = a_{ji}.$$

(a) If  $a_{11} \neq 0$ , show that the substitution

$$x_1 = y_1 - \frac{1}{a_{11}}(a_{12}y_2 + \dots + a_{1n}y_n), \quad x_2 = y_2, \quad \dots, \quad x_n = y_n$$

yields the equation  $q(x_1, \dots, x_n) = a_{11}y_1^2 + q'(y_2, \dots, y_n)$ , where  $q'$  is also a quadratic polynomial.

(b) If  $a_{11} = 0$  but, say,  $a_{12} \neq 0$ , show that the substitution  $x_1 = y_1 + y_2$ ;

$$x_1 = y_1 + y_2, \quad x_2 = y_1 - y_2, \quad x_3 = y_3, \quad \dots, \quad x_n = y_n$$

yields the equation  $q(x_1, \dots, x_n) = P \sum_{i,j} b_{ij} y_i y_j$ , where  $b_{11} \neq 0$ , which reduces this case to case (a).

**Remark:** This method of diagonalizing  $q$  is known as completing the square.

## **Positive Definite Quadratic Forms**

---

**12.40.** Determine whether or not each of the following quadratic forms is positive definite:

- (a)  $q(x, y) = 4x^2 + 5xy + 7y^2$ ,      (c)  $q(x, y, z) = x^2 + 4xy + 5y^2 + 6xz + 2yz + 4z^2$   
 (b)  $q(x, y) = 2x^2 - 3xy - y^2$ ,      (d)  $q(x, y, z) = x^2 + 2xy + 2y^2 + 4xz + 6yz + 7z^2$

**12.41.** Find those values of  $k$  such that the given quadratic form is positive definite:

- (a)  $q(x, y) = 2x^2 - 5xy + ky^2$ ,      (b)  $q(x, y) = 3x^2 - kxy + 12y^2$   
 (c)  $q(x, y, z) = x^2 + 2xy + 2y^2 + 2xz + 6yz + kz^2$

**12.42.** Suppose  $A$  is a real symmetric positive definite matrix. Show that  $A = P^T P$  for some nonsingular matrix  $P$ .

## Hermitian Forms

---

**12.43.** Modify Algorithm 12.1 so that, for a given Hermitian matrix  $H$ , it finds a nonsingular matrix  $P$  for which  $D = P^T H \bar{P}$  is diagonal.

**12.44.** For each Hermitian matrix  $H$ , find a nonsingular matrix  $P$  such that  $D = P^T H P$  is diagonal:

(a)  $H = \begin{bmatrix} 1 & i \\ -i & 2 \end{bmatrix}$ ,      (b)  $H = \begin{bmatrix} 1 & 2+3i \\ 2-3i & -1 \end{bmatrix}$ ,      (c)  $H = \begin{bmatrix} 1 & i & 2+i \\ -i & 2 & 1-i \\ 2-i & 1+i & 2 \end{bmatrix}$

Find the rank and signature in each case.

**12.45.** Let  $A$  be a complex nonsingular matrix. Show that  $H = A^* A$  is Hermitian and positive definite.

**12.46.** We say that  $B$  is Hermitian congruent to  $A$  if there exists a nonsingular matrix  $P$  such that  $B = P^T A \bar{P}$  or, equivalently, if there exists a nonsingular matrix  $Q$  such that  $B = Q^* A Q$ . Show that Hermitian congruence is an equivalence relation. (Note: If  $P = \bar{Q}$  then  $P^T A \bar{P} = Q^* A Q$ .)

**12.47.** Prove [Theorem 12.7](#): Let  $f$  be a Hermitian form on  $V$ . Then there is a basis  $S$  of  $V$  in which  $f$  is represented by a diagonal matrix, and every such diagonal representation has the same number  $p$  of positive entries and the same number  $n$  of negative entries.

## Miscellaneous Problems

---

**12.48.** Let  $e$  denote an elementary row operation, and let  $e^*$  denote the corresponding conjugate column operation (where each scalar  $k$  in  $e$  is replaced by  $\bar{k}$  in  $e^*$ ). Show that the elementary matrix corresponding to  $e^*$  is the conjugate transpose of the elementary matrix corresponding to  $e$ .

**12.49.** Let  $V$  and  $W$  be vector spaces over  $K$ . A mapping  $f: V \times W \rightarrow K$  is called a bilinear form on  $V$  and  $W$  if

- (i)  $f(av_1 + bv_2, w) = af(v_1, w) + bf(v_2, w)$ ,
- (ii)  $f(v, aw_1 + bw_2) = af(v, w_1) + bf(v, w_2)$

for every  $a, b \in K$ ,  $v_i \in V$ ;  $w_j \in W$ . Prove the following:

- (a) The set  $B(V; W)$  of bilinear forms on  $V$  and  $W$  is a subspace of the vector space of functions from  $V \times W$  into  $K$ .
- (b) If  $\{\phi_1, \dots, \phi_m\}$  is a basis of  $V^*$  and  $\{\sigma_1, \dots, \sigma_n\}$  is a basis of  $W^*$ , then  $\{f_{ij} : i = 1; \dots; m; j = 1; \dots; n\}$  is a basis of  $B(V; W)$ , where  $f_{ij}$  is defined by  $f_{ij}(v; w) = \phi_i(v)\sigma_j(w)$ . Thus,  $\dim B(V; W) = \dim V \dim W$ . [Note that if  $V = W$ , then we obtain the space  $B(V)$  investigated in this chapter.]

**12.50.** Let  $V$  be a vector space over  $K$ . A mapping

$f: \overbrace{V \times V \times \dots \times V}^{m \text{ times}} \rightarrow K$  is called a multilinear (or  $m$ -linear) form on  $V$  if  $f$  is linear in each variable; that is, for  $i = 1; \dots; m$ ,

$$f(\dots, \widehat{au + bv}, \dots) = af(\dots, \hat{u}, \dots) + bf(\dots, \hat{v}, \dots)$$

where  $\widehat{i}$  denotes the  $i$ th element, and other elements are held fixed. An  $m$ -linear form  $f$  is said to be alternating if  $f(v_1, \dots, v_m) = 0$  whenever  $v_i = v_j$  for  $i \neq j$ . Prove the following:

- (a) The set  $B_m(V)$  of  $m$ -linear forms on  $V$  is a subspace of the vector space of functions from  $V \times V \times \dots \times V$  into  $K$ .
- (b) The set  $A_m(V)$  of alternating  $m$ -linear forms on  $V$  is a subspace of  $B_m(V)$ .

**Remark 1:** If  $m = 2$ , then we obtain the space  $B(V)$  investigated in this chapter.

**Remark 2:** If  $V = K^m$ , then the determinant function is an alternating  $m$ -linear form on  $V$ .

## ANSWERS TO SUPPLEMENTARY PROBLEMS

*Notation:*  $M = [R_1, R_2, \dots]$  denotes a matrix  $M$  with rows  $R_1, R_2, \dots$

**12.24.** (a) yes, (b) no, (c) yes, (d) no, (e) no, (f) yes

**12.25.** (a)  $A = \frac{1}{2} \begin{bmatrix} 4 & 1 & 7 & 3 \end{bmatrix}$ , (b)  $B = \frac{1}{2} \begin{bmatrix} 0 & 4 & 20 & 32 \end{bmatrix}$ , (c)  $P = \frac{1}{2} \begin{bmatrix} 3 & 5 & 2 \end{bmatrix} \in$

**12.26.** (b)  $\frac{1}{2} \begin{bmatrix} 1 & 0 & 2 & 0 & 0 & 1 & 0 & 2 & 3 & 0 & 5 & 0 & 0 & 3 & 0 & 5 \end{bmatrix}$

**12.33.**

(a)  $[2, -4, 8; -4, 1, 7; -8, 7, 5]$ ,

(b)  $[1, 0, -\frac{1}{2}; 0, 1, 0; -\frac{1}{2}, 0, 0]$ ,

(c)  $[0, \frac{1}{2}, 2; \frac{1}{2}, 1, 0; 2, 0, 1]$ ,

(d)  $[0, \frac{1}{2}, 0; \frac{1}{2}, 0, 1; \frac{1}{2}, 0, \frac{1}{2}; 0, \frac{1}{2}, 0]$

- 12.34.** (a)  $P = \frac{1}{2}1; 0; 2; 0; 1; 2; 0; 0; 1$ ;  $D = \text{diag}(1; 3; 9)$ ;  
 (b)  $P = \frac{1}{2}1; 2; 11; 0; 1; 5; 0; 0; 1$ ;  $D = \text{diag}(1; 1; 28)$ ;  
 (c)  $P = \frac{1}{2}1; 1; 1; 4; 0; 1; 1; 2; 0; 0; 1; 0; 0; 0; 0; 1$ ;  $D = \text{diag}(1; 1; 0; 9)$
- 12.35.**  $A = \frac{1}{2}2; 3; 3; 3$ ,  $P = [1; 2; 3; 1]$ ,  $q(s; t) = 43s^2 - 4st + 17t^2$
- 12.36.** (a)  $x = r - 3s - 19t$ ,  $y = s + 7t$ ,  $z = t$ ;  $q(r, s, t) = r^2 s^2 + 36t^2$ ;  
 (b)  $x = r - 2t$ ,  $y = s + 2t$ ,  $z = t$ ;  $q(r, s, t) = 2r^2 - 3s^2 + 29t^2$ ;  
 (c)  $x = r - st$ ,  $y = st$ ,  $z = t$ ;  $q(r, s, t) = r^2 - 2s^2$
- 12.37.**  $q(x, y) = x^2 - y^2$ ,  $u = (1; 1)$ ,  $v = (1; 1)$
- 12.40.** (a) yes, (b) no, (c) no, (d) yes
- 12.41.** (a)  $k > \frac{25}{8}$ , (b)  $12 < k < 12$ , (c)  $k > 5$
- 12.44.** (a)  $P = [1, i, 0, 1]$ ,  $D = I$ ;  $s = 2$ ; (b)  $P = [1 - i + 3i, 0, 1, D = \text{diag}(1, 14)$ ,  $s = 0$ , (c)  $P = \frac{1}{2}1; i, 3 + i, 0, 1, i, 0, 0, 1$ ,  $D = \text{diag}(1, 1, 4)$ ,  $s = 1$